Gestalten

University of Regensburg, Summer term 2026

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Regensburg, May 26, 2026

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1 Presentable categories

Talk given by Sil.

1.1 Preliminaries

Concepts we should know:

(-2) *Regular cardinal* κ: any union of sets of cardinality $< \kappa$ indexed by a set of cardinality $< \kappa$ has cardinality $< \kappa$.

(-1) *Essentially κ-small (∞-)category C*: (If κ uncountable) The set of isomorphism classes of C has cardinality $< \kappa$, and for all $x, y \in C$ the homotopy groups $\pi_i \hom_C(x, y)$ have cardinality $< \kappa$ (For $\kappa = \aleph_0$ one would take the smallest subcategory of Cat_∞ generated under finite colimits by $[0]$ and $[1]$);

(0) *κ-filtered category I*: Every κ-small diagram in I admits a cocone;

(1) *κ-compact object $X \in C$*: $\mathrm{Hom}_C(X, -): C \to \mathrm{An}$ preserves κ-filtered colimits;

(2) *κ-presentable categories*: A category C which has small colimits and such that every object is a κ-filtered colimit of κ-compact objects;

(3) *Presentable categories*: A category C which is κ-presentable for some κ.

Remark 1.1. C is presentable iff it "admits a presentation:" There is a small category S such that $C \simeq \langle S \rangle[R^{-1}]$, where $\langle S \rangle = \mathrm{PSh}(S) = \mathrm{Fun}(S^{\mathrm{op}}, \mathrm{An})$ is the free cocompletion of S, where R is a (small) set of morphisms in $\langle S \rangle$, and where $\langle S \rangle[R^{-1}] \subseteq \langle S \rangle$ is the full subcategory spanned by those $X \in \langle S \rangle$ such that $\mathrm{Hom}_{\langle S \rangle}(f, X): \mathrm{Hom}_{\langle S \rangle}(B, X) \to \mathrm{Hom}_{\langle S \rangle}(A, X)$ is an isomorphism for every morphism $f: A \to B$ in R.

Similarly, C is κ-presentable if we can choose S and R above such that R consists of maps between κ-compact objects.

Remark 1.2. The description of $\langle S \rangle[R^{-1}]$ as a full subcategory of $\langle S \rangle$ is an instance of the principle that "free constructions in Pr^{L} correspond to cofree constructions in Cat, via passing to right adjoints." Indeed, the localization functor $\langle S \rangle \to \langle S \rangle[R^{-1}]$ is a morphism in Pr^{L}, hence preserves colimits and admits a right adjoint; this right adjoint is automatically fully faithful and identifies $\langle S \rangle[R^{-1}]$ with the subcategory described above. See [Lur09, Proposition 5.5.4.20].

Definition 1.3. We denote by Pr^{L} the category of presentable categories and cocontinuous functors. We let $\mathrm{Pr}^{\mathrm{L}}_{\kappa} \subseteq \mathrm{Pr}^{\mathrm{L}}$ denote the (non-full) subcategory of κ-presentable categories and those cocontinuous functors which preserve κ-compact objects.

Example 1.4. The categories An, Sp, $D(R)$ are all in $\mathrm{Pr}^{\mathrm{L}}_{\aleph_0}$. For X a locally compact Hausdorff space, $\mathrm{Shv}(X) \in \mathrm{Pr}^{\mathrm{L}}_{\aleph_1}$. Also the category of almost modules is $\aleph_1$-presentable.

Lemma 1.5. *Let κ be an uncountable regular cardinal. If D is κ-presentable and I is a κ-small category, then also* $\mathrm{Fun}(I, D)$ *is κ-presentable. Its κ-compact generators are given by* $\mathrm{Fun}(I, D^{\kappa})$.

Proof. This follows from the argument given in [Lur09, Proposition 5.4.4.3]; note that no enlargement of the regular cardinal is needed: since I is κ-small, the argument only uses that κ-filtered colimits of anima commute with κ-small limits. □

Definition 1.6. Let Rex_{κ} be the category of categories with κ-small colimits. (In the case $\kappa = \aleph_0$ we assume the categories are additionally idempotent complete).

Proposition 1.7. *Let C be a κ-presentable category. Then the subcategory $C^{\kappa} \subseteq C$ is essentially small, and the inclusion $C^{\kappa} \hookrightarrow C$ uniquely extends to an equivalence $\mathrm{Ind}_{\kappa}(C^{\kappa}) \xrightarrow{\sim} C$. Moreover, the assignments $C \mapsto C^{\kappa}$ and $D \mapsto \mathrm{Ind}_{\kappa}(D)$ define an equivalence*

$$\mathrm{Rex}_{\kappa} \rightleftarrows \mathrm{Pr}^{\mathrm{L}}_{\kappa}.$$

Theorem 1.8. *The category $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ is κ-presentable. (In other words: $\mathrm{Pr}^{\mathrm{L}}_{\kappa} \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$.)*

Proof. By the previous proposition, $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ is equivalent to Rex_{κ}, so we need to show Rex_{κ} is κ-presentable. To this end, we consider the free-forgetful adjunction

$$\mathrm{fgt}\colon \mathrm{Rex}_{\kappa} \rightleftarrows \mathrm{Cat}\colon \langle - \rangle^{\kappa}.$$

We will first show that this adjunction is monadic. By the Barr–Beck theorem, we need to show:

- The functor fgt is conservative. This is clear.
- The category Rex_{κ} admits fgt-split simplicial colimits, and $\mathrm{fgt}\colon \mathrm{Rex}_{\kappa} \to \mathrm{Cat}$ preserves these. Here we say that a functor $C_{\bullet}\colon \mathbf{\Delta}^{\mathrm{op}} \to \mathrm{Rex}_{\kappa}$ is fgt-*split* if its image $\mathrm{fgt}(C_{\bullet})\colon \mathbf{\Delta}^{\mathrm{op}} \to \mathrm{Cat}$ admits a splitting.

For the second claim, consider such a diagram $C_{\bullet}$. Let $C_{-1} := \mathrm{colim}_{[n] \in \mathbf{\Delta}^{\mathrm{op}}} C_n$ be its colimit in Cat. Then one can show that C_{-1} admits κ-small colimits, and moreover each functor $C_n \to C_{-1}$ preserves κ-small colimits. In particular, the augmented simplicial object in Cat lifts to one in Rex_{κ}. We now claim that this diagram is a colimit in Rex_{κ}. To see this, let $D \in \mathrm{Rex}_{\kappa}$ be any other object. We need to show that $\mathrm{Hom}_{\mathrm{Rex}_{\kappa}}(C_{-1}, D) \xrightarrow{\sim} \lim_{[n] \in \mathbf{\Delta}} \mathrm{Hom}_{\mathrm{Rex}_{\kappa}}(C_n, D)$. **To do.**

We now know that Rex_{κ} is monadic over Cat. Moreover, Cat is $\aleph_0$-presentable, as it is a Bousfield localization of $\mathrm{sAn} = \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{An})$ at finitely many morphisms. We conclude that Rex_{κ} admits all colimits, and it is generated by the objects $\langle [n] \rangle^{\kappa}$. So it suffices to show that these categories are κ-compact in Rex_{κ}. Since $[n] \in \mathrm{Cat}$ is κ-compact (even $\aleph_0$-compact), it suffices to show that $\langle - \rangle^{\kappa}\colon \mathrm{Cat} \to \mathrm{Rex}_{\kappa}$ preserves κ-compact objects. This is equivalent to the claim that its right adjoint $\mathrm{fgt}\colon \mathrm{Rex}_{\kappa} \to \mathrm{Cat}$ preserves κ-filtered colimits, which is an easy computation. □

Proposition 1.9. *The object* $\mathrm{Pr}^{\mathrm{L}}_\kappa$ *of* $\mathrm{Pr}^{\mathrm{L}}_\kappa$ *is* κ*-compact if and only if* κ *is not a limit cardinal. (In particular, this applies to* $\kappa = \aleph_1$*.)*

Lemma 1.10. *Let* $F : C \to D$ *be a morphism in* $\mathrm{Pr}^{\mathrm{L}}_\kappa$*, and let* G *denote its right adjoint. Then the following are equivalent:*

(1) The functor F *is fully faithful;*

(2) The induced functor
$$F^\kappa : C^\kappa \to D^\kappa$$
is fully faithful;

(3) The unit $\eta_X : X \to GFX$ *is an equivalence for every* $X \in C^\kappa$*.*

Proof. The implication (1) $\Rightarrow$ (3) is the usual adjunction criterion for a functor to be fully faithful. The implication (3) $\Rightarrow$ (2) follows from the adjunction: for $X, Y \in C^\kappa$, we have

$$\mathrm{Hom}_D(FX, FY) \simeq \mathrm{Hom}_C(X, GFY) \simeq \mathrm{Hom}_C(X, Y).$$

Conversely, assume that F^κ is fully faithful. For $Y \in C^\kappa$, the map $\eta_Y : Y \to GFY$ induces, for every $X \in C^\kappa$, the map
$$\mathrm{Hom}_C(X, Y) \to \mathrm{Hom}_C(X, GFY) \simeq \mathrm{Hom}_D(FX, FY),$$
which is an equivalence by assumption. Since the κ-compact objects generate C under colimits, this implies that η_Y is an equivalence. Thus (2) $\Rightarrow$ (3).

It remains to see that (2) implies (1). Consider the subcategory $C' \subseteq C$ consisting of those objects X such that the map $\mathrm{Hom}_C(X, Y) \to \mathrm{Hom}_D(FX, FY)$ is an isomorphism for each $Y \in C$. Note that we have $C^\kappa \subseteq C'$: if X is κ-compact, then both $\mathrm{Hom}_C(X, -)$ and $\mathrm{Hom}_D(FX, F(-))$ preserve κ-filtered colimits, and every $Y \in C$ may be written as a κ-filtered colimit of κ-compact objects. Also note that C' is closed under arbitrary colimits, since both $\mathrm{Hom}_C(-, Y)$ and $\mathrm{Hom}_D(F(-), FY)$ send colimits in X to limits. It follows that $C' = C$, showing that F is fully faithful. □

1.2 Basic properties of Pr^{L}

Theorem 1.11. *Let* $F : C \to D$ *be a functor between* $C, D \in \mathrm{Pr}^{\mathrm{L}}$*.*

(1) *Then* F *admits a right adjoint if and only if it preserves colimits.*

(2) *Moreover,* F *admits a left adjoint if and only if it preserves limits and it is* κ*-accessible for some regular cardinal* κ *(i.e. it preserves* κ*-filtered colimits).*

Corollary 1.12. *There is an equivalence* $(\mathrm{Pr}^{\mathrm{L}})^{\mathrm{op}} \simeq \mathrm{Pr}^{\mathrm{R}}$.

Proposition 1.13. (1) *The category* Pr^{L} *admits small limits, which are preserved by* $\mathrm{fgt} : \mathrm{Pr}^{\mathrm{L}} \to \mathrm{Cat}$.

(2) *The category* $\mathrm{Pr}^{\mathrm{L}}_\kappa$ *admits limits, preserved by the composite*

$$\mathrm{Pr}^{\mathrm{L}}_\kappa \simeq \mathrm{Rex}_\kappa \to \mathrm{Cat}.$$

(3) *If κ is uncountable, then the inclusion $\mathrm{Pr}^{\mathrm{L}}_{\kappa} \hookrightarrow \mathrm{Pr}^{\mathrm{L}}$ preserves κ-small limits.*

Proof. For (1) and (3) we refer to Lurie's HTT. For (2), one needs to use that $\mathrm{Rex}_{\kappa} \to \mathrm{Cat}$ is monadic. □

Proposition 1.14. *The categories Pr^{R} and $\mathrm{Pr}^{\mathrm{R}}_{\kappa}$ admit small limits, and the functors $\mathrm{Pr}^{\mathrm{R}}_{\kappa} \to \mathrm{Pr}^{\mathrm{R}} \to \mathrm{Cat}$ preserve small limits.*

Corollary 1.15. *The categories Pr^{L} and $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ admit colimits (which are computed by passing to adjoints and computing the limit in Pr^{R} resp. $\mathrm{Pr}^{\mathrm{R}}_{\kappa}$.)*

Example 1.16. Let I be a set.

(1) The product $\prod_{i\in I} C_i$ in Pr^{L} is just the product of underlying categories. But the product in $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ would be $\mathrm{Ind}_{\kappa}(\prod_{i\in I} C_i^{\kappa})$.

(2) The coproduct $\coprod_{i\in I} C_i$ in Pr^{L} is also the product $\prod_{i\in I} C_i$ in Cat. So is the coproduct in $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$.

(3) The pushout of $*/H$ and $*/G$ over $*$ is $*/(G \star H)$, where $G \star H$ is the free product of G and H, and where $*/G$ denotes the classifying anima of G. But the pushout $\langle */H\rangle \sqcup_{\langle * \rangle} \langle */G\rangle$ is given by the pullback $\langle */H\rangle \times_{\mathrm{An}} \langle */G\rangle$ in Cat.

Remark 1.17. Item (2) of the previous example is striking: the natural map

$$\coprod_{i\in I} C_i \to \prod_{i\in I} C_i$$

in Pr^{L} is an isomorphism for *arbitrary* indexing sets I. Adapting the notion of *semiadditivity* from [Lur17, Definition 6.1.6.13] to arbitrary cardinals, this says that Pr^{L} is κ-semiadditive for every cardinal κ. This is an extremely unusual property of Pr^{L}.

The analogous statement in $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ requires $|I| < \kappa$: the coproduct $\coprod^{\mathrm{Pr}^{\mathrm{L}}_{\kappa}}_{i\in I} C_i$ is always the naive product $\prod_i C_i$, whereas the product $\prod^{\mathrm{Pr}^{\mathrm{L}}_{\kappa}}_{i\in I} C_i = \mathrm{Ind}_{\kappa}(\prod_i C_i^{\kappa})$ generally differs from it; the two agree precisely when $|I| < \kappa$.

1.3 Tensor products in Pr^{L}

Proposition 1.18. *Let $C, D \in \mathrm{Pr}^{\mathrm{L}}$. There exists a universal presentable category $C \otimes D$ with a functor $C \times D \to C \otimes D$ which preserves colimits in both variables. If $C, D \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$, then $C \otimes D \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$, and the functor $C \times D \to C \otimes D$ preserves κ-compact objects.*

Proof. We may set $\langle S\rangle \otimes \langle T\rangle = \langle S \times T\rangle$. Moreover, we may set $\langle S\rangle[R^{-1}] \otimes \langle T\rangle[Q^{-1}] = \langle S \times T\rangle[(R \times Q)^{-1}]$. □

Corollary 1.19. *The categories Pr^{L} and $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ admit symmetric monoidal structures $\mathrm{Pr}^{L,\otimes}$, $\mathrm{Pr}^{L,\otimes}_{\kappa}$.*

Proof. We define these as suboperads of $\mathrm{CAT}^{\times}$. □

Example 1.20. (1) For $C = \langle S\rangle$, we get $C \otimes D \simeq \mathrm{Fun}(S^{\mathrm{op}}, D)$.

(2) For $C \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$ and $D \in \mathrm{Pr}^{\mathrm{L}}$, then $C \otimes D \simeq \mathrm{Fun}^{\kappa\text{-lex}}((C^{\kappa})^{\mathrm{op}}, D)$.

(3) Let A be a (static) ring. Then $D_{\geq 0}(A)$ has compact projective generators A^n for $n \geq 0$, so $D_{\geq 0}(A) \simeq \mathrm{Fun}^{\times}(\{A^n\}^{\mathrm{op}}, \mathrm{An})$. It follows that

$$D_{\geq 0}(A) \otimes C \simeq \mathrm{Fun}^{\times}(\{A^n\}^{\mathrm{op}}, C).$$

(Objects of the RHS are also known as *A-module objects in C*.)

Proposition 1.21. *The symmetric monoidal category $\mathrm{Pr}^{L,\otimes}$ is closed monoidal, with internal hom given by*

$$[C, D] = \mathrm{Fun}^{\mathrm{L}}(C, D),$$

the full subcategory of $\mathrm{Fun}(C, D)$ *spanned by the colimit-preserving functors.*

Moreover, also $\mathrm{Pr}^{L,\otimes}_{\kappa}$ is closed monoidal, with internal hom

$$[C, D] = \mathrm{Ind}_{\kappa}(\mathrm{Fun}^{\kappa\text{-rex}}(C^{\kappa}, D^{\kappa})).$$

Proof. A colimit-preserving functor $E \to \mathrm{Fun}^{\mathrm{L}}(C, D)$ is the same as a functor $E \times C \to D$ which preserves colimits in both variables, which by universality is the same as a colimit-preserving functor $E \otimes C \to D$. It remains to show that $\mathrm{Fun}^{\mathrm{L}}(C, D)$ is presentable. Since κ-small limits in Pr^{L} and $\mathrm{Pr}^{\mathrm{L}}_{\kappa}$ agree, we may reduce to the case $C = \langle [n] \rangle^{\kappa}$. Unwinding the statement in this case, we must prove that for every $D \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$, the functor

$$\mathrm{Ind}_{\kappa}(\mathrm{Fun}([n], D^{\kappa})) \to \mathrm{Fun}([n], \mathrm{Ind}_{\kappa}(D^{\kappa}))$$

is an equivalence. This is HTT.5.3.5.15. □

Proposition 1.22. *We have* $(\mathrm{Pr}^{\mathrm{L}}_{\kappa})^{\otimes} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$.

Proof. Since $(\mathrm{Pr}^{\mathrm{L}}_{\kappa})^{\otimes} \simeq \mathrm{Rex}^{\otimes}_{\kappa}$, it suffices to prove the claim for the latter. The only thing that remains is to show that the functor

$$- \otimes - \colon \mathrm{Rex}_{\kappa} \times \mathrm{Rex}_{\kappa} \to \mathrm{Rex}_{\kappa}$$

preserves κ-compact objects. This in turn reduces to the claim that the tensor product of $\langle [n] \rangle^{\kappa}$ and $\langle [m] \rangle^{\kappa}$ is κ-compact. But this tensor product is $\langle [n] \times [m] \rangle^{\kappa}$, which is indeed κ-compact. □

1.4 C-linear presentable categories

Consider $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$. For $A \in \mathrm{CAlg}(C)$, we may consider the module category $\mathrm{Mod}_A(C) \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$. Moreover, the forgetful functor $\mathrm{Mod}_A(C) \to C$ admits a left adjoint

$$\mathrm{free} \colon C \to \mathrm{Mod}_A(C), \qquad X \mapsto A \otimes X,$$

which defines a functor in $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$.

Given two A-modules M, M', we obtain a *relative tensor product over A*

$$M \otimes_A M' = \mathrm{colim}_{\mathbf{\Delta}^{\mathrm{op}}} \left(\dots \substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow} M \otimes A \otimes A \otimes M' \substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow} M \otimes A \otimes M' \substack{\longrightarrow\\ \longrightarrow} M \otimes M' \right).$$

Definition 1.23. Let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$. Then we may form the module category $\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$ (where we are using that $\mathrm{Pr}^{\mathrm{L}}_{\kappa} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$). We refer to the objects of $\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$ as *C-linear κ-presentable categories* and to its morphisms as *C-linear functors*.

Given $D \in \mathrm{Pr}^{\mathrm{L}}_{\kappa}$ and $B \in \mathrm{CAlg}(D)$, there is always an equivalence

$$\mathrm{CAlg}(D)_{B/} \simeq \mathrm{CAlg}(\mathrm{Mod}_B(D)).$$

Theorem 1.24. *Given $A \in \mathrm{CAlg}(C)$ as before, the relative tensor product defines a symmetric monoidal structure on $\mathrm{Mod}_A(C)$, turning the module category into an object*

$$\mathrm{Mod}_A(C) \in \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa})) \simeq \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})_{C/}.$$

Moreover, the assignment $A \mapsto \mathrm{Mod}_A(C)$ defines a fully faithful functor

$$\mathrm{Mod}_{(-)}(C) \colon \mathrm{CAlg}(C) \hookrightarrow \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa}))$$

which is moreover a morphism in $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$. Its right adjoint

$$\mathrm{End}_{(-)}(\mathbb{1}) \colon \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa})) \to \mathrm{CAlg}(C)$$

sends an object $D \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})_{C/}$ to the endomorphism object of its monoidal unit. This functor preserves κ-filtered colimits.

Notation 1.25. For $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$, we refer to the adjunction of Theorem 1.24 as the *module-endomorphism adjunction*

$$\begin{tikzcd} \mathrm{CAlg}(C) \arrow[r, hook, shift left, "\mathrm{Mod}_{(-)}(C)"] & \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa})), \arrow[l, shift left, "\mathrm{End}_{(-)}(\mathbb{1})"] \end{tikzcd}$$

Thus $R \in \mathrm{CAlg}(C)$ is sent to $\mathrm{Mod}_R(C)$, and the right adjoint sends $D \in \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}^{\mathrm{L}}_{\kappa}))$ to $\mathrm{End}_D(\mathbb{1}_D) \in \mathrm{CAlg}(C)$.

Convention 1.26. From now on, we fix our cutoff cardinal to be $\kappa = \aleph_1$. This choice is somewhat arbitrary: we could have worked with any uncountable regular cardinal. The reason why we need uncountability is to make sure that various countable colimits of compact objects remain compact. For example, given two modules M and M' over an algebra A, the relative tensor product $M \otimes_A M'$ defined above is computed as the geometric realization of the bar complex, hence as a *countable* colimit. Moreover, while κ-compact objects in a presheaf category are generally *retracts* of κ-small colimits of representables, for κ uncountable we may absorb the retract into the colimit.

For these reasons, working with an uncountable cutoff cardinal κ gives better behavior of the κ-compact objects. We will record some basic facts about κ-compact objects in Chapter A.

1.5 Stefanich rings

We are now ready to define the main object of study in these notes.

Definition 1.27. Set

$$0\mathrm{Pr} := \mathrm{An}, \qquad 1\mathrm{Pr} := \mathrm{Pr}^{\mathrm{L}}_{\aleph_1}.$$

For $n \geq 1$, define inductively

$$(n+1)\mathrm{Pr} := \mathrm{Mod}_{n\mathrm{Pr}}(1\mathrm{Pr}) \in \mathrm{CAlg}(1\mathrm{Pr}).$$

Notation 1.28. For every $n \geq 1$, the module-endomorphism adjunction gives an adjunction

$$\mathrm{Mod}^{\mathbb{P}}_n \colon \mathrm{CAlg}((n-1)\mathrm{Pr}) \rightleftarrows \mathrm{CAlg}(n\mathrm{Pr}) : \mathrm{End}^{\mathbb{P}}_n,$$

where

$$\mathrm{Mod}^{\mathbb{P}}_n(R) := \mathrm{Mod}_R((n-1)\mathrm{Pr}), \qquad \mathrm{End}^{\mathbb{P}}_n(D) := \mathrm{End}_D(\mathbb{1}_D).$$

Definition 1.29. We define the category of *Stefanich rings* as the colimit

$$\mathrm{StRing} \;:=\; \mathrm{colim}_{1\mathrm{Pr}}\big(\mathrm{CAlg}(\mathrm{An}) \xrightarrow{\mathrm{Mod}^{\mathbb{P}}_1} \mathrm{CAlg}(1\mathrm{Pr}) \xrightarrow{\mathrm{Mod}^{\mathbb{P}}_2} \mathrm{CAlg}(2\mathrm{Pr}) \xrightarrow{\mathrm{Mod}^{\mathbb{P}}_3} \ldots\big).$$

Equivalently, passing to the right adjoints gives

$$\mathrm{StRing} \simeq \lim_{\mathrm{CAT}}\big(\mathrm{CAlg}(\mathrm{An}) \xleftarrow{\mathrm{End}^{\mathbb{P}}_1} \mathrm{CAlg}(1\mathrm{Pr}) \xleftarrow{\mathrm{End}^{\mathbb{P}}_2} \mathrm{CAlg}(2\mathrm{Pr}) \xleftarrow{\mathrm{End}^{\mathbb{P}}_3} \ldots\big).$$

Notation 1.30. The initial Stefanich ring is denoted by

$$\mathbb{P} := (*, \mathrm{An}, 1\mathrm{Pr}, 2\mathrm{Pr}, \ldots).$$

The notations $\mathrm{Mod}^{\mathbb{P}}_n$ and $\mathrm{End}^{\mathbb{P}}_n$ are used in anticipation of later generalizations for arbitrary Stefanich rings in place of $\mathbb{P}$.

2 Quasicoherent sheaves of higher categories I

Talk given by Christoph.

The goal of the next three talks is to construct higher analogues of quasicoherent sheaves on fpqc-stacks. In this talk we treat the affine case, meaning stacks of the form $\mathrm{Spec}(A)$ for a commutative ring spectrum $A \in \mathrm{CAlg}(\mathrm{Sp})$. We write

$$\mathrm{Aff}_{\mathbb{S}} := \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{op}},$$

and denote by $\mathrm{Spec}(A)$ the object of $\mathrm{Aff}_{\mathbb{S}}$ corresponding to A. A map $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ in $\mathrm{Aff}_{\mathbb{S}}$ is the same thing as a map $A \to B$ in $\mathrm{CAlg}(\mathrm{Sp})$.

For each such affine $\mathrm{Spec}(A)$ we will define presentable categories

$$0\mathrm{Pr}_A,\ 1\mathrm{Pr}_A,\ 2\mathrm{Pr}_A, \ldots$$

which should be thought of as higher versions of $\mathrm{QCoh}(\mathrm{Spec}(A))$. These categories assemble into a Stefanich ring $\mathrm{St}(A)$, resulting in a functor

$$\mathrm{St}\colon\ \mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \hookrightarrow \mathrm{StRing}.$$

The main goals of the talk are then the following:

- We show that the resulting assignments $\mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$ can be upgraded to a six-functor formalism

$$\mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}, \mathrm{all}, \text{countably presented maps}) \to (n+1)\mathrm{Pr}, \qquad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A.$$

- We show that these assignments satisfy *ambidexterity* in sufficiently high degrees.

2.1 Affine Stefanich rings

We start by constructing the categories $n\mathrm{Pr}_A$.

Definition 2.1. Consider a commutative ring spectrum $A \in \mathrm{CAlg}(\mathrm{Sp})$. We denote its category of modules alternatively by

$$0\mathrm{Pr}_A := \mathrm{Mod}_A(\mathrm{Sp}) \in \mathrm{CAlg}(1\mathrm{Pr}).$$

For $n \geq 1$, we inductively define the presentable category $n\mathrm{Pr}_A$ by

$$n\mathrm{Pr}_A := \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr}) \quad \in \quad \mathrm{CAlg}(1\mathrm{Pr}).$$

Observe that $n\mathrm{Pr}_A$ comes equipped with a canonical functor $n\mathrm{Pr} \to n\mathrm{Pr}_A$. This is defined inductively: for $n = 0$, this is the unique symmetric monoidal left adjoint $\mathrm{An} \to \mathrm{Mod}_A(\mathrm{Sp}) = 0\mathrm{Pr}_A$, while for $n \geq 1$ it is defined as the base change functor

$$n\mathrm{Pr} = \mathrm{Mod}_{(n-1)\mathrm{Pr}}(1\mathrm{Pr}) \xrightarrow{(n-1)\mathrm{Pr}_A \otimes_{(n-1)\mathrm{Pr}} -} \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr}) = n\mathrm{Pr}_A.$$

We may thus regard $n\mathrm{Pr}_A$ as an object

$$n\mathrm{Pr}_A \quad \in \quad \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/} \;\simeq\; \mathrm{CAlg}(\mathrm{Mod}_{n\mathrm{Pr}}(1\mathrm{Pr})) \;\simeq\; \mathrm{CAlg}((n+1)\mathrm{Pr}).$$

Observation 2.2. Recall from Notation 1.28 the fully faithful functor

$$\mathrm{Mod}_n^{\mathbb{P}} \colon \mathrm{CAlg}((n-1)\mathrm{Pr}) \hookrightarrow \mathrm{CAlg}(n\mathrm{Pr}), \qquad R \mapsto \mathrm{Mod}_R((n-1)\mathrm{Pr}),$$

with right adjoint $\mathrm{End}_n^{\mathbb{P}}(D) := \mathrm{End}_D(\mathbb{1}_D)$. Note that for $A \in \mathrm{CAlg}(\mathrm{Sp})$, there is a canonical equivalence

$$n\mathrm{Pr}_A = \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr}) \simeq \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(\mathrm{Mod}_{(n-1)\mathrm{Pr}}(1\mathrm{Pr})) = \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(n\mathrm{Pr}) = \mathrm{Mod}_{n+1}^{\mathbb{P}}((n-1)\mathrm{Pr}_A).$$

In particular, since $\mathrm{Mod}_{n+1}^{\mathbb{P}}$ is fully faithful, there is a canonical equivalence

$$\mathrm{End}_{n+1}^{\mathbb{P}}(n\mathrm{Pr}_A) \simeq (n-1)\mathrm{Pr}_A. \tag{2.1}$$

Definition 2.3. Given a commutative ring spectrum A, we define its associated Stefanich ring

$$\mathrm{St}(A) \quad \in \quad \mathrm{StRing} \;=\; \lim_{\mathrm{CAT}}\big(\mathrm{CAlg}(\mathrm{An}) \xleftarrow{\mathrm{End}_1^{\mathbb{P}}} \mathrm{CAlg}(1\mathrm{Pr}) \xleftarrow{\mathrm{End}_2^{\mathbb{P}}} \mathrm{CAlg}(2\mathrm{Pr}) \xleftarrow{\mathrm{End}_3^{\mathbb{P}}} \dots\big)$$

to be the sequence

$$\mathrm{St}(A) := (\Omega^\infty(A), \mathrm{Mod}_A, 1\mathrm{Pr}_A, 2\mathrm{Pr}_A, \dots),$$

equipped with the structure maps from (2.1).

Lemma 2.4. *Let $(C_n)_{n \geq n_0}$ be a sequence of objects of $1\mathrm{Pr}$, together with fully faithful left adjoints $F_n \colon C_n \to C_{n+1}$ with right adjoints $G_n \colon C_{n+1} \to C_n$. Let*

$$C := \mathrm{colim}_{1\mathrm{Pr}}(C_{n_0} \xhookrightarrow{F_{n_0}} C_{n_0+1} \xhookrightarrow{F_{n_0+1}} \dots).$$

Then for every $n \geq n_0$ the canonical functor $\iota_n \colon C_n \to C$ is fully faithful.

Proof. Since the F_n are left adjoints, the colimit defining C may equivalently be computed as the limit along the right adjoints,

$$C \simeq \lim_{\mathrm{CAT}}(\dots \xrightarrow{G_{n+1}} C_{n+1} \xrightarrow{G_n} C_n),$$

and under this identification ι_n is left adjoint to the projection $q_n \colon C \to C_n$. We must show that the unit map $R \to q_n\iota_n(R)$ is an isomorphism for each $R \in C_n$. Since the functors F_m are fully faithful, the sequence of objects

$$X_j = F_{j-1}F_{j-2}\cdots F_n(R) \quad \in \quad C_j$$

for $j \geq n$ defines an object $X := (X_j)_{j\geq n}$ in this limit. We claim that the identification $R = X_n = q_n(X)$ exhibits X as a left-adjoint object to R under q_n (that is, it induces an isomorphism $X \cong \iota_n(R)$). Indeed, for any object $B = (B_j)_{j\geq n}$ of the limit, we have

$$\begin{aligned}\mathrm{Hom}_C(X, B) &\simeq \lim_{j\geq n} \mathrm{Hom}_{C_j}(X_j, B_j)\\ &\simeq \lim_{j\geq n} \mathrm{Hom}_{C_n}(R, G_n \cdots G_{j-1}(B_j))\\ &\simeq \mathrm{Hom}_{C_n}(R, B_n).\end{aligned}$$

Here the second equivalence uses the adjunctions $F_m \dashv G_m$, and the last equivalence uses the compatibility equivalences $G_i(B_{i+1}) \simeq B_i$ defining the object B of the limit. This shows that the unit of the adjunction $\iota_n \dashv q_n$ at $R \in C_n$ is the identity map $R \to R$, hence is an isomorphism, showing that ι_n is fully faithful. □

Lemma 2.5. *The assignment* $A \mapsto \mathrm{St}(A)$ *defines a fully faithful functor*

$$\mathrm{St}\colon \mathrm{CAlg}(\mathrm{Sp}) \hookrightarrow \mathrm{StRing}.$$

Proof. Since we have $n\mathrm{Pr}_A = \mathrm{Mod}^{\mathbb{P}}_{n+1}((n-1)\mathrm{Pr}_A)$ for all n, this assignment is equivalent to the following composite:

$$\mathrm{CAlg}(\mathrm{Sp}) \xrightarrow{\mathrm{Mod}_{(-)}(\mathrm{Sp})} \mathrm{CAlg}(1\mathrm{Pr}) \hookrightarrow \mathrm{StRing},$$

where the second functor is fully faithful by Lemma 2.4, being a colimit over the fully faithful left adjoints

$$\mathrm{Mod}^{\mathbb{P}}_n\colon \mathrm{CAlg}((n-1)\mathrm{Pr}) \hookrightarrow \mathrm{CAlg}(n\mathrm{Pr}).$$

But also the first functor is fully faithful, as it factors as

$$\mathrm{CAlg}(\mathrm{Sp}) \xhookrightarrow{\mathrm{Mod}_{(-)}(\mathrm{Sp})} \mathrm{CAlg}(\mathrm{Mod}_{\mathrm{Sp}}(1\mathrm{Pr})) \hookrightarrow \mathrm{CAlg}(1\mathrm{Pr}),$$

where we use that $\mathrm{Sp} \in \mathrm{CAlg}(1\mathrm{Pr})$ is an idempotent algebra. □

Notation 2.6. Consider a map $f\colon A \to B$ in $\mathrm{CAlg}(\mathrm{Sp})$. By the functoriality of the associated Stefanich ring, this results in a map of Stefanich rings

$$\mathrm{St}(f)\colon \mathrm{St}(A) \to \mathrm{St}(B).$$

The n-th component is a symmetric monoidal functor of the form

$$f_n^*\colon n\mathrm{Pr}_A \to n\mathrm{Pr}_B,$$

which we refer to as the *n-th pullback functor*. Concretely, $f_0^*\colon \mathrm{Mod}_A(\mathrm{Sp}) \to \mathrm{Mod}_B(\mathrm{Sp})$ is extension of scalars along f, and for $n > 0$ the functor $f_n^*\colon \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr}) \to \mathrm{Mod}_{(n-1)\mathrm{Pr}_B}(1\mathrm{Pr})$ is extension of scalars along

$$f_{n-1}^*\colon (n-1)\mathrm{Pr}_A \to (n-1)\mathrm{Pr}_B.$$

2.2 Extension and restriction of scalars

In preparation for the construction of the six-functor formalism $\mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$, we recall some basic properties of extension and restriction of scalars in presentably symmetric monoidal categories. Consider some $\mathcal{V} \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$. Given a map $g\colon R \to S$ in $\mathrm{CAlg}(\mathcal{V})$, we obtain an adjunction

$$g^*\colon \mathrm{Mod}_R(\mathcal{V}) \rightleftarrows \mathrm{Mod}_S(\mathcal{V})\colon g_*$$

given by extension/restriction of scalars. The functor g^* is given on underlying objects in $\mathcal{V}$ by $M \mapsto M \otimes_R S$. It is symmetric monoidal.

Remark 2.7. We recall some basic properties of this adjunction:

(a) The functor $g_*\colon \mathrm{Mod}_S(\mathcal{V}) \to \mathrm{Mod}_R(\mathcal{V})$ commutes with the forgetful functors to $\mathcal{V}$. It is functorial in g: we have $(g'g)_* \simeq g_* g'_*$. Consequently, g^* is functorial in g.

(b) The functor g^* preserves free modules. By definition, an S-module is called *free* if it is in the image of the functor $f^*\colon \mathcal{V} \to \mathrm{Mod}_S(\mathcal{V})$, $M \mapsto S \otimes M$ defined as extension of scalars along $f\colon \mathbb{1} \to R$. So the claim follows from the fact that the composite $\mathbb{1} \to R \to S$ is the unit map $\mathbb{1} \to S$.

(c) The functor $g_*\colon \mathrm{Mod}_S(\mathcal{V}) \to \mathrm{Mod}_R(\mathcal{V})$ preserves colimits. This holds because the map $\mathrm{Mod}_R(\mathcal{V}) \to \mathcal{V}$ creates colimits.

(d) The functor g_* satisfies the **projection formula**: for $N \in \mathrm{Mod}_S(\mathcal{V})$ and $M \in \mathrm{Mod}_R(\mathcal{V})$, the map

$$M \otimes_R g_*N \to g_*(g^*M \otimes_S N)$$

adjoint to $g^*(M \otimes_R g_*N) \simeq g^*M \otimes_S g^*g_*N \to g^*M \otimes_S N$ is an isomorphism. Since both sides preserve colimits in M, we may reduce the claim to the case of a free module $M = f^*M' = R \otimes M'$, in which case we see that both sides reduce to $M' \otimes g_*N$ by an explicit computation of the formula for the relative tensor product.

(e) The functors g_* satisfy **base change**: Consider a pushout square in $\mathrm{CAlg}(\mathcal{V})$ of the form

$$\begin{tikzcd} R \arrow[r] \arrow[d] & S \arrow[d] \\ T \arrow[r] & U, \end{tikzcd}$$

meaning that $U \simeq T \otimes_R S$ as commutative algebras. By passing to extension of scalar functions, we obtain a commutative diagram in $\mathrm{CAlg}(1\mathrm{Pr})$ of the form

$$\begin{tikzcd} \mathrm{Mod}_R(\mathcal{V}) \arrow[r] \arrow[d] & \mathrm{Mod}_S(\mathcal{V}) \arrow[d] \\ \mathrm{Mod}_T(\mathcal{V}) \arrow[r] & \mathrm{Mod}_U(\mathcal{V}). \end{tikzcd}$$

The horizontal functors have right adjoints, and we need to show that both composites $\mathrm{Mod}_S(\mathcal{V}) \to \mathrm{Mod}_T(\mathcal{V})$ agree, along the canonical map. Computing both sides, we get $T \otimes_R M$ versus $(T \otimes_R S) \otimes_S M$, and these agree. (Details omitted.)

(f) Finally, assume that $\mathcal{V} \in \mathrm{CAlg}(1\mathrm{Pr})$, and let $g: R \to S$ be a countably presented map, meaning that S is $\aleph_1$-compact when regarded as an object of $\mathrm{CAlg}_R(\mathcal{V})$. Then the functor

$$g_*: \mathrm{Mod}_S(\mathcal{V}) \to \mathrm{Mod}_R(\mathcal{V})$$

preserves $\aleph_1$-compact objects by Corollary A.3, applied with $\kappa = \aleph_1$ and $C = \mathcal{V}$. In combination with (c), we conclude that g_* is a right adjoint to g^* internal to the 2-category 1Pr.

Lemma 2.8. *Let $\mathcal{V} \in \mathrm{CAlg}(1\mathrm{Pr})$, and let $g: R \to S$ be a countably presented map in $\mathrm{CAlg}(\mathcal{V})$. Then also the induced functor*

$$g^*: \mathrm{Mod}_R(\mathcal{V}) \to \mathrm{Mod}_S(\mathcal{V})$$

is countably presented in $\mathrm{CAlg}(\mathrm{Mod}_{\mathcal{V}}(1\mathrm{Pr})) \simeq \mathrm{CAlg}_{\mathcal{V}}(1\mathrm{Pr})$.

Proof. Recall from Theorem 1.24 that the functor

$$\mathrm{CAlg}(\mathrm{Mod}_{\mathcal{V}}(1\mathrm{Pr})) \to \mathrm{CAlg}(\mathcal{V}), \qquad D \mapsto \mathrm{End}_D(\mathbb{1}_D)$$

preserves $\aleph_1$-filtered colimits. It follows that its left adjoint

$$\mathrm{CAlg}(\mathcal{V}) \hookrightarrow \mathrm{CAlg}(\mathrm{Mod}_{\mathcal{V}}(1\mathrm{Pr})), \qquad R \mapsto \mathrm{Mod}_R(\mathcal{V})$$

preserves $\aleph_1$-compact objects. Replacing $\mathcal{V}$ by $\mathrm{Mod}_R(\mathcal{V})$ then gives the claim. □

Corollary 2.9. *Let $g: A \to B$ be a countably presented morphism of commutative ring spectra. Then for each $n \geq 0$, the functor*

$$g_n^*: n\mathrm{Pr}_A \to n\mathrm{Pr}_B$$

is countably presented as a map in $\mathrm{CAlg}((n+1)\mathrm{Pr})$.

Proof. By induction this is immediate from Lemma 2.8. □

Corollary 2.10. *Let $f: A \to B$ be a countably presented map in $\mathrm{CAlg}(\mathrm{Sp})$. Then, for every $n \geq 0$, the object $n\mathrm{Pr}_B \in (n+1)\mathrm{Pr}_A \simeq \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$ is $\aleph_1$-compact.*

Proof. By the previous corollary, $n\mathrm{Pr}_B$ is $\aleph_1$-compact as an object of $\mathrm{CAlg}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$. By Lemma A.1 this is equivalent to it being $\aleph_1$-compact as an object of $\mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$. □

2.3 Construction of the six-functor formalism

We will now construct the desired six-functor formalism on $\mathrm{Aff}_{\mathbb{S}}$ we are after.

Definition 2.11. A map $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ in $\mathrm{Aff}_{\mathbb{S}}$ is called *countably presented* if B is countably presented as an A-algebra.

This class of maps satisfies the usual closure properties.

Lemma 2.12. *Let $A \in \mathrm{CAlg}(\mathrm{Sp})$.*

(i) Countably presented A-algebras are closed under countable colimits, retracts, and finite colimits in $\mathrm{CAlg}_A(\mathrm{Sp})$.

(ii) *Countably presented maps of affine spectral schemes are stable under base change and composition.*

(iii) *If $A \to B$ and $A \to C$ are countably presented and $B \to C$ is a map over A, then $B \to C$ is countably presented.*

(iv) *If* $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *is countably presented and admits a section* $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$*, then the section is countably presented.*

Proof. Part (i) is Lemma A.4(1), applied with $\kappa = \aleph_1$ and $C = \mathrm{Sp}$.

For base change in (ii), suppose $A \to B$ is countably presented and $A \to A'$ is any map. The base change is $\mathrm{Spec}(A' \otimes_A B) \to \mathrm{Spec}(A')$, and $A' \to A' \otimes_A B$ is countably presented by Lemma A.4(2). For composition, if $A \to B$ and $B \to C$ are countably presented, then $A \to C$ is countably presented by Lemma A.4(3).

Part (iii) is Lemma A.4(4). Finally, a section $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ corresponds to an A-algebra retraction $B \to A$, so (iv) is exactly Lemma A.5. ◻

We now come to the main result of this talk:

Theorem 2.13. *For every $n \geq 0$, there exists a six-functor formalism*

$$\begin{array}{ccc} \mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}, \text{all}, \text{countably presented maps}) & \to & (n+1)\mathrm{Pr} \\ \mathrm{Spec}(A) & \mapsto & n\mathrm{Pr}_A \\ \mathrm{Spec}(A) \xleftarrow{f} \mathrm{Spec}(B) \xrightarrow{g} \mathrm{Spec}(C) & \mapsto & n\mathrm{Pr}_A \xrightarrow{f_n^*} n\mathrm{Pr}_B \xrightarrow{g_{n,*}} n\mathrm{Pr}_C, \end{array}$$

where $g_{n,*}$ is right adjoint to g_n^*.

Proof. We start with the case $n = 0$. The functoriality of module categories provides a functor

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \to 1\mathrm{Pr}, \qquad \mathrm{Spec}(A) \mapsto \mathrm{Mod}_A(\mathrm{Sp}), \qquad f \mapsto f^*.$$

To extend this to the span category via the right adjoints, it suffices to check the following properties:

- For every countably presented map $g\colon \mathrm{Spec}(A) \to \mathrm{Spec}(B)$, the functor $g^*\colon \mathrm{Mod}_B(\mathrm{Sp}) \to \mathrm{Mod}_A(\mathrm{Sp})$ admits a right adjoint g_* internal to the 2-category $1\mathrm{Pr}$. This was verified in Remark 2.7(f).
- These right adjoints g_* satisfy base change. This was verified in Remark 2.7(e).
- For every such f, the right adjoint f_* satisfies the projection formula. This was verified in Remark 2.7(d).

The proof for $n > 0$ is essentially identical. Functoriality of the assignment $A \mapsto n\mathrm{Pr}_A$ provides a functor

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \to (n+1)\mathrm{Pr}, \qquad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A, \qquad f \mapsto f_n^*.$$

If $g\colon \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is countably presented, then by Corollary 2.9 so is the functor

$$g_{n-1}^*\colon (n-1)\mathrm{Pr}_A \to (n-1)\mathrm{Pr}_B$$

as a morphism in CAlg(nPr). It then follows from Remark 2.7(f) that the associated restriction of scalars functor

$$g_{n,*} = (g_{n-1}^*)_* \colon n\mathrm{Pr}_B \to n\mathrm{Pr}_A$$

is an internal right adjoint to g_n^* in $(n+1)$Pr. Moreover, the other parts of Remark 2.7 show that these right adjoints satisfy the required base change and projection formula conditions. □

Remark 2.14. If we drop the countable presentation hypotheses, we still obtain a presentable six-functor formalism:

$$\mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}) \to \mathrm{Pr}^{\mathrm{L}}, \qquad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$$

2.4 Ambidexterity

It turns out that for $n \geq 1$ the symmetric monoidal pullback functor f_n^* admits not only a well-behaved right adjoint $f_{n,*}$, but also a left adjoint, and the two coincide. This phenomenon, of a functor having identical left and right adjoints, is known as *ambidexterity*, a term introduced by Hopkins–Lurie [HL13]. Before applying this to our functors f_n^*, we set up the general framework.

Definition 2.15. Let C be an $(\infty, 2)$-category, and let $F \colon X \to Y$ be a 1-morphism in C.

- We say F is a *left adjoint* if there exist a 1-morphism $F^R \colon Y \to X$ together with 2-morphisms

  $$\eta \colon \mathrm{id}_X \to F^R F \quad \text{in } \mathrm{End}_C(X), \qquad \varepsilon \colon F F^R \to \mathrm{id}_Y \quad \text{in } \mathrm{End}_C(Y),$$

  such that the composites

  $$F \xrightarrow{F\eta} F F^R F \xrightarrow{\varepsilon F} F \qquad \text{and} \qquad F^R \xrightarrow{\eta F^R} F^R F F^R \xrightarrow{F^R \varepsilon} F^R$$

  are equivalent to the identity.

- Dually, F is a *right adjoint* if there exist a 1-morphism $F^L \colon Y \to X$ and 2-morphisms $\eta \colon \mathrm{id}_X \to F F^L$, $\varepsilon \colon F^L F \to \mathrm{id}_Y$ satisfying the analogous triangle identities.

For the present discussion the reader may take C to be an object of 2Pr, though the definition makes sense for any reasonable notion of $(\infty, 2)$-category. If F is a left adjoint, the right adjoint F^R is determined up to canonical isomorphism, and the left adjoint of F^R is again F (with the same η and ε).

Remark 2.16. In our formalism of presentable (∞, n)-categories, there are forgetful functors $n\mathrm{Pr} \to k\mathrm{Pr}$ for any $k \leq n$. Indeed, the symmetric monoidal functor An $\to$ 1Pr (since An is the unit object) yields, by passing to modules, symmetric monoidal functors $n\mathrm{Pr} \to (n+1)\mathrm{Pr}$ inductively, whose right adjoints are the desired forgetful functors. In particular, Definition 2.15 applies to 1-morphisms in nPr for any $n \geq 2$.

We can now state the general ambidexterity result.

Proposition 2.17. *Let C be an $(\infty, 3)$-category, and let $F \colon X \to Y$ be a 1-morphism in C. The following are equivalent.*

(i) The morphism F admits a right adjoint $F^R\colon Y \to X$, and both

$$\eta\colon \mathrm{id}_X \to F^R F \quad \textit{in } \mathrm{End}_C(X) \qquad \textit{and} \qquad \varepsilon\colon FF^R \to \mathrm{id}_Y \quad \textit{in } \mathrm{End}_C(Y)$$

admit right *adjoints (in the respective $(\infty,2)$-categories).*

(ii) The morphism F admits a left adjoint $F^L\colon Y \to X$, and both

$$\eta\colon \mathrm{id}_X \to FF^L \quad \textit{in } \mathrm{End}_C(X) \qquad \textit{and} \qquad \varepsilon\colon F^L F \to \mathrm{id}_Y \quad \textit{in } \mathrm{End}_C(Y)$$

admit left *adjoints.*

Moreover, when these conditions hold, there is a canonical identification $F^L \simeq F^R$.

Remark 2.18. There is a dual version of Proposition 2.17, in which the words "right" and "left" are swapped in (i) and (ii) respectively; it again yields a canonical identification $F^L \simeq F^R$. However, in any given application, η and ε must be assumed to admit the *same* type of adjoint for the result to apply.

Proof. We show that (i) implies (ii) with $F^L = F^R$; the dual implication is similar. Given (F^R, η, ε) as in (i), define

$$\eta'\colon \mathrm{id}_X \to FF^R \qquad \text{and} \qquad \varepsilon'\colon F^R F \to \mathrm{id}_Y$$

by $\eta' := \varepsilon^R$ and $\varepsilon' := \eta^R$. The triangle identities for $(F, F^R, \eta', \varepsilon')$ follow from those of $(F, F^R, \eta, \varepsilon)$ together with the fact that forming right adjoints is compatible with composition. By construction, η' and ε' admit left adjoints (namely ε and η themselves), giving (ii). □

We now establish an ambidexterity property for the functors $f_{n,*}$.

Proposition 2.19. *Let $n \geq 1$, and let $f\colon A \to B$ be a countably presented map in* $\mathrm{CAlg}(\mathrm{Sp})$.

- *The right adjoint $f_{n,*}\colon n\mathrm{Pr}_B \to n\mathrm{Pr}_A$ is also* left *adjoint to f_n^*. Moreover, it satisfies base change and the projection formula as a left adjoint.*
- *The category $n\mathrm{Pr}_B$ is self-dual as an object of* $(n+1)\mathrm{Pr}_A \simeq \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$.

Proof. Since we have the adjunction $f_n^* \dashv f_{n,*}$, we obtain a unit and counit

$$\eta\colon \mathrm{id} \to f_{n,*}f_n^* \in \mathrm{End}_{(n+1)\mathrm{Pr}_A}(n\mathrm{Pr}_A) \qquad\qquad \varepsilon\colon f_n^* f_{n,*} \to \mathrm{id} \in \mathrm{End}_{(n+1)\mathrm{Pr}_A}(n\mathrm{Pr}_B).$$

We will deduce the proposition from Proposition 2.17, applied in the $(\infty,3)$-category $C = (n+1)\mathrm{Pr}_A$ (which lies in $(n+2)\mathrm{Pr}$ and hence in $3\mathrm{Pr}$, since $n \geq 1$); concretely, we need to verify that η and ε admit right adjoints. For η, note that evaluation at the unit induces an equivalence

$$\mathrm{End}_{(n+1)\mathrm{Pr}_A}(n\mathrm{Pr}_A) \xrightarrow{\sim} n\mathrm{Pr}_A.$$

Under this equivalence, the unit η corresponds to the map

$$f_{n-1}^*\colon (n-1)\mathrm{Pr}_A \to (n-1)\mathrm{Pr}_B,$$

which indeed has a right adjoint $f_{n-1,*}$ in $n\mathrm{Pr}_A$.

For ε, note that there is an equivalence

$$\begin{aligned}\mathrm{End}_{(n+1)\mathrm{Pr}_A(1\mathrm{Pr})}(n\mathrm{Pr}_B) &= \mathrm{End}_{\mathrm{Mod}_{n\mathrm{Pr}_A}}(\mathrm{Mod}_{(n-1)\mathrm{Pr}_B}(n\mathrm{Pr}_A), \mathrm{Mod}_{(n-1)\mathrm{Pr}_B}(n\mathrm{Pr}_A)) \\ &\simeq \mathrm{Mod}_{(n-1)\mathrm{Pr}_B \otimes_{(n-1)\mathrm{Pr}_A} (n-1)\mathrm{Pr}_B}(n\mathrm{Pr}_A).\end{aligned}$$

Under this equivalence, the counit ε corresponds to the 'multiplication map'

$$(n-1)\mathrm{Pr}_B \otimes_{(n-1)\mathrm{Pr}_A} (n-1)\mathrm{Pr}_B \to (n-1)\mathrm{Pr}_B.$$

The left-hand side is equivalent to $(n-1)\mathrm{Pr}_{B\otimes_A B}$, and under this equivalence the map agrees with ∇^*_{n-1} induced by the codiagonal map $\nabla\colon B \otimes_A B \to B$. For this we also know it admits a $(n-1)\mathrm{Pr}_{B\otimes_A B}$-linear right adjoint.

For the claim about base change, we apply the same result inside the 3-category $\mathrm{Ar}((n+1)\mathrm{Pr}_A)$, the arrow category of $(n+1)\mathrm{Pr}_A$. Since adjoints in this arrow category are precisely the adjointable squares, this gives the claim.

Finally, for the duality data on $n\mathrm{Pr}_B$, we are using the following two maps:

$$n\mathrm{Pr}_A \xrightarrow{f_n^*} n\mathrm{Pr}_B \xrightarrow{\nabla_{n,*}} n\mathrm{Pr}_B \otimes_{n\mathrm{Pr}_A} n\mathrm{Pr}_B$$

$$n\mathrm{Pr}_B \otimes_{n\mathrm{Pr}_A} n\mathrm{Pr}_B \xrightarrow{\nabla_n^*} n\mathrm{Pr}_B \xrightarrow{f_{n,*}} n\mathrm{Pr}_A.$$

The triangle identities are an immediate consequence of base change. □

3 Descent for quasicoherent sheaves

Talk given by Melvin.

We use the following output of the previous talk:

- For $A \in \mathrm{CAlg}(\mathrm{Sp})$, we defined categories $n\mathrm{Pr}_A$ by
$$0\mathrm{Pr}_A := \mathrm{D}(A) := \mathrm{Mod}_A(\mathrm{Sp}), \qquad (n+1)\mathrm{Pr}_A := \mathrm{Mod}_{n\mathrm{Pr}_A}((n+1)\mathrm{Pr}) \simeq \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr}).$$
- The assignment $\mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$ defines a six-functor formalism on $\mathrm{Aff}_{\mathbb{S}}$ valued in $1\mathrm{Pr}$.
- For $n \geq 1$, we have ambidexterity: if $f \colon \mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is countably presented, then the right adjoint $f_{n,*} \colon n\mathrm{Pr}_A \to n\mathrm{Pr}_B$ is also a left adjoint to f_n^* in a canonical way.

3.1 Hyperdescent

The goal of this chapter is the following hyperdescent statement:

Theorem 3.1 (cf. [Mat16, Corollary 3.33, Proposition 3.45])**.** *The functor $n\mathrm{Pr}^!_{(-)} \colon \mathrm{Aff}_{\mathbb{S}} \to 1\mathrm{Pr}$ (defined on maps by the right adjoints $f^!$) is a hypersheaf with respect to the fpqc topology for each $n \geq 0$.*

Remark 3.2. For $n \geq 1$, the maps $f^!$ agree with f^* by ambidexterity.

To make sense of this statement, we recall the following two definitions:

Definition 3.3. Let C be a site with finite limits. A *hypercover* of $A \in C$ is a simplicial object $A_\bullet \in \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, C)$ together with a map $A_0 \to A$ which is a cover in C, such that if we right Kan extend $A_\bullet$ to $A(-) \in \mathrm{Fun}(\mathrm{PSh}_{\mathrm{fin}}(\mathbf{\Delta})^{\mathrm{op}}, C)$, the map $A(\Delta^n) \to A(\partial\Delta^n)$ is a cover in C.

Definition 3.4. Recall the following basic definitions:

- For a static ring $B_0 \in \mathrm{CAlg}(\mathrm{Ab})$ and a static B_0-module M_0, we say that M_0 is *flat* if the functor $M_0 \otimes_{B_0} - \colon \mathrm{Mod}_{B_0}(\mathrm{Ab}) \to \mathrm{Ab}$ is exact. It is called *faithfully flat* if, in addition, the functor $M_0 \otimes_{B_0} -$ is conservative, i.e. detects isomorphisms. (Equivalently it detects exact sequences.)
- A morphism $B_0 \to A_0$ in $\mathrm{CAlg}(\mathrm{Ab})$ is called *(faithfully) flat* if A_0 is (faithfully) flat as a B_0-module.

- A morphism $B \to A$ in $\mathrm{CAlg}(\mathrm{Sp})$ is *(faithfully) flat* if the map $\pi_0 B \to \pi_0 A$ in $\mathrm{CAlg}(\mathrm{Ab})$ is (faithfully) flat and for each $i \in \mathbb{Z}$ the multiplication map $\pi_i B \otimes_{\pi_0 B} \pi_0 A \to \pi_i A$ is an isomorphism.
- A morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ in $\mathrm{Aff}_{\mathbb{S}}$ is called (faithfully) flat if its associated map $B \to A$ in $\mathrm{CAlg}(\mathrm{Sp})$ is.

Definition 3.5. The *fpqc topology* on $\mathrm{Aff}_{\mathbb{S}}$ is the Grothendieck topology in which a sieve over $\mathrm{Spec}(B)$ is declared to be a covering sieve if it contains a finite collection of morphisms $\mathrm{Spec}(A_i) \to \mathrm{Spec}(B)$ such that the induced map $\mathrm{Spec}(\prod_{i=1}^n A_i) \to \mathrm{Spec}(B)$ is faithfully flat.

Observe that a functor $F\colon \mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \to C$ is a hypersheaf with respect to the fpqc topology if and only if the following two properties are satisfied:

- The functor F preserves finite products.
- For an fpqc-hypercover $\mathrm{Spec}(A_\bullet) \to \mathrm{Spec}(A)$, the induced map

  $$F(\mathrm{Spec}(A)) \to \lim_{[n]\in\mathbf{\Delta}} F(\mathrm{Spec}(A_n))$$

  is an isomorphism.

The proof has three parts:

- The main argument treats countably presented fpqc-hypercovers; the final section reduces arbitrary fpqc-hypercovers to this case.
- We first prove the theorem for $n = 0$.
- We then prove the general case by induction on n.

3.2 The base case $n = 0$

We will start by showing that the assignment $\mathrm{Spec}(A) \mapsto \mathrm{D}(A)$ satisfies fpqc-hyperdescent. Since we have $\mathrm{D}(\prod_i A_i) \xrightarrow{\sim} \prod_i \mathrm{D}(A_i)$, it remains to show that for an fpqc-hypercover $\mathrm{Spec}(A_\bullet) \to \mathrm{Spec}(A)$ in $\mathrm{Aff}_{\mathbb{S}}$ the functor

$$\mathrm{D}(A) \xrightarrow{\sim} \lim^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_\bullet)$$

is an equivalence. Here the limit on the right is taken along the functors $f^!$.

3.2.1 Step 1: The functor is fully faithful

We start by showing that this comparison functor is fully faithful.

Note that the functor $\mathrm{D}(A) \xrightarrow{\sim} \lim^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_\bullet)$ admits a left adjoint, given by sending an object $(A_n)_{[n]\in\mathbf{\Delta}}$ to the colimit $\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} (f_n)_* A_n$, where we denote by $f_n\colon \mathrm{Spec}(A_n) \to \mathrm{Spec}(A)$ the maps constituting the hypercover. So we must show that for any object $M \in \mathrm{D}(A)$, the counit map

$$\mathrm{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} (f_n)_* f_n^! M \to M$$

is an isomorphism of A-modules. Note that we may rewrite the left-hand side as

$$\operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}}(f_n)_* f_n^! M = \operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}}(f_n)_* \underline{\mathrm{Hom}}_{A_n}(A_n, f_n^! M) \simeq \operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}}(f_n)_* \underline{\mathrm{Hom}}_{A}(A_n, M).$$

We may now filter $\mathbf{\Delta}$ by the subcategories $\mathbf{\Delta}_{\leq n}$ to rewrite this colimit in turn as

$$\operatorname{colim}_{m\geq 0} \operatorname{colim}_{[n]\in(\mathbf{\Delta}_{\leq m})^{\mathrm{op}}} \underline{\mathrm{Hom}}_A(A_n, M) \simeq \operatorname{colim}_{m\geq 0} \underline{\mathrm{Hom}}_A(\lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_\bullet, M),$$

where we use that a colimit over $\mathbf{\Delta}_{\leq m}$ is a finite colimit and is thus preserved by the exact functor $\underline{\mathrm{Hom}}_A(-, M)\colon \mathrm{D}(A) \to \mathrm{D}(A)$. If we similarly write $M \simeq \underline{\mathrm{Hom}}_A(A, M) \simeq \operatorname{colim}_{m\geq 0} \underline{\mathrm{Hom}}_A(A, M)$, then the comparison map is obtained by applying $\underline{\mathrm{Hom}}_A(-, M)$ to the canonical map of pro-systems $A \to \{\lim_{\mathbf{\Delta}_{\leq m}} A_\bullet\}_{m\geq 0}$ in $\mathrm{D}(A)$. So we need to show that this map is a pro-isomorphism in $\mathrm{D}(A)$.

We first justify the reduction to the connective case.

Lemma 3.6. *Let $A \to A_\bullet$ be an fpqc-hypercover of commutative ring spectra. Put $A^+ := \tau_{\geq 0} A$ and $A_n^+ := \tau_{\geq 0} A_n$. Then $A^+ \to A_\bullet^+$ is an fpqc-hypercover. Moreover, for every $n \geq 0$ we have a canonical equivalence*

$$A_n \simeq A \otimes_{A^+} A_n^+,$$

and for every $m \geq 0$ the canonical map

$$A \otimes_{A^+} \lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_n^+ \longrightarrow \lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_n$$

is an equivalence in $\mathrm{D}(A)$. Consequently, to show that

$$A \longrightarrow \{\lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_n\}_{m\geq 0}$$

is a pro-isomorphism in $\mathrm{D}(A)$, it suffices to prove the analogous statement for $A^+ \to A_\bullet^+$.

Proof. We first record the basic flatness input. If $R \to S$ is flat, then $R^+ := \tau_{\geq 0} R \to S^+ := \tau_{\geq 0} S$ is flat and the canonical map

$$R \otimes_{R^+} S^+ \to S$$

is an equivalence. More generally, for every R-algebra T there is a canonical equivalence

$$T^+ \otimes_{R^+} S^+ \simeq \tau_{\geq 0}(T \otimes_R S).$$

Indeed, both statements follow by comparing homotopy groups, using the defining condition for flatness:

$$\pi_i S \simeq \pi_i R \otimes_{\pi_0 R} \pi_0 S.$$

Let K be a finite simplicial set, and write A_K for the commutative ring spectrum corresponding to the value of the right Kan extension of $\mathrm{Spec}(A_\bullet)$ on K; similarly, write A_K^+ for the corresponding object associated with $\mathrm{Spec}(A_\bullet^+)$. We claim, by induction over the cells of K, that

$$A_K^+ \simeq \tau_{\geq 0} A_K,$$

and that A_K is flat over A.

The claim is clear for $K = \emptyset$, where $A_K = A$. Suppose that L is obtained from K by attaching an r-simplex,

$$L = K \cup_{\partial\Delta^r} \Delta^r.$$

Then

$$A_L \simeq A_K \otimes_{A_{\partial\Delta^r}} A_r.$$

The matching map $A_{\partial\Delta^r} \to A_r$ is faithfully flat by the hypercover condition. Hence $A_K \to A_L$ is flat, and so A_L is flat over A. Moreover, by the flatness input above and the induction hypothesis, we get

$$\tau_{\geq 0}A_L \simeq \tau_{\geq 0}A_K \otimes_{\tau_{\geq 0}A_{\partial\Delta^r}} \tau_{\geq 0}A_r \simeq A_K^+ \otimes_{A^+_{\partial\Delta^r}} A_r^+ \simeq A_L^+.$$

This proves the induction.

Applying this to $K = \partial\Delta^r$, we see that the r-th matching map for $A^+ \to A^+_\bullet$ is the connective cover of the faithfully flat map

$$A_{\partial\Delta^r} \to A_r.$$

It is therefore faithfully flat, so $A^+ \to A^+_\bullet$ is again an fpqc-hypercover.

Since A_n is flat over A, the flatness input also gives $A_n \simeq A \otimes_{A^+} A_n^+$ for every n. Finally, the last displayed equivalence follows because the functor

$$A \otimes_{A^+} -\colon \mathrm{D}(A^+) \to \mathrm{D}(A)$$

is exact and $\mathbf{\Delta}_{\leq m}$ is finite:

$$A \otimes_{A^+} \lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_n^+ \simeq \lim_{[n]\in\mathbf{\Delta}_{\leq m}} (A \otimes_{A^+} A_n^+) \simeq \lim_{[n]\in\mathbf{\Delta}_{\leq m}} A_n.$$

The final assertion follows by applying this exact base-change functor to the pro-isomorphism for the connective hypercover. $\square$

By Lemma 3.6, we may assume from now on that both A and all of the A_i are connective.

Now, to show that the map $A \to \{\lim_{\mathbf{\Delta}_{\leq m}} A_\bullet\}_{m\geq 0}$ is an isomorphism in $\mathrm{Pro}(\mathrm{D}(A))$, we may equivalently show its cofiber is zero. The cofiber is given by the pro-system $\{C_m\}_{m\geq 0}$, where we set

$$C_m := \mathrm{cofib}(A \to \lim_{\mathbf{\Delta}_{\leq m}} A_\bullet).$$

It will thus suffice to show that the map $C_{m+2} \to C_m$ is the zero map in $\mathrm{D}(A)$ for each $m \geq 0$. For this, recall the following definition:

Definition 3.7. A morphism $X \to Y$ in $\mathrm{D}(A)$ is called a *phantom map* if the induced map $\pi_i(X) \to \pi_i(Y)$ is the zero map for each $i \in \mathbb{Z}$. Equivalently, for every perfect A-module Z and every map $Z \to X$, the composite $Z \to X \to Y$ is nullhomotopic.

Observation 3.8. Every A-module X may be written as a filtered colimit of perfect A-modules: simply consider the collection of maps $\{X' \to X\}$ with X' perfect. If we write $X = \mathrm{colim}_{i\in I} X_i$, then we see that a map $X \to Y$ is phantom if and only if it lies in the kernel of the map

$$\pi_0(\lim_i \hom_A(X_i, Y)) \cong \pi_0 \hom_A(X, Y) \to \lim_i \pi_0 \hom_A(X_i, Y).$$

This kernel is $\lim^1_i \pi_{-1} \hom_A(X_i, Y)$.

Lemma 3.9 ([HPS97, Theorem 4.2.5])**.** *The composite of two phantom maps is nullhomotopic.*

Proof sketch. Consider two phantom maps $W \to X \to Y$. By the above observation, both maps come from $\lim^1$-terms. Their composite would then be an element of some $\lim^2$. But $\lim^2$ of abelian groups are zero. $\square$

In light of the lemma, it remains to show that each map $C_{m+1} \to C_m$ is a phantom map. We now make the following claim:

Claim 3.10. For each $m \geq 0$, the A-module $C_m := \mathrm{cofib}(A \to \lim_{\mathbf{\Delta}_{\leq m}} A_\bullet)$ is a countably presented flat A-module in degree $m-1$, i.e. $C_m[-(m-1)]$ is a flat A-module.

We will prove this claim below, but let us first explain why it implies that $C_{m+1} \to C_m$ is phantom. For this, recall the following spectral analogue of Lazard's theorem:

Theorem 3.11 (Lazard, HA 7.2.2.15)**.** *Let A be a connective ring spectrum. Then every connective countably presented flat A-module M is a sequential colimit of finite rank free A-modules:*

$$M \simeq \mathrm{colim}_{i \geq 0} A^{\oplus n_i}.$$

Combining this with Claim 3.10, we may thus write $C_{m+1} \simeq \mathrm{colim}_{i\geq 0} A^{\oplus n_i}[-m]$. Now, if Z is any perfect A-module and $Z \to C_{m+1}$ is any map, then it must factor through some finite stage $Z \to A^{\oplus n_i}[-m]$. So it suffices to show that any map $A[-m] \to C_m$ is nullhomotopic in $\mathrm{D}(A)$. But homotopy classes of such maps correspond to elements of $\pi_{-m}(C_m)$, which is 0 since by Claim 3.10 C_m is concentrated in degree $m-1$.

We now move towards a proof of Claim 3.10. For this, recall that an A-module M is flat if and only if $M \otimes_A \pi_0(A)$ is a flat static $\pi_0 A$-module. So we may check the claim after base changing to $\pi_0 A$. Observe that base change commutes with the cofiber and with the finite limit $\lim_{\leq m} A_\bullet$. So we may assume that A and $A_\bullet$ are static.

The augmented cosimplicial object $\{A_\bullet\}$ may be turned into a cochain complex

$$A \to A_0 \to A_1 \to \dots$$

by taking the alternating sums of the face maps (the unreduced version of Dold–Kan). We claim that this is an exact sequence. We may test this after base change along faithfully flat maps $A \to B$. Since our cosimplicial object is an fpqc-hypercover, we may then base change to objects after which the sequence becomes split, hence exact. Details are omitted here.

It then follows (why?) that

$$C_m = \mathrm{cofib}(A \to \lim_{\mathbf{\Delta}_{\leq m}} A_\bullet) \simeq \mathrm{coker}(A_{m-1} \to A_m)[m-1].$$

Thus, Claim 3.10 reduces to the claim that $\mathrm{coker}(A_{m-1} \to A_m)$ is a countably presented static flat A-module.

Scholze says the following about this: "Moreover, the same holds true after any base change in A, in particular along a quotient map $A \to A/I$; this shows that this cokernel is flat over A."

All in all, we have now shown that the functor $\mathrm{D}(A) \to \lim^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_n)$ is fully faithful.

3.2.2 Step 2: Essential surjectivity

We now show the functor $\mathrm{D}(A) \to \lim^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_n)$ is an equivalence.

Step 2a: We first reduce to the case where $A_\bullet$ is a Čech cover. To this end, consider the following commutative square:

$$\begin{tikzcd} A \arrow[r] \arrow[d] & A_i \arrow[d] \\ A_0 \arrow[r] & A_0 \otimes_A A_i. \end{tikzcd}$$

Since the cosimplicial object at the bottom is split, we have

$$\mathrm{D}(A_0) \xrightarrow{\sim} \lim\nolimits^!_{[i]\in\mathbf{\Delta}} \mathrm{D}(A_0 \otimes_A A_i),$$

and similarly $\mathrm{D}(A_0^{\otimes_A n}) \xrightarrow{\sim} \lim^!_{[i]\in\mathbf{\Delta}} \mathrm{D}(A_0^{\otimes_A n} \otimes_A A_i)$ for all $n \geq 0$. So we are done if we can show that

$$\mathrm{D}(A) \xrightarrow{\sim} \lim\nolimits^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_0^{\otimes_A n}) \qquad \text{and} \qquad \mathrm{D}(A_i) \xrightarrow{\sim} \lim\nolimits^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_0^{\otimes_A n} \otimes_A A_i),$$

which is the special case of Čech descent.

Step 2b: Assume now that $A_i \cong A_0^{\otimes_A i}$ is the Čech cover of $A \to A_0$. To show that the map $\mathrm{D}(A) \to \lim^!_{[n]\in\mathbf{\Delta}} \mathrm{D}(A_n)$ is an isomorphism in Pr^{R}, we may pass to left adjoints and show that

$$\operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{D}(A_n) \to \mathrm{D}(A)$$

is an isomorphism in Pr^{L}. Here the colimit is taken along the functors $(f_n)_*\colon \mathrm{D}(A_n) \to \mathrm{D}(A)$. Recall that each of these functors lives in $\mathrm{Pr}_{\mathrm{D}(A)} = \mathrm{Mod}_{\mathrm{D}(A)}(\mathrm{Pr})$, and hence the colimit may be computed in $\mathrm{Mod}_{\mathrm{D}(A)}(\mathrm{Pr})$. Also note that if we apply the functor $\mathrm{D}(A_0) \otimes_{\mathrm{D}(A)} -\colon \mathrm{Mod}_{\mathrm{D}(A)}(\mathrm{Pr}) \to \mathrm{Mod}_{\mathrm{D}(A_0)}(\mathrm{Pr})$, the augmented simplicial diagram admits a splitting, since $\mathrm{D}(A_0) \otimes_{\mathrm{D}(A)} \mathrm{D}(A_n) \cong \mathrm{D}(A_{n+1})$. It follows that the map $\operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{D}(A_n) \to \mathrm{D}(A)$ becomes an isomorphism after applying $\mathrm{D}(A_i) \otimes_{\mathrm{D}(A)} -$ for each i, and thus by passing to colimits also after applying $(\operatorname{colim}_i \mathrm{D}(A_i)) \otimes_{\mathrm{D}(A)} -$.

Now, by Step 1 we know that the functor $\mathrm{D}(A) \to \lim^!_{\mathbf{\Delta}} \mathrm{D}(A_\bullet) \simeq \operatorname{colim}_* \mathrm{D}(A_\bullet)$ is fully faithful. In particular, if we denote the image of the monoidal unit $\mathbb{1} \in \mathrm{D}(A)$ by M, then we see that M is sent to $\mathbb{1}$ under the $\mathrm{D}(A)$-linear left adjoint $\operatorname{colim}_* \mathrm{D}(A_\bullet)$. If we consider the $\mathrm{D}(A)$-linear functor $\mathrm{D}(A) \to \operatorname{colim}_* \mathrm{D}(A_\bullet)$ determined by M, it thus follows that the composite

$$\mathrm{D}(A) \to \operatorname{colim}_* \mathrm{D}(A_\bullet) \to \mathrm{D}(A)$$

is the identity in $\mathrm{Pr}_{\mathrm{D}(A)}$, showing that $\mathrm{D}(A)$ is a retract of this colimit. It thus follows that the map $\operatorname{colim}_{[n]\in\mathbf{\Delta}^{\mathrm{op}}} \mathrm{D}(A_n) \to \mathrm{D}(A)$ becomes an isomorphism after applying $\mathrm{D}(A) \otimes_{\mathrm{D}(A)} -$. Since this is the identity functor, we are done.

This finishes the proof in the countably presented case, for $n = 0$.

3.3 The induction step for $n > 0$

We now assume that $n > 0$, and that we have already proved the claim for $n - 1$. Observe that we again have an adjunction

$$F_n^*\colon n\mathrm{Pr}_A \rightleftarrows \lim n\mathrm{Pr}_{A_\bullet}\colon F_{n,\sharp}.$$

Also, in this case each of the individual adjunctions $n\mathrm{Pr}_A \rightleftarrows n\mathrm{Pr}_{A_\bullet}$ is an ambidextrous adjunction.

We first show that F_n^* is fully faithful, or equivalently that the counit map

$$F_{n,\sharp}F_n^* \to \mathrm{id}$$

is an isomorphism. Note that this adjunction is $n\mathrm{Pr}_A$-linear, so it suffices to show this claim on the monoidal unit: we need to show that the map $F_{n,\sharp}F_n^*(n-1)\mathrm{Pr}_A \to (n-1)\mathrm{Pr}_A$ is an equivalence.

For $n = 1$, this map is $F_{n,\sharp}F_n^*\,\mathrm{D}(A) \xrightarrow{\sim} \mathrm{D}(A)$, which is the base case we just showed.

For $n \geq 2$, we need to show that $F_{n,\sharp}F_n^*(n-1)\mathrm{Pr}_A \xrightarrow{\sim} (n-1)\mathrm{Pr}_A$. One can show that the LHS is dualizable, so we may equivalently show that the dual map $(n-1)\mathrm{Pr}_A \to F_{n,\sharp}F_n^*(n-1)\mathrm{Pr}_A$ is an isomorphism. But this is a linear map, so it agrees with the map one level lower. This concludes the induction.

Essential surjectivity is proved similarly to the previous situation, by reducing to the Čech-cover. (It's even a bit easier, as we have ambidexterity.)

This finishes the proof.

3.4 Reduction to the countably presented case

Finally, we show that the claim for $n = 0$ can be reduced to the countably presented case. We use the following lemma:

Lemma 3.12. *Every faithfully flat map $A \to B$ of commutative ring spectra is an $\aleph_1$-filtered colimit of countably presented such maps.*

Proof. First, if A is static, then so is B. This case is due to Waterhouse;[1] let us quickly recall the argument. By Lazard's theorem, the flat A-modules are the Ind-category of the finite projective A-modules, so we can write the underlying module of B as an $\aleph_1$-filtered colimit of countably presented flat modules M_i. Since each M_i is countably presented, it is $\aleph_1$-compact as an A-module, so $\mathrm{Hom}_A(M_i, -)$ commutes with $\aleph_1$-filtered colimits. We will use two consequences of this: any map from a countably presented A-module into $B = \mathrm{colim}_i\, M_i$ factors through some M_i, and any two such maps that agree after composing to B already agree at some stage M_i.

The tensor product $M_i \otimes_A M_i$ is again countably presented, so the composite $M_i \otimes_A M_i \to B \otimes_A B \to B$ of the canonical map with the multiplication of B factors through some M_j, $j = j(i)$, compatibly with the maps to B; similarly the unit $A \to B$ factors through some M_{i_0}. Choosing inductively a sequence $i_0, i_1, \ldots$ with $i_{n+1} \geq j(i_n)$ yields compatible maps $M_{i_n} \otimes_A M_{i_n} \to M_{i_{n+1}}$, and the colimit $M := \mathrm{colim}(M_{i_0} \to M_{i_1} \to \ldots)$ acquires a multiplication

$$M \otimes_A M = \mathrm{colim}_n (M_{i_n} \otimes_A M_{i_n}) \to \mathrm{colim}_n\, M_{i_{n+1}} = M$$

and a unit $A \to M_{i_0} \to M$, both compatible with $M \to B$.

The commutativity, associativity, and unitality of this multiplication each amount to an equality of two maps out of a countably presented module ($M_{i_n}^{\otimes 2}$, resp. $M_{i_n}^{\otimes 3}$, resp. M_{i_n}) that holds after composing into the commutative A-algebra B. By the second consequence above, each such identity

[1] W. C. Waterhouse, *Basically bounded functors and flat sheaves*, Pacific J. Math. **57** (1975), no. 2, 597–610.

holds after passing to a later stage; as there are only countably many of them, they can all be arranged to hold on a single cofinal subsystem. On such a subsystem M is a countably presented flat A-subalgebra of B, and these subalgebras form an $\aleph_1$-filtered system with colimit B. This writes B in the desired form.

In general, write the map $A \to B$ as the $\aleph_1$-filtered colimit of all countably presented A-algebras B_i mapping to B. We want to see that the set of i for which $A \to B_i$ is faithfully flat is cofinal. We get a similar presentation of $\pi_0 A$ as the colimit of the $\pi_0 B_i$, and by the static case, for any i there is some j so that $\pi_0 B_i \to \pi_0 B_j$ factors over a faithfully flat $\pi_0 A$-algebra. Taking a sequence $i_0, i_1, \ldots$ so that each map $\pi_0 B_{i_0} \to \pi_0 B_{i_1} \to \ldots$ factors over a faithfully flat $\pi_0 A$-algebra, the colimit $B_j = \mathrm{colim}(B_{i_0} \to B_{i_1} \to \ldots)$ is still countably presented, and has the property that $\pi_0 A \to \pi_0 B_j$ is faithfully flat. Using a similar argument, we can arrange that $\pi_0 B_i \otimes_{\pi_0 A} \pi_n A \to \pi_n B_i$ is an isomorphism for any given i, and hence for all i by a further countable colimit. □

As a consequence, every fpqc-hypercover is an $\aleph_1$-filtered colimit of countably presented fpqc-hypercovers.

Now, given any hypercover $\mathrm{Spec}(A_\bullet) \to \mathrm{Spec}(A)$, we have

$$\lim^!_{\mathbf{\Delta}} n\mathrm{Pr}_{A_\bullet} \simeq \lim^!_{\mathbf{\Delta}} \mathrm{colim}_r\, n\mathrm{Pr}_{A_{r,\bullet}} \simeq \mathrm{colim}_r \lim^!_{\mathbf{\Delta}} n\mathrm{Pr}_{A_{n},\bullet} \simeq n\mathrm{Pr}_A,$$

where we use that $1\mathrm{Pr}$ is $\aleph_1$-compactly generated so that $\aleph_1$-filtered colimits commute with countable limits. This finishes the proof of the general form of the theorem.

4 Quasicoherent sheaves of higher categories II

Talk given by Bastiaan.

We now pass from affine spectral schemes ($\mathrm{Aff}_{\mathbb{S}} := \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{op}}$) to fpqc-stacks.

On affines, Talk 2 introduced the functors

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \to \mathrm{CAlg}((n+1)\mathrm{Pr}) = \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/}, \qquad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$$

for all $n \geq 0$, and showed they canonically extend to six-functor formalisms in which every countably presented map is cohomologically proper. Denote by

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}}) \;\subseteq\; \mathrm{PSh}(\mathrm{Aff}_{\mathbb{S}})$$

the subcategory of *small* fpqc-sheaves. Here *small* means that the sheaf is a small colimit of representable objects. (In fact, we will mostly restrict to sheaves which are countable colimits of affines.) By the main result of Talk 3, we obtain a unique limit-preserving extension

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\mathrm{op}} \to \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/}, \qquad X \mapsto n\mathrm{Pr}_X.$$

Since the endomorphism functor $\mathrm{End}\colon (n+1)\mathrm{Pr} \to n\mathrm{Pr}$ commutes with limits, this means that every fpqc-stack X defines a Stefanich ring

$$\mathrm{St}(X) := (\Omega^\infty \mathcal{O}(X), \mathrm{D}(X), 1\mathrm{Pr}_X, 2\mathrm{Pr}_X, \ldots),$$

with n-th component $\mathrm{St}(X)_n := (n-1)\mathrm{Pr}_X$ (using the low-degree conventions $0\mathrm{Pr}_X := \mathrm{D}(X)$ and $(-1)\mathrm{Pr}_X := \Omega^\infty\mathcal{O}(X)$). This extends the affine assignment $A \mapsto \mathrm{St}(A)$ to a *Stefanich ring functor*

$$\mathrm{St}\colon \mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\mathrm{op}} \to \mathrm{StRing},$$

restricting to $\mathrm{St}\colon \mathrm{CAlg}(\mathrm{Sp}) \hookrightarrow \mathrm{StRing}$ on affines.

For $n = 0$, the value $0\mathrm{Pr}_X = \mathrm{D}(X)$ is the usual derived ∞-category of quasicoherent sheaves on X, at least when X is a countable colimit of affines.

Remark 4.1. In general, there is a difference between taking the limit defining $n\mathrm{Pr}_X$ inside $1\mathrm{Pr} = \mathrm{Pr}^{\mathrm{L}}_{\aleph_1}$ or inside the larger category Pr^{L} of all presentable categories. Usually, $\mathrm{QCoh}(X)$ is defined using limits in Pr^{L}. For countable colimits of affines the two limits agree.

The rest of the chapter has four goals:

- For $n \geq 1$, we extend the assignment $X \mapsto n\mathrm{Pr}_X$ to a six-functor formalism, with !-able maps given by so-called 'bi-countably presented maps';
- We show that "every bi-countably presented map is étale", and for $n \geq 2$ that in fact "every bi-countably presented map is proper";
- We deduce a Künneth formula for sheaves of $(n-1)$-presentable categories on a fiber product;
- We conclude that the Stefanich ring functor preserves countable colimits and finite limits.

4.1 Bi-countably presented maps

We will now extend the assignment $X \mapsto n\mathrm{Pr}_X$ to a six-functor formalism. We may think of this as a 'higher' version of the six-functor formalism on $X \mapsto D(X)$. A first question is: what are the !-able maps? It turns out that at the higher categorical level, the class of !-able maps contains essentially *all* maps. The relevant class of maps is the following:

Definition 4.2. Let $A \in \mathrm{CAlg}(\mathrm{Sp})$. A map $Z \to \mathrm{Spec}(A)$ of fpqc-sheaves is *bi-countably presented over* $\mathrm{Spec}(A)$ if Z is a countable colimit, in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})_{/\mathrm{Spec}(A)}$, of affines $\mathrm{Spec}(B_i)$ with $A \to B_i$ countably presented.

A map $f\colon Y \to X$ of fpqc-stacks is *bi-countably presented* if for every affine $\mathrm{Spec}(A) \to X$, the fiber product $Y \times_X \mathrm{Spec}(A)$ is bi-countably presented over $\mathrm{Spec}(A)$. Let E denote the class of bi-countably presented maps.

Lemma 4.3. *Let $S = \mathrm{Spec}(A)$, and let $p\colon Z \to S$ be bi-countably presented. Then every section $s\colon S \to Z$ of p is bi-countably presented as a map to Z.*

Proof. We first record two elementary fpqc-local reductions. Let $S' = \mathrm{Spec}(A') \to S = \mathrm{Spec}(A)$ be faithfully flat and countably presented. A map $W \to S$ is bi-countably presented if and only if its base change $W_{S'} \to S'$ is bi-countably presented. The forward implication is base change. Conversely, write $S'_\bullet$ for the Čech nerve of $S' \to S$. Since $A \to A'$ is countably presented, each $S'_m \to S$ is countably presented by Lemma 2.12. If $W_{S'}$ is a countable colimit of affines countably presented over S', then each $W_{S'_m}$ is a countable colimit of affines countably presented over S. By fpqc-descent,

$$W \simeq |W_{S'_\bullet}|,$$

and the indexing category $\mathbf{\Delta}^{\mathrm{op}}$ is countable. Hence W is again a countable colimit of affines countably presented over S.

Second, let

$$Z \simeq \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i)$$

be a countable affine presentation over S, and let $T = \mathrm{Spec}(C) \to S$ be affine. Any finite collection of sections $T \to Z_T := Z \times_S T$ factors, after a faithfully flat countably presented base change $T' \to T$, through stages of the induced presentation

$$Z_T \simeq \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i \otimes_A C).$$

Indeed, by the definition of sheafification, this factorization exists after some fpqc-cover of T. By Lemma 3.12, and because the stages are countably presented over T, this cover can be refined by a faithfully flat countably presented affine cover. For finitely many sections we take a common refinement.

We now prove the lemma. Let $T = \mathrm{Spec}(C) \to Z$ be an affine test object. We must show that

$$T \times_Z S \to T$$

is bi-countably presented. After base change along $T \to S$, the map $Z_T \to T$ has two sections: the section t induced by the given map $T \to Z$, and the section s_T obtained by pulling back s. By the fpqc-local reductions above, we may pass to a faithfully flat countably presented cover of T and assume that both sections factor through stages of a countable presentation of Z_T.

Let $\mathrm{Aff}_T^{\mathrm{cp}}$ be the full subcategory of affine T-schemes which are countably presented over T. It has finite limits by Lemma 2.12. Let $\widetilde{Z}$ be the corresponding countable colimit of representables in $\mathrm{PSh}(\mathrm{Aff}_T^{\mathrm{cp}})$. The two sections $s_T, t \colon T \to Z_T$ now come from maps

$$* \rightrightarrows \widetilde{Z}$$

in this presheaf category, where $*$ denotes the terminal representable. Consider the presheaf fiber product

$$P := * \times_{\widetilde{Z}} *.$$

The objects $*$ and $\widetilde{Z}$ are $\aleph_1$-compact in $\mathrm{PSh}(\mathrm{Aff}_T^{\mathrm{cp}})$, and Proposition A.10 shows that P is again $\aleph_1$-compact. Since presheaf categories are generated by representables, P is a countable colimit of representables from $\mathrm{Aff}_T^{\mathrm{cp}}$. Sheafification preserves finite limits, so the sheafification of P is precisely

$$T \times_{Z_T} T \simeq T \times_Z S.$$

Thus, fpqc-locally on T, the map $T \times_Z S \to T$ is bi-countably presented. The first reduction descends this conclusion to T itself. □

Proposition 4.4. *The class E is closed under composition, base change, sections, and formation of diagonals.*

Proof. Closure under base change is immediate from the definition. For closure under composition, let $Z \to Y \to X$ be two maps in E and let $\mathrm{Spec}(A) \to X$ be affine. Write

$$Y_A := Y \times_X \mathrm{Spec}(A) \simeq \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i),$$

where I is countable and each $A \to B_i$ is countably presented. By base change, $Z_A \to Y_A$ is bi-countably presented. Applying the definition of bi-countably presented to the affine test objects $\mathrm{Spec}(B_i) \to Y_A$, each

$$Z_A \times_{Y_A} \mathrm{Spec}(B_i)$$

is a countable colimit of affines $\mathrm{Spec}(C_{i,k})$ with $B_i \to C_{i,k}$ countably presented. Since countably presented maps are closed under composition, each $A \to C_{i,k}$ is countably presented. Universality of colimits in sheaves gives

$$Z_A \simeq \mathrm{colim}_{i \in I} \left(Z_A \times_{Y_A} \mathrm{Spec}(B_i) \right) \simeq \mathrm{colim}_{(i,k)} \mathrm{Spec}(C_{i,k}),$$

and the final indexing category is still countable. Hence $Z \to X$ lies in E.

It remains to discuss sections and diagonals. The diagonal $\Delta_f\colon Y \to Y \times_X Y$ is a section of the first projection
$$Y \times_X Y \to Y.$$
This projection is a base change of f, hence lies in E. Since E is already closed under base change, it is therefore enough to know that E is closed under sections.

Let $p\colon Y \to X$ lie in E, let $s\colon X \to Y$ be a section, and test s against an affine map $a\colon \operatorname{Spec}(A) \to Y$. Base change p along the composite $\operatorname{Spec}(A) \xrightarrow{a} Y \xrightarrow{p} X$. Since $p \in E$, the resulting map $Y_A := Y \times_X \operatorname{Spec}(A) \to \operatorname{Spec}(A)$ is bi-countably presented over $\operatorname{Spec}(A)$. The map a gives one section of $Y_A \to \operatorname{Spec}(A)$, and $s \circ p \circ a$ gives another. Moreover
$$X \times_Y \operatorname{Spec}(A) \simeq \operatorname{Spec}(A) \times_{Y_A} \operatorname{Spec}(A)$$
with respect to these two sections. By Lemma 4.3, the first section $\operatorname{Spec}(A) \to Y_A$ is bi-countably presented as a map to Y_A. Pulling it back along the second section shows that
$$\operatorname{Spec}(A) \times_{Y_A} \operatorname{Spec}(A) \to \operatorname{Spec}(A)$$
is bi-countably presented over $\operatorname{Spec}(A)$. Thus $s \in E$, so E is closed under sections and therefore under diagonals. □

4.2 A six-functor formalism for nPr at $n \geq 1$

We fix $n \geq 1$. Given a map $f\colon Y \to X$ of fpqc-stacks, we denote the resulting symmetric monoidal pullback functor by
$$f_n^*\colon n\mathrm{Pr}_X \to n\mathrm{Pr}_Y,$$
and its right adjoint by $f_{n,*}$. We denote by $\otimes_{n\mathrm{Pr}_X}$ the tensor product of $n\mathrm{Pr}_X$.

Remark 4.5. Throughout this talk, we will frequently abuse notation and regard $n\mathrm{Pr}_Y$ as an object of $(n+1)\mathrm{Pr}_X$ by setting
$$n\mathrm{Pr}_Y := f_{n+1,*}(\mathbb{1}_Y) \quad \in \quad (n+1)\mathrm{Pr}_X.$$
For $p_X : X \to \mathrm{pt}$, we get a 'forgetful functor' $p_{X,n+1,*}\colon (n+1)\mathrm{Pr}_X \to (n+1)\mathrm{Pr}$, and we claim that the image of $f_{n+1,*}(\mathbb{1}_Y)$ under this functor is $n\mathrm{Pr}_Y$. Indeed, if we write $Y \simeq \operatorname{colim}_{\operatorname{Spec}(A)\to Y} \operatorname{Spec}(A)$, then we may write $p_{X,n+1,*}f_{n+1,*}(\mathbb{1}) = p_{Y,n+1,*}(\mathbb{1})$ as a limit over $p_{A,n+1}(\mathbb{1}) = n\mathrm{Pr}_A$, which is $n\mathrm{Pr}_Y$ by definition.

Theorem 4.6. *Let $f\colon Y \to X$ be a bi-countably presented map of fpqc-stacks, and let $n \geq 1$.*

(i) The functor
$$f_n^*\colon n\mathrm{Pr}_X \to n\mathrm{Pr}_Y$$
admits a left adjoint $f_{n,\sharp}$ in $(n+1)\mathrm{Pr}_X$. In particular, $f_{n,\sharp}$ commutes with all small colimits and with tensoring.

(ii) The object $n\mathrm{Pr}_Y \in (n+1)\mathrm{Pr}_X$ is $\aleph_1$-compact.

(iii) The formation of $f_{n,\sharp}$ commutes with base change: for every cartesian square

$$\begin{tikzcd} Y' \arrow[r, "v"] \arrow[d, "f'"'] & Y \arrow[d, "f"] \\ X' \arrow[r, "u"'] & X \end{tikzcd}$$

the Beck–Chevalley transformation

$$u_n^* f_{n,\sharp} \to f'_{n,\sharp} v_n^*$$

is an equivalence.

(iv) For $n \geq 2$, the object $f_{n,\sharp}(\mathbb{1}) \in n\mathrm{Pr}_X$ is dualizable.

(v) For $n \geq 2$, the left adjoint $f_{n,\sharp}$ is also right *adjoint to f_n^*, i.e. $f_{n,\sharp} \simeq f_{n,*}$. In particular, $f_{n,*}$ also commutes with small colimits, tensoring, and base change.*

In other words: at level $n \geq 1$, "all maps are étale", and at level $n \geq 2$, "all maps are proper".

Proof. **Step 1: The affine case.** We first assume that $X = \mathrm{Spec}(A)$ is affine and prove (i)-(iv). By assumption $Y = \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i)$ is a countable colimit of affines $Y_i = \mathrm{Spec}(B_i)$, where each B_i is countably presented over A. Consider

$$f_n^* \colon n\mathrm{Pr}_X = n\mathrm{Pr}_A \to n\mathrm{Pr}_Y = \lim_{i \in I^{\mathrm{op}}} n\mathrm{Pr}_{B_i}.$$

Since $n \geq 1$ and all transition maps are countably presented, the result from Talk 2 shows that all the pullback functors involved admit linear left adjoints. In particular, the limit defining $n\mathrm{Pr}_Y$ is along right adjoints, and hence agrees with the colimit along the corresponding left adjoints:

$$n\mathrm{Pr}_Y \simeq \mathrm{colim}_{i \in I, \sharp}\, n\mathrm{Pr}_{B_i}.$$

(We use countability of I here so that this colimit takes place inside the $\aleph_1$-presentable setting 1Pr.) In this colimit description, the desired left adjoint is given as the colimit of the lower-$\sharp$ maps $n\mathrm{Pr}_{B_i} \to n\mathrm{Pr}_A$:

$$f_{n,\sharp} \colon n\mathrm{Pr}_Y \simeq \mathrm{colim}_{i \in I, \sharp}\, n\mathrm{Pr}_{B_i} \to n\mathrm{Pr}_A.$$

(See Horev and Yanovski [HY17, Corollary 1.3].) This is a map in $(n+1)\mathrm{Pr}_A = \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$, and in particular it commutes with all small colimits and with tensoring. This gives (i). We may explicitly write the resulting functor as

$$f_{n,\sharp} : \lim_{i \in I^{\mathrm{op}}} n\mathrm{Pr}_{B_i} \to n\mathrm{Pr}_A, \qquad (Z_i) \mapsto \mathrm{colim}_{i \in I} (g_i)_{n,\sharp}(Z_i),$$

with $g_i : \mathrm{Spec}(B_i) \to \mathrm{Spec}(A)$ the affine structure map.

For (ii), since $\aleph_1$-compact objects in $(n+1)\mathrm{Pr}_A$ are closed under countable colimits, the $\aleph_1$-compactness of $n\mathrm{Pr}_Y$ follows from the $\aleph_1$-compactness of the $n\mathrm{Pr}_{B_i}$ established in Talk 2.

For (iii), let $A \to A'$ be a base change, and write $B'_i := B_i \otimes_A A'$ and $Y' := Y \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A')$. Since colimits in sheaves are universal,

$$Y' \simeq \mathrm{colim}_{i \in I} \mathrm{Spec}(B'_i).$$

Hence the same construction applied over A' gives

$$f'_{n,\sharp}\colon n\mathrm{Pr}_{Y'} \to n\mathrm{Pr}_{A'}, \qquad (Z'_i) \mapsto \mathrm{colim}_{i\in I}(g'_i)_{n,\sharp}(Z'_i),$$

where $g'_i : \mathrm{Spec}(B'_i) \to \mathrm{Spec}(A')$ is the base change of $g_i\colon \mathrm{Spec}(B_i) \to \mathrm{Spec}(A)$. The base-change property is known from Talk 2:

$$u_n^*(g_i)_{n,\sharp} \simeq (g'_i)_{n,\sharp}(v_i)_n^*.$$

Thus base change carries the diagram whose colimit defines $f_{n,\sharp}$ to the diagram whose colimit defines $f'_{n,\sharp}$. Since the indexing category I is countable, this colimit is taken inside 1Pr, and we obtain the desired Beck–Chevalley equivalence

$$u_n^* f_{n,\sharp} \simeq f'_{n,\sharp} v_n^*.$$

For (iv), we evaluate at the unit. In the affine case, we have $(g_i)_{n,\sharp} = (g_i)_{n,*}$ by ambidexterity from Talk 2, so

$$f_{n,\sharp}(\mathbb{1}) \simeq \mathrm{colim}_{i\in I}(g_i)_{n,*}(\mathbb{1}) \simeq \mathrm{colim}_{i\in I,*}(n-1)\mathrm{Pr}_{B_i},$$

where the colimit is along the lower-$*$ functors. For $n \geq 2$ (so that $n-1 \geq 1$), the lower-$*$ functors $(n-1)\mathrm{Pr}_{B_i} \to (n-1)\mathrm{Pr}_{B_j}$ are themselves linear left adjoints. Hence the colimit above is a countable colimit of dualizable objects along internal left adjoint functors. Such a colimit is itself dualizable, by the following more general result:

Claim: For $\mathcal{V} \in \mathrm{CAlg}(1\mathrm{Pr})$, countable colimits of dualizable objects along internal left adjoints in $\mathrm{Pr}_{\mathcal{V}}$ are again dualizable.

This is a result by Ramzi. In short, the reason is that the dualizable objects can be characterized as those C that are retracts of $\mathcal{P}_{\mathcal{V}}(C^{\aleph_1})$ in a canonical way. Since this retract diagram is functorial in C, it shows closure under countable colimits.

Step 2: The general case. For general X, we may write X as a colimit of affine spectral schemes, providing an equivalence

$$(n+1)\mathrm{Pr}_X \simeq \lim_{\mathrm{Spec}(A)\to X}(n+1)\mathrm{Pr}_A.$$

This is a limit in $1\mathrm{Pr} = \mathrm{Pr}^{\mathrm{L}}_{\aleph_1} \simeq \mathrm{Rex}_{\aleph_1}$, so on the level of underlying $(\infty,1)$-categories we have

$$((n+1)\mathrm{Pr}_X)^{\aleph_1} = \lim_{\mathrm{Spec}(A)\to X}((n+1)\mathrm{Pr}_A)^{\aleph_1}.$$

Letting $Y_A := \mathrm{Spec}(A) \times_X Y$, we have already established that $n\mathrm{Pr}_{Y_A}$ is $\aleph_1$-compact in $(n+1)\mathrm{Pr}_A$, thus giving the $\aleph_1$-compactness of $n\mathrm{Pr}_Y$ in $(n+1)\mathrm{Pr}_X$. Moreover, to check that $f_n^*\colon n\mathrm{Pr}_X \to n\mathrm{Pr}_Y$ admits a left adjoint in $((n+1)\mathrm{Pr}_X)^{\aleph_1}$, it will suffice to show that all maps $(f_A)_n^* : n\mathrm{Pr}_A \to n\mathrm{Pr}_{Y_A}$ admit left adjoints compatible with base change, which we established in Step 1. Similarly, the dualizability of $f_{n,\sharp}(\mathbb{1}) \in n\mathrm{Pr}_X$ may be checked after pullback to affines.

Step 3: Ambidexterity. For (v), we use the general ambidexterity result again: it suffices to check that the unit and counit of the adjunction $f_n^* \dashv f_{n,\sharp}$ admit right adjoints. For the counit $f_{n,\sharp}f_n^* \to \mathrm{id}$, we use that under the equivalence $\mathrm{End}_{(n+1)\mathrm{Pr}_X}(n\mathrm{Pr}_X, n\mathrm{Pr}_X) \simeq n\mathrm{Pr}_X$ this map corresponds to the map $f_{n,\sharp}(\mathbb{1}) \to \mathbb{1}$, so we need to show this admits a right adjoint in $n\mathrm{Pr}_X$. By (iv), the object $f_{n,\sharp}(\mathbb{1})$ is dualizable, so this is equivalent to its dual $\mathbb{1} \to f_{n,*}(\mathbb{1})$ admitting a left adjoint in $n\mathrm{Pr}_X$. But this dual map is the map $(n-1)\mathrm{Pr}_X \to (n-1)\mathrm{Pr}_Y$, which has a left adjoint in $n\mathrm{Pr}_X$ by part (i) applied one level down (which uses $n-1 \geq 1$).

For the unit $\eta\colon \mathrm{id} \to f_n^* f_{n,\sharp}$ in $\mathrm{End}_{(n+1)\mathrm{Pr}_X}(n\mathrm{Pr}_Y, n\mathrm{Pr}_Y)$ we use a base-change argument. Consider the cartesian square

$$\begin{tikzcd} Y \times_X Y \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "f"] \\ Y \arrow[r, "f"'] & X \end{tikzcd}$$

together with the diagonal $\delta\colon Y \to Y \times_X Y$. Since the class E is closed under formation of diagonals, δ is bi-countably presented, so the conclusions of parts (i)–(iv) are available for δ at level n. By base change (part (iii)), there is a canonical equivalence

$$f_n^* f_{n,\sharp} \simeq p_{1,n,\sharp}\, p_{2,n}^*.$$

The level-n counit $\varepsilon_\delta\colon \delta_{n,\sharp}\delta_n^* \to \mathrm{id}_{n\mathrm{Pr}_{Y\times_X Y}}$ of $\delta_{n,\sharp} \dashv \delta_n^*$ then induces a 2-morphism

$$p_{1,n,\sharp}\, \varepsilon_\delta\, p_{2,n}^*\colon\ \mathrm{id}_{n\mathrm{Pr}_Y} = p_{1,n,\sharp}\, \delta_{n,\sharp}\, \delta_n^*\, p_{2,n}^* \ \to p_{1,n,\sharp}\, p_{2,n}^* \simeq f_n^* f_{n,\sharp}.$$

This 2-morphism agrees with the unit η_f, which one verifies by checking that together with ε_f it satisfies the triangle identities (a consequence of base change). But since we already showed that the counit ε_δ is a left adjoint and whiskering with the functors $p_{1,n,\sharp}$ and $p_{2,n}^*$ preserves adjunctions, this finishes the proof. $\square$

4.3 The Künneth formula

As a consequence of the theorem, we obtain the following Künneth formula:

Corollary 4.7. *Let $f\colon Y \to X$ and $f'\colon Y' \to X$ be bi-countably presented maps, and let $n \geq 2$. Then the objects $(n-1)\mathrm{Pr}_Y$ and $(n-1)\mathrm{Pr}_{Y'}$ in $n\mathrm{Pr}_X$ are dualizable, and the* Künneth formula

$$(n-1)\mathrm{Pr}_Y \otimes_{n\mathrm{Pr}_X} (n-1)\mathrm{Pr}_{Y'} \xrightarrow{\sim} (n-1)\mathrm{Pr}_{Y\times_X Y'}$$

holds, where the equivalence is implemented via the external tensor product.

Proof. By definition, $(n-1)\mathrm{Pr}_Y$ as an object of $n\mathrm{Pr}_X$ is $f_{n,*}(\mathbb{1}_{n\mathrm{Pr}_Y})$. Since $n \geq 2$, Theorem 4.6 gives ambidexterity $f_{n,*} \simeq f_{n,\sharp}$, and hence

$$(n-1)\mathrm{Pr}_Y \simeq f_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Y}).$$

The same theorem then implies that this object is dualizable in $n\mathrm{Pr}_X$; similarly for Y'.

For the Künneth formula, write $Z := Y \times_X Y'$, with projections

$$\begin{tikzcd} Z \arrow[r, "q"] \arrow[d, "p"'] & Y' \arrow[d, "f'"] \\ Y \arrow[r, "f"'] & X. \end{tikzcd}$$

The maps p, q, and $f \circ p = f' \circ q$ are again bi-countably presented by Proposition 4.4, so the relevant $\sharp$-functors are available. Using the projection formula for $f_{n,\sharp}$, base change for the cartesian square above, and then the projection formula for $p_{n,\sharp}$, we then compute

$$\begin{aligned}
(n-1)\mathrm{Pr}_Y \otimes_{n\mathrm{Pr}_X} (n-1)\mathrm{Pr}_{Y'} &\simeq f_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Y}) \otimes_{n\mathrm{Pr}_X} f'_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_{Y'}}) \\
&\simeq f_{n,\sharp}\big(\mathbb{1}_{n\mathrm{Pr}_Y} \otimes_{n\mathrm{Pr}_Y} f_n^* f'_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_{Y'}})\big) \\
&\simeq f_{n,\sharp}\big(\mathbb{1}_{n\mathrm{Pr}_Y} \otimes_{n\mathrm{Pr}_Y} p_{n,\sharp} q_n^*(\mathbb{1}_{n\mathrm{Pr}_{Y'}})\big) \\
&\simeq f_{n,\sharp} p_{n,\sharp}\big(p_n^*(\mathbb{1}_{n\mathrm{Pr}_Y}) \otimes_{n\mathrm{Pr}_Z} q_n^*(\mathbb{1}_{n\mathrm{Pr}_{Y'}})\big) \\
&\simeq (f \circ p)_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Z}) \\
&\simeq (n-1)\mathrm{Pr}_Z.
\end{aligned}$$

In the penultimate line we used that pullback preserves monoidal units. Since $Z = Y \times_X Y'$, this is the desired equivalence. □

In particular, for $n \geq 2$, sheaves of presentable $(n-1)$-categories on a fiber product are determined by what they are on the factors and what sheaves of presentable n-categories are on the base. This is the first genuinely surprising statement about quasicoherent sheaves of higher categories: *both* parts of Corollary 4.7 fail at lower categorical levels:

- At level $n = 0$, it would say that $\mathcal{O}(Y) \in \mathrm{D}(X)$ is dualizable (i.e. a perfect complex), or that $\mathcal{O}(Y) \otimes_{\mathrm{D}(X)} \mathcal{O}(Y') \simeq \mathcal{O}(Y \times_X Y')$, both of which are absurd without strong assumptions on Y and Y'.

- At level $n = 1$, it is often the case that $\mathrm{D}(Y) \in 1\mathrm{Pr}_X$ is dualizable and the Künneth formula holds (cf. work by Gaitsgory), but one still needs some assumption such as "Y is a geometric stack over X".

- For $n \geq 2$, these results are essentially *always* true.

However, we cannot stay on a fixed categorical level (say 1Pr): the Künneth formula always uses the next categorical level on the base X.

4.4 Dropping the bi-countable-presentation hypothesis

We will now see that for some of the previous assertions we may drop the condition on the map being bi-countably presented if instead we require sufficient compactness conditions on X and Y.

Proposition 4.8. *Let $f \colon Y \to X$ be a map between $\aleph_1$-compact objects of* $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})$ *(so both X and Y are countable colimits of affine spectral schemes by Lemma 4.10 below). For $n \geq 2$, the right adjoint*

$$f_{n,*} \colon n\mathrm{Pr}_Y \to n\mathrm{Pr}_X$$

of f_n^* admits a further right adjoint in $\mathrm{Mod}_{n\mathrm{Pr}_X}(\mathrm{Pr}^{\mathrm{L}})$*, which is compatible with base change. Moreover, for any $f' \colon Y' \to X$ satisfying the same hypothesis, the Künneth formula*

$$(n-1)\mathrm{Pr}_Y \otimes_{n\mathrm{Pr}_X} (n-1)\mathrm{Pr}_{Y'} \simeq (n-1)\mathrm{Pr}_{Y \times_X Y'}$$

holds.

Remark 4.9. If X is affine, the conclusion says that $f_{n,*}$ is a map in $\mathrm{Mod}_{n\mathrm{Pr}_A}(\mathrm{Pr}^{\mathrm{L}})$, but *not* in general in $\mathrm{Mod}_{n\mathrm{Pr}_A}(\mathrm{Pr}^{\mathrm{L}}_{\aleph_1}) = \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$: it fails only to preserve $\aleph_1$-compact objects.

The proof of the proposition requires various auxiliary lemmas, which will be given after the proof of the proposition.

Proof. We first reduce to the case where X is affine. Since X is $\aleph_1$-compact, by Lemma 4.10 it is a countable colimit of affines,

$$X \simeq \mathrm{colim}_j \operatorname{Spec}(A_j).$$

By Lemma 4.10, $\aleph_1$-compact objects of $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})$ are closed under finite limits; hence each base change $Y_j := Y \times_X \operatorname{Spec}(A_j)$ is again $\aleph_1$-compact. For such countable colimits, the descent formula

$$n\mathrm{Pr}_X \simeq \lim_j n\mathrm{Pr}_{A_j}$$

holds not only in $1\mathrm{Pr} = \mathrm{Pr}^{\mathrm{L}}_{\aleph_1}$, but also in Pr^{L}, hence on the level of underlying $(\infty,1)$-categories. Thus the desired assertions may be checked after pullback to the affine pieces $\operatorname{Spec}(A_j) \to X$. We may therefore assume that $X = \operatorname{Spec}(A)$.

By Lemma 4.11 below, we may choose a diagram $(Y_i)_{i\in I}$ over X, together with a compatible cone $Y \to Y_i$, such that each $f_i \colon Y_i \to X$ is bi-countably presented and

$$n\mathrm{Pr}_Y \simeq \mathrm{colim}_{i\in I}\, n\mathrm{Pr}_{Y_i},$$

where the transition maps are given by pullback. The pullback f_n^* is a morphism in $1\mathrm{Pr}$, hence preserves $\aleph_1$-compact objects. For each i, the map f_i is bi-countably presented. Hence, by Theorem 4.6, for $n \geq 2$ the functor

$$f_{i,n,*} \colon n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_X$$

preserves colimits, is $n\mathrm{Pr}_X$-linear, and satisfies base change. By Lemma 4.12 below, if $\alpha_i \colon n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_Y$ denotes the colimit structure functor and $p_{ij} \colon Y_j \to Y_i$ denotes the transition map associated to a morphism $i \to j$ in I, then

$$f_{n,*}(\alpha_i E) \simeq \mathrm{colim}_{j\in I_{i/}}\, f_{j,n,*}(p^*_{ij,n}E).$$

Thus $f_{n,*}$ is assembled from the functors $f_{j,n,*}$ by taking $\aleph_1$-filtered colimits. Since filtered colimits of functors preserve colimits, $n\mathrm{Pr}_X$-linearity, and base change whenever the functors in the system do, it follows that $f_{n,*}$ also commutes with small colimits, tensors by $n\mathrm{Pr}_X$, and base change.

With these properties established, the Künneth formula is proved formally just like in Corollary 4.7. □

Lemma 4.10. *The inclusion*

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}}) \hookrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathbb{S}})$$

preserves $\aleph_1$-filtered colimits. The same holds for the corresponding fpqc-sheaf categories on affines over any fixed affine base.

Moreover, these sheaf categories are $\aleph_1$-compactly generated by affines, their $\aleph_1$-compact objects are closed under finite limits, and sheafification sends $\aleph_1$-compact presheaves to $\aleph_1$-compact sheaves. In particular, for every affine $X = \operatorname{Spec}(A)$, every affine object $\operatorname{Spec}(B) \to X$ is $\aleph_1$-compact in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})_{/X}$, and every $\aleph_1$-compact object in this slice is a countable colimit of affine objects.

Proof. By definition of the fpqc topology on affines, covering sieves are tested on finite covering families. Since finite sets are $\aleph_1$-small, Lemma A.12 shows that fpqc-sheaves are closed under $\aleph_1$-filtered colimits in presheaves. The same argument applies on the fpqc site of affines over a fixed affine base.

Now apply Corollary A.13 with $\kappa = \aleph_1$, both on $\mathrm{Aff}_{\mathbb{S}}$ and on the corresponding site over an affine base. This gives $\aleph_1$-compact generation, closure of $\aleph_1$-compact objects under finite limits, and preservation of $\aleph_1$-compact objects under sheafification. Since representable presheaves are compact, affine sheaves are $\aleph_1$-compact.

The sheaf category over an affine X is generated under colimits by affine objects over X. Hence its $\aleph_1$-compact objects are retracts of countable colimits of affine objects; absorbing the retract by the usual countable telescope gives the stated countable affine presentation. □

Lemma 4.11. *Let $X = \mathrm{Spec}(A)$ be affine. Then every $\aleph_1$-compact object $Y \in \mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})_{/X}$ admits a compatible cone $Y \to Y_i$ over an $\aleph_1$-filtered category I, where each $Y_i \to X$ is bi-countably presented, such that, for every $n \geq 1$, the induced functor*

$$\mathrm{colim}_{i\in I}\, n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_Y$$

along pullback is an equivalence in $1\mathrm{Pr} = \mathrm{Pr}^{\mathrm{L}}_{\aleph_1}$.

Proof. By Lemma 4.10, the affine objects over X are $\aleph_1$-compact. Since affine objects generate $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})_{/X}$ under colimits, this sheaf category is $\aleph_1$-compactly generated by affines. Thus an $\aleph_1$-compact object over X admits a presentation by countably many affines: we have

$$Y \simeq \mathrm{colim}_{m\in M}\, \mathrm{Spec}(B_m)$$

for some countable diagram $B\colon M^{\mathrm{op}} \to \mathrm{CAlg}_A(\mathrm{Sp})$. Since $\mathrm{CAlg}_A(\mathrm{Sp})$ is $\aleph_1$-compactly generated, Lemma 1.5 shows that also $\mathrm{Fun}(M^{\mathrm{op}}, \mathrm{CAlg}_A(\mathrm{Sp}))$ is $\aleph_1$-compactly generated, with $\aleph_1$-compact objects given by the functors $M^{\mathrm{op}} \to \mathrm{CAlg}_A(\mathrm{Sp})^{\aleph_1}$, i.e. the diagrams of countably presented A-algebras. Applying this to the diagram B, we obtain an $\aleph_1$-filtered category I and diagrams $B_i\colon M^{\mathrm{op}} \to \mathrm{CAlg}_A(\mathrm{Sp})$ such that each $B_i(m)$ is countably presented over A, and we have

$$B \simeq \mathrm{colim}_{i\in I}\, B_i$$

in $\mathrm{Fun}(M^{\mathrm{op}}, \mathrm{CAlg}_A(\mathrm{Sp}))$. Define

$$Y_i := \mathrm{colim}_{m\in M}\, \mathrm{Spec}(B_i(m)).$$

Since M is countable and each $A \to B_i(m)$ is countably presented, each $Y_i \to X$ is bi-countably presented. The maps $B_i \to B$ induce a compatible cone $Y \to Y_i$ over X.

It remains to check the asserted formula after applying $n\mathrm{Pr}_{(-)}$. First consider the affine case: if $B \simeq \mathrm{colim}_i\, B_i$ in $\mathrm{CAlg}_A(\mathrm{Sp})$, then

$$n\mathrm{Pr}_B \simeq \mathrm{colim}_i\, n\mathrm{Pr}_{B_i}.$$

The base case $n = 0$ follows from Theorem 1.24 applied to $C = \mathrm{Sp}$: the functor $\mathrm{CAlg}_A(\mathrm{Sp}) \to \mathrm{CAlg}(1\mathrm{Pr})_{0\mathrm{Pr}_A/}$, $B \mapsto 0\mathrm{Pr}_B$, is a left adjoint and hence preserves colimits. For higher n the same argument applies inductively, applying the theorem with $C = n\mathrm{Pr}$ and base algebra $(n-1)\mathrm{Pr}_A$.

Now apply descent to the countable affine presentations of Y and the Y_i. Since $B \simeq \operatorname{colim}_i B_i$ in the functor category, we have $B_m \simeq \operatorname{colim}_i B_i(m)$ for every $m \in M$. Hence

$$\begin{aligned} n\mathrm{Pr}_Y &\simeq \lim_{m \in M^{\mathrm{op}}} n\mathrm{Pr}_{B_m} \\ &\simeq \lim_{m \in M^{\mathrm{op}}} \operatorname{colim}_i n\mathrm{Pr}_{B_i(m)}. \end{aligned}$$

Since M is countable and 1Pr is $\aleph_1$-compactly generated, $\aleph_1$-filtered colimits commute with this countable limit in 1Pr. Thus

$$\begin{aligned} n\mathrm{Pr}_Y &\simeq \operatorname{colim}_i \lim_{m \in M^{\mathrm{op}}} n\mathrm{Pr}_{B_i(m)} \\ &\simeq \operatorname{colim}_i n\mathrm{Pr}_{Y_i}. \end{aligned}$$

□

Lemma 4.12. *Let $X = \mathrm{Spec}(A)$, and let $(Y_i)_{i \in I}$ be a diagram over X, indexed by an $\aleph_1$-filtered category I, equipped with a compatible cone $Y \to Y_i$ over X. Assume that*

$$n\mathrm{Pr}_Y \simeq \operatorname{colim}_{i \in I} n\mathrm{Pr}_{Y_i}$$

in 1Pr *along pullback. In particular, the structure functors $\alpha_i \colon n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_Y$ preserve $\aleph_1$-compact objects, and compact generators of $n\mathrm{Pr}_Y$ are obtained from compact generators of the stages under countable colimits. Write $f_i \colon Y_i \to X$ and $f \colon Y \to X$ for the structure maps. For a morphism $i \to j$ in I, write $p_{ij} \colon Y_j \to Y_i$ for the induced transition map. Assume that f_n^* preserves $\aleph_1$-compact objects and that each right adjoint*

$$f_{i,n,*} \colon n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_X$$

preserves small colimits. Then $f_{n,*}$ preserves small colimits, and, if $\alpha_i \colon n\mathrm{Pr}_{Y_i} \to n\mathrm{Pr}_Y$ denotes the colimit structure functor, then

$$f_{n,*}(\alpha_i E) \simeq \operatorname{colim}_{j \in I_{i/}} f_{j,n,*}(p_{ij,n}^* E)$$

for every $E \in n\mathrm{Pr}_{Y_i}$.

Proof. For fixed i, define

$$G_i(E) := \operatorname{colim}_{j \in I_{i/}} f_{j,n,*}(p_{ij,n}^* E), \qquad E \in n\mathrm{Pr}_{Y_i}.$$

Since each $f_{j,n,*}$ preserves colimits, so does G_i. If $i \to k$ is a morphism in I, then the inclusion $I_{k/} \to I_{i/}$ is cofinal, and hence

$$G_k(p_{ik,n}^* E) \simeq G_i(E).$$

Thus the restrictions of the G_i to compact objects are compatible with the 1Pr-colimit presentation of $n\mathrm{Pr}_Y$. Since compact generators of $n\mathrm{Pr}_Y$ are generated under countable colimits by the objects $\alpha_i E$ with $E \in (n\mathrm{Pr}_{Y_i})^{\aleph_1}$, these compatible restrictions determine a functor on $(n\mathrm{Pr}_Y)^{\aleph_1}$ preserving countable colimits. Extending it cocontinuously gives a colimit-preserving functor

$$G \colon n\mathrm{Pr}_Y \to n\mathrm{Pr}_X$$

with $G(\alpha_i E) \simeq G_i(E)$ for compact E. Since both sides of this formula preserve colimits in E, the same formula holds for all $E \in n\mathrm{Pr}_{Y_i}$.

We claim that G is right adjoint to f_n^*. It suffices to test this on $\aleph_1$-compact generators $K \in (n\mathrm{Pr}_X)^{\aleph_1}$ and on $\aleph_1$-compact generators of $n\mathrm{Pr}_Y$ represented by some stage $n\mathrm{Pr}_{Y_i}$. Here we use the 1Pr-colimit presentation and the assumption that f_n^* preserves compact objects. Using compactness of K and the standard mapping-space formula for filtered colimits of categories, we compute

$$\begin{aligned}
\mathrm{Hom}_{n\mathrm{Pr}_X}(K, G(\alpha_i E)) &\simeq \mathrm{colim}_{j \in I_{i/}} \mathrm{Hom}_{n\mathrm{Pr}_X}\big(K, f_{j,n,*}(p_{ij,n}^* E)\big) \\
&\simeq \mathrm{colim}_{j \in I_{i/}} \mathrm{Hom}_{n\mathrm{Pr}_{Y_j}}\big(f_{j,n}^* K, p_{ij,n}^* E\big) \\
&\simeq \mathrm{colim}_{j \in I_{i/}} \mathrm{Hom}_{n\mathrm{Pr}_{Y_j}}\big(p_{ij,n}^* f_{i,n}^* K, p_{ij,n}^* E\big) \\
&\simeq \mathrm{Hom}_{n\mathrm{Pr}_Y}\big(f_n^* K, \alpha_i E\big).
\end{aligned}$$

Since $n\mathrm{Pr}_X$ is $\aleph_1$-compactly generated, this identifies G with the right adjoint $f_{n,*}$. □

4.5 The Stefanich ring functor

We can summarize what we have shown in the language of Stefanich rings:

Corollary 4.13. *The functor*

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\aleph_1} \to (\mathrm{StRing}_{\mathbb{S}/})^{\mathrm{op}}, \qquad X \mapsto \mathrm{St}(X)$$

commutes with countable colimits and finite limits.

In other words, if the target were itself an ∞-topos, then this functor would have all the properties of a morphism of ∞-topoi (except that we restrict to countable colimits rather than to all small colimits).

Sketch of proof. Commutation with countable colimits holds essentially by definition. To prove commutation with finite limits, it suffices to handle the terminal object (which works by definition of $\mathrm{StRing}_{\mathbb{S}/}$) and pullbacks. So consider a pullback diagram

$$\begin{tikzcd}
Y \times_X Y' \arrow[r] \arrow[d] & Y' \arrow[d, "f'"] \\
Y \arrow[r, "f"'] & X
\end{tikzcd}$$

of $\aleph_1$-compact fpqc-stacks. A priori, to compute the pushout of Stefanich rings, we would start by forming the infinite tuple

$$\big(\mathcal{O}(Y) \otimes_{\mathcal{O}(X)} \mathcal{O}(Y'), \quad \mathrm{D}(Y) \otimes_{\mathrm{D}(X)} \mathrm{D}(Y'), \quad 1\mathrm{Pr}_Y \otimes_{1\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \quad \ldots\big),$$

consisting of all the levelwise pushouts. These come with natural maps from each entry to endomorphisms of the unit of the next, but these are typically *not* isomorphisms; one needs to "spectrify", i.e. apply the left adjoint to the inclusion of Stefanich rings into the larger category that allows for non-invertible comparison maps.

In the next talk, we will explain a convenient way of computing such tensor products. What we will see there is that we can make a 'first approximation' to the spectrification, given by:

$$\big(\mathcal{O}(Y) \otimes_{\mathrm{D}(X)} \mathcal{O}(Y'), \quad \mathrm{D}(Y) \otimes_{1\mathrm{Pr}_X} \mathrm{D}(Y'), \quad 1\mathrm{Pr}_Y \otimes_{2\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \quad \ldots\big).$$

Here on each level the tensor product is taken at the next-higher categorical level. (For example, in the zeroth slot we form the tensor product not over $O(X)$, i.e. in $O(X)$-modules, but rather inside $\mathrm{D}(X)$; this is finer, since there is a canonical functor $\mathrm{Mod}_{O(X)}(\mathrm{D}(\mathbb{S})) \to \mathrm{D}(X)$.) By Proposition 4.8, starting from the slot $1\mathrm{Pr}_Y \otimes_{2\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}$, this expression agrees with

$$(\Omega^\infty O(Y \times_X Y'),\ \mathrm{D}(Y \times_X Y'),\ 1\mathrm{Pr}_{Y \times_X Y'},\ \ldots),$$

which is already a Stefanich ring, namely the one corresponding to $Y \times_X Y'$. As the first few terms do not affect the Stefanich ring structure, this gives the desired result. □

Remark 4.14. A general slogan in this area is that "everything becomes affine under sufficient categorification". The Künneth formulas above, which hold from level 1Pr onwards, justify this slogan in some form. The slogan is a bit imprecise, however: stacks like $B^n\mathbb{G}_m$ are n-affine (in a sense to be made precise in the next talk) but not k-affine for $k < n$, so in particular $\bigsqcup_k B^k\mathbb{G}_m$ is not n-affine for any n. We will later see that even the infinite-dimensional affine space $\mathbb{A}^\infty = \mathrm{colim}_n\, \mathbb{A}^n$ is not n-affine for any n, although all its terms are 0-affine and the transition maps are cohomologically smooth (since complete intersections are cohomologically smooth in the quasicoherent six-functor formalism).

5 Stefanich rings

Talk given by Yiming.

Recall that we defined the $\aleph_1$-presentable category StRing of *Stefanich rings* as

$$\begin{aligned}\mathrm{StRing} &:= \mathrm{colim}_{1\mathrm{Pr}}(\mathrm{CAlg}(\mathrm{An}) \xrightarrow{\mathrm{Mod}_1^{\mathbb{P}}} \mathrm{CAlg}(1\mathrm{Pr}) \xrightarrow{\mathrm{Mod}_2^{\mathbb{P}}} \mathrm{CAlg}(2\mathrm{Pr}) \xrightarrow{\mathrm{Mod}_3^{\mathbb{P}}} \dots) \\ &\simeq \lim_{\mathrm{CAT}}(\mathrm{CAlg}(\mathrm{An}) \xleftarrow{\mathrm{End}_1^{\mathbb{P}}} \mathrm{CAlg}(1\mathrm{Pr}) \xleftarrow{\mathrm{End}_2^{\mathbb{P}}} \mathrm{CAlg}(2\mathrm{Pr}) \xleftarrow{\mathrm{End}_3^{\mathbb{P}}} \dots),\end{aligned}$$

where we write $\mathrm{Mod}_n^{\mathbb{P}}(R) := \mathrm{Mod}_R((n-1)\mathrm{Pr})$ and $\mathrm{End}_n^{\mathbb{P}}(D) := \mathrm{End}_D(\mathbb{1}_D)$.

We denote the objects of this category as $A = (A_0, A_1, A_2, \dots)$, where $A_n \in \mathrm{CAlg}(n\mathrm{Pr})$ and the structure isomorphisms $A_n \simeq \mathrm{End}_{n+1}^{\mathbb{P}}(A_{n+1})$ are left implicit in the notation. Similarly, morphisms from A to B are collections of maps $f_{n-1}^*\colon A_n \to B_n$ compatible with the structural isomorphisms.

Remark 5.1. Given an fpqc-stack X, the associated Stefanich ring has $A_n = (n-1)\mathrm{Pr}_X$. The indexing convention for $f_{n-1}^*\colon A_n \to B_n$ reflects the convention from the previous lecture.

We may equip StRing with the cocartesian monoidal structure; we will have a look at the resulting "tensor product" of Stefanich rings in more detail below. Since the functors $\mathrm{End}_n^{\mathbb{P}}$ preserve the monoidal unit, we see that the monoidal unit in StRing is the initial Stefanich ring $\mathbb{P} := (*, \mathrm{An}, 1\mathrm{Pr}, 2\mathrm{Pr}, \dots)$, with n-th component $\mathbb{P}_n := (n-1)\mathrm{Pr}$.

Since $\mathbb{P}$ is initial, every Stefanich ring A receives a unique map $a\colon \mathbb{P} \to A$. The individual maps $a_{n-1}^*\colon (n-1)\mathrm{Pr} \to A_n$ in $\mathrm{CAlg}(\mathrm{Pr})$ are obtained using the identification $\mathrm{CAlg}(n\mathrm{Pr}) \simeq \mathrm{CAlg}(\mathrm{Pr})_{(n-1)\mathrm{Pr}/}$. By passing to right adjoints, we obtain 'forgetful functors'

$$a_{n-1,*}\colon A_n \to (n-1)\mathrm{Pr}, \qquad B_{n-1} \mapsto \underline{\mathrm{Hom}}_{A_n}(\mathbb{1}_{A_n}, B_{n-1}).$$

Given $B_{n-1} \in A_n$, we refer to $a_{n-1,*}(B_{n-1})$ as its *underlying* $(n-1)$*-category.*

Notation 5.2. We will denote the monoidal unit of A_n by

$$(A/A)_{n-1} \;:=\; \mathbb{1}_{A_n} \;\in\; \mathrm{CAlg}(A_n).$$

Observe that we have $a_{n-1,*}((A/A)_{n-1}) = \mathrm{End}_{A_n}(\mathbb{1}_{A_n}) = A_{n-1}$.

5.1 Internal Stefanich rings

We will now provide a limit description of the slice category $\mathrm{StRing}_{A/}$ under any Stefanich ring A, exhibiting A-algebras in Stefanich rings as some form of internal Stefanich rings in A.

Construction 5.3. Let $f\colon A \to B$ be a morphism of Stefanich rings. For any $n \geq 1$, the symmetric monoidal functor $f_{n-1}^*\colon A_n \to B_n$ admits a lax symmetric monoidal right adjoint $f_{n-1,*}\colon B_n \to A_n$, which therefore induces a functor of the form

$$f_{n-1,*}\colon \mathrm{CAlg}(B_n) \to \mathrm{CAlg}(A_n).$$

Observe that the composite $\mathrm{CAlg}(B_n) \xrightarrow{f_{n-1,*}} \mathrm{CAlg}(A_n) \xrightarrow{\mathrm{fgt}} \mathrm{CAlg}((n-1)\mathrm{Pr})$ is the forgetful functor $\mathrm{CAlg}(B_n) \to \mathrm{CAlg}((n-1)\mathrm{Pr})$.

We then define

$$(B/A)_{n-1} \;:=\; f_{n-1,*}(\mathbb{1}_{B_n}) \quad \in \quad \mathrm{CAlg}(A_n).$$

Note that the underlying $(n-1)$-presentable category of $(B/A)_{n-1}$ is B_{n-1}. So we may think of $(B/A)_{n-1}$ as a lift of B_{n-1} to a commutative algebra in A_n. For $f = \mathrm{id}_A$ this recovers Notation 5.2.

Construction 5.4. We now construct an internal form of the module-endomorphism adjunction: For a Stefanich ring A and $m \geq 1$, we will produce an adjunction

$$\mathrm{CAlg}(A_m) \underset{\mathrm{End}_m^A}{\overset{\mathrm{Mod}_m^A}{\rightleftarrows}} \mathrm{CAlg}(A_{m+1}).$$

Recall from Notation 1.25 that, more generally, for $n \geq 1$ and $C \in \mathrm{CAlg}(n\mathrm{Pr})$ we have an adjunction

$$\mathrm{CAlg}(C) \underset{\mathrm{End}_{(-)}(\mathbb{1})}{\overset{\mathrm{Mod}_{(-)}(C)}{\rightleftarrows}} \mathrm{CAlg}(\mathrm{Mod}_C(n\mathrm{Pr})),$$

where the left adjoint sends R to $\mathrm{Mod}_R(C)$ and the right adjoint sends D to $\mathrm{End}_D(\mathbb{1}_D)$. Here we are using that C is an algebra over $(n-1)\mathrm{Pr}$ in $1\mathrm{Pr}$, so that

$$\mathrm{Mod}_C(1\mathrm{Pr}) \simeq \mathrm{Mod}_C(\mathrm{Mod}_{(n-1)\mathrm{Pr}}(1\mathrm{Pr})) \simeq \mathrm{Mod}_C(n\mathrm{Pr}),$$

giving the $(n-1)\mathrm{Pr}$-linear structure for free.

Now let A be a Stefanich ring. Applying this adjunction with $C = m\mathrm{Pr}$, we see that the structural equivalence

$$A_m \simeq \mathrm{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}})$$

in $\mathrm{CAlg}(m\mathrm{Pr})$ corresponds to a symmetric monoidal functor

$$L_m^A\colon \mathrm{Mod}_{A_m}(m\mathrm{Pr}) \to A_{m+1}$$

in $\mathrm{CAlg}((m+1)\mathrm{Pr})$. It admits a right adjoint, which we denote by

$$U_m^A\colon A_{m+1} \to \mathrm{Mod}_{A_m}(m\mathrm{Pr}), \qquad X \mapsto \underline{\mathrm{Hom}}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}, X).$$

We use the same notation for the induced functors on commutative algebra objects. Finally, we write

$$\mathrm{Mod}_m^A := L_m^A \circ \mathrm{Mod}_{(-)}(A_m)\colon \mathrm{CAlg}(A_m) \to \mathrm{CAlg}(A_{m+1})$$

for the internal module functor, and

$$\mathrm{End}_m^A := \mathrm{End}_{(-)}(\mathbb{1}) \circ U_m^A\colon \mathrm{CAlg}(A_{m+1}) \to \mathrm{CAlg}(A_m)$$

for its right adjoint. Explicitly, we have

$$\mathrm{Mod}_m^A(R) = L_m^A(\mathrm{Mod}_R(A_m)) \qquad \text{and} \qquad \mathrm{End}_m^A(D) = \mathrm{End}_{U_m^A(D)}(\mathbb{1}).$$

Observation 5.5. The functor $U_m^A\colon \mathrm{CAlg}(A_{m+1}) \to \mathrm{Mod}_{A_m}(m\mathrm{Pr})$ preserves the monoidal unit: the unit of the adjunction $L_m^A \dashv U_m^A$ takes the form

$$A_m \to U_m^A(\mathbb{1}_{A_{m+1}}) = \mathrm{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}),$$

which is the structure isomorphism of the Stefanich ring A.

Proposition 5.6. *Let* $A \in \mathrm{StRing}$ *and* $m \geq 1$*. The internal module functor*

$$\mathrm{Mod}_m^A\colon \mathrm{CAlg}(A_m) \to \mathrm{CAlg}(A_{m+1})$$

is fully faithful.

Proof. Since Mod_m^A is a morphism in 1Pr, by Lemma 1.10 it suffices to check that the unit of the adjunction $\mathrm{Mod}_m^A \dashv \mathrm{End}_m^A$ is an equivalence on $\aleph_1$-compact objects, i.e. on countably presented objects of $\mathrm{CAlg}(A_m)$. Let $R \in \mathrm{CAlg}(A_m)$ be countably presented. Unwinding definitions, we get

$$\mathrm{End}_m^A(\mathrm{Mod}_m^A(R)) \simeq \mathrm{End}_{U_m^A L_m^A(\mathrm{Mod}_R(A_m))}(\mathbb{1}) = \mathrm{End}_{T(\mathrm{Mod}_R(A_m))}(\mathbb{1}),$$

where we use the abbreviated notation

$$T := U_m^A L_m^A\colon \mathrm{Mod}_{A_m}(m\mathrm{Pr}) \to \mathrm{Mod}_{A_m}(m\mathrm{Pr}).$$

To analyze this endomorphism category, note:

- The functors U_m^A and L_m^A are both maps in $(m+1)\mathrm{Pr}$, hence so is T. Since $m \geq 1$, this means that T is (at least) a 2-functor, so it preserves adjunctions.
- By Observation 5.5, the unit map $A_m \to T(A_m)$ is an isomorphism.

We now consider the free R-module functor

$$F_R\colon A_m \to \mathrm{Mod}_R(A_m), \qquad X \mapsto R \otimes X.$$

It admits an A_m-linear right adjoint $G_R\colon \mathrm{Mod}_R(A_m) \to A_m$, given by forgetting the R-module structure. Since R is countably presented, the functor G_R preserves $\aleph_1$-compact objects, and so this adjunction lives inside $\mathrm{Mod}_{A_m}(m\mathrm{Pr})$. So applying the 2-functor T to the adjunction $F_R \dashv G_R$, we obtain an adjunction

$$T(F_R)\colon T(A_m) \rightleftarrows T(\mathrm{Mod}_R(A_m)) :T(G_R)$$

in $\mathrm{Mod}_{A_m}(m\mathrm{Pr})$. Since T is lax monoidal, the functor $T(F_R)$ is still symmetric monoidal, and so the unit object of $T(\mathrm{Mod}_R(A_m))$ is $T(F_R)(\mathbb{1}_{T(A_m)})$. Combining all of this, we may now compute:

$$\begin{aligned}
\mathrm{End}_m^A(\mathrm{Mod}_m^A(R)) &= \underline{\mathrm{End}}_{T(\mathrm{Mod}_R(A_m))}(\mathbb{1}) \\
&\simeq \underline{\mathrm{End}}_{T(\mathrm{Mod}_R(A_m))}(T(F_R)(\mathbb{1}_{T(A_m)})) \\
&\simeq \underline{\mathrm{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(G_R)T(F_R)(\mathbb{1}_{T(A_m)})) \\
&\simeq \underline{\mathrm{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(G_R \circ F_R)(\mathbb{1}_{T(A_m)})) \\
&\simeq \underline{\mathrm{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(R \otimes -)(\mathbb{1}_{T(A_m)})) \\
&\simeq \underline{\mathrm{Hom}}_{A_m}(\mathbb{1}_{A_m}, (R \otimes -)(\mathbb{1}_{A_m})) \\
&\simeq R,
\end{aligned}$$

where in the second-to-last equivalence we use that under the equivalence $A_m \xrightarrow{\sim} T(A_m)$, the endofunctor $T(R \otimes -)$ corresponds to $R \otimes -\colon A_m \to A_m$. Under this identification, the unit map $R \to \mathrm{End}_m^A(\mathrm{Mod}_m^A(R))$ is the map induced by the action of R on the free rank-one R-module $F_R(\mathbb{1}_{A_m})$. Under the adjunction $T(F_R) \dashv T(G_R)$, this action corresponds to the identity of $G_R F_R(\mathbb{1}_{A_m}) \simeq R$. Thus the unit is an equivalence on countably presented R, and Mod_m^A is fully faithful. □

Definition 5.7. Let $A \in \mathrm{StRing}$. The category of *internal Stefanich rings in A* is the colimit

$$\mathrm{StRing}(A) := \mathrm{colim}_{1\mathrm{Pr}}(\mathrm{CAlg}(A_1) \xhookrightarrow{\mathrm{Mod}_1^A} \mathrm{CAlg}(A_2) \xhookrightarrow{\mathrm{Mod}_2^A} \dots)$$

formed along the internal module functors. Equivalently, since each Mod_m^A is a left adjoint with right adjoint End_m^A, it may be computed as the limit

$$\mathrm{StRing}(A) \simeq \lim_{\mathrm{CAT}}(\mathrm{CAlg}(A_1) \xleftarrow{\mathrm{End}_1^A} \mathrm{CAlg}(A_2) \xleftarrow{\mathrm{End}_2^A} \dots).$$

We now show that A-algebras in Stefanich rings are the same as Stefanich rings internal to A.

Proposition 5.8. *Let $A \in \mathrm{StRing}$. The assignments $B \mapsto (B/A)_n$, for $n \geq 0$, determine an equivalence*

$$\mathrm{StRing}_{A/} \xrightarrow{\sim} \mathrm{StRing}(A), \qquad B \mapsto ((B/A)_0, (B/A)_1, (B/A)_2, \dots).$$

Proof. By passing to slices, we obtain

$$\mathrm{StRing}_{A/} = \lim_{\mathrm{CAT}}(\dots \xrightarrow{\mathrm{End}_3^{\mathbb{P}}} \mathrm{CAlg}(2\mathrm{Pr})_{A_2/} \xrightarrow{\mathrm{End}_2^{\mathbb{P}}} \mathrm{CAlg}(1\mathrm{Pr})_{A_1/} \xrightarrow{\mathrm{End}_1^{\mathbb{P}}} \mathrm{CAlg}(\mathrm{An})_{A_0/}).$$

The entry $\mathrm{CAlg}(\mathrm{An})_{A_0/}$ is redundant for the limit, and we will ignore it. Under the equivalences $\mathrm{CAlg}(m\mathrm{Pr})_{A_m/} \simeq \mathrm{CAlg}(\mathrm{Mod}_{A_m}(m\mathrm{Pr}))$, the transition maps $\mathrm{CAlg}((m+1)\mathrm{Pr})_{A_{m+1}/} \to \mathrm{CAlg}(m\mathrm{Pr})_{A_m/}$ admit the following simple description. An object D of the source is an A_{m+1}-linear monoidal category; the transition sends it to $\mathrm{End}_D(\mathbb{1}_D)$, regarded over $\mathrm{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}) \simeq A_m$. Thus the transition factors as

$$\mathrm{CAlg}(\mathrm{Mod}_{A_{m+1}}((m+1)\mathrm{Pr})) \xrightarrow{\mathrm{End}_{(-)}(\mathbb{1})} \mathrm{CAlg}(A_{m+1}) \xrightarrow{U_m^A} \mathrm{CAlg}(\mathrm{Mod}_{A_m}(m\mathrm{Pr})),$$

where we use the notation from Construction 5.4. Hence we may alternatively compute $\mathrm{StRing}_{A/}$ as the limit of the chain

$$\dots \to \mathrm{CAlg}(\mathrm{Mod}_{A_{m+1}}((m+1)\mathrm{Pr})) \xrightarrow{\mathrm{End}_{(-)}(\mathbb{1})} \mathrm{CAlg}(A_{m+1}) \xrightarrow{U_m^A} \mathrm{CAlg}(\mathrm{Mod}_{A_m}(m\mathrm{Pr})) \xrightarrow{\mathrm{End}_{(-)}(\mathbb{1})} \mathrm{CAlg}(A_m) \to \dots.$$

Removing the intermediate $\mathrm{CAlg}(\mathrm{Mod}_{A_m}(m\mathrm{Pr}))$-vertices gives the limit over the composites

$$\mathrm{End}_m^A = \mathrm{End}_{(-)}(\mathbb{1}) \circ U_m^A\colon \mathrm{CAlg}(A_{m+1}) \to \mathrm{CAlg}(A_m),$$

hence the identification

$$\mathrm{StRing}_{A/} \simeq \lim_{\mathrm{CAT}}(\mathrm{CAlg}(A_1) \xleftarrow{\mathrm{End}_1^A} \mathrm{CAlg}(A_2) \xleftarrow{\mathrm{End}_2^A} \dots) = \mathrm{StRing}(A),$$

where the last equality is the limit description of Definition 5.7. Under this equivalence, an object $f\colon A \to B$ is sent in degree $n \geq 1$ to

$$f_{n-1,*}(\mathbb{1}_{B_n}) = (B/A)_{n-1} \quad \in \quad \mathrm{CAlg}(A_n),$$

that is, to the tuple $((B/A)_0, (B/A)_1, (B/A)_2, \dots)$, as claimed. □

Corollary 5.9. *The functor* $\mathrm{StRing}_{A/} \to \mathrm{CAlg}(A_n), B \mapsto (B/A)_n$ *admits a fully faithful left adjoint*

$$\mathrm{CAlg}(A_n) \hookrightarrow \mathrm{StRing}_{A/}.$$

Proof. By Proposition 5.6, the internal module functors Mod^A_m are fully faithful left adjoints. Hence Lemma 2.4 shows that, for every $n \geq 1$, the canonical functor

$$\mathrm{CAlg}(A_n) \hookrightarrow \mathrm{StRing}(A)$$

is fully faithful and left adjoint to the projection $\mathrm{StRing}(A) \to \mathrm{CAlg}(A_n)$. The claim now follows from the equivalence $\mathrm{StRing}_{A/} \simeq \mathrm{StRing}(A)$ of Proposition 5.8, under which this projection corresponds to the functor $B \mapsto (B/A)_n$. □

In Proposition 5.6 we showed that the composite

$$\mathrm{Mod}^A_m \colon \mathrm{CAlg}(A_m) \xhookrightarrow{\mathrm{Mod}_{(-)}(A_m)} \mathrm{CAlg}(\mathrm{Mod}_{A_m}(m\mathrm{Pr})) \xrightarrow{L^A_m} \mathrm{CAlg}(A_{m+1})$$

is fully faithful. While the functor $\mathrm{Mod}_{(-)}(A_m)$ is also always fully faithful, the functor L^A_m is usually not. Nevertheless, we still have the following statement:

Lemma 5.10. *Let* $A \in \mathrm{StRing}$ *and* $m \geq 1$. *The functor*

$$L^A_m \colon \mathrm{Mod}_{A_m}(m\mathrm{Pr}) \to A_{m+1}$$

is fully faithful on dualizable objects. (In particular, it is fully faithful on $\aleph_1$*-filtered colimits of dualizable objects, cf. Lemma 1.10.)*

Proof. Set $C := \mathrm{Mod}_{A_m}(m\mathrm{Pr})$ and $D := A_{m+1}$, and let U^A_m denote the right adjoint of L^A_m. Let $X, Y \in C$ be dualizable. Since L^A_m is symmetric monoidal, it preserves duals, and we have a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}_C(X \otimes Y^\vee, \mathbb{1}_C) \arrow[r, "\sim"] \arrow[d] & \mathrm{Hom}_C(X, Y) \arrow[d] \\
\mathrm{Hom}_D(L^A_m(X \otimes Y^\vee), \mathbb{1}_D) \arrow[r, "\sim"] & \mathrm{Hom}_D(L^A_m(X), L^A_m(Y)).
\end{tikzcd}$$

The horizontal maps are the usual identifications coming from duality, given by tensoring with Y and precomposing with the coevaluation $\mathbb{1}_C \to Y^\vee \otimes Y$. Thus it suffices to show that the left vertical map is an equivalence. Composing this map with the adjunction equivalence

$$\mathrm{Hom}_D(L^A_m(X \otimes Y^\vee), \mathbb{1}_D) \simeq \mathrm{Hom}_C(X \otimes Y^\vee, U^A_m(\mathbb{1}_D))$$

we see that it suffices to show that the unit map $\mathbb{1}_C \to U^A_m L^A_m \mathbb{1}_C = U^A_m \mathbb{1}_D$ is an isomorphism. But this map is the canonical comparison map $A_m \xrightarrow{\sim} \mathrm{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}})$, which is an isomorphism as A is a Stefanich ring. □

5.2 Shifting

We will now define the *shift functor* on StRing.

Construction 5.11. The forgetful functors $\mathrm{CAlg}((n+1)\mathrm{Pr}) \to \mathrm{CAlg}(n\mathrm{Pr})$ induce a functor

$$\begin{tikzcd} \mathrm{StRing} \arrow[r, "\simeq"] \arrow[d, "\mathrm{shift}"'] & \lim(\cdots \to \mathrm{CAlg}(3\mathrm{Pr}) \to \mathrm{CAlg}(2\mathrm{Pr}) \to \mathrm{CAlg}(1\mathrm{Pr})) \arrow[d] \\ \mathrm{StRing} \arrow[r, "\simeq"] & \lim(\cdots \to \mathrm{CAlg}(2\mathrm{Pr}) \to \mathrm{CAlg}(1\mathrm{Pr}) \to \mathrm{CAlg}(\mathrm{An})) \end{tikzcd}$$

given on objects by $A = (A_1, A_2, \dots) \mapsto (A_2, A_3, \dots)$.

Lemma 5.12. *The shift functor induces an equivalence*

$$\mathrm{shift}\colon \mathrm{StRing} \xrightarrow{\sim} \mathrm{StRing}_{(1\mathrm{Pr},2\mathrm{Pr},\dots)/}.$$

In particular, for any $n \geq 0$, iterated shifting induces an equivalence

$$\mathrm{shift}^n\colon \mathrm{StRing} \xrightarrow{\sim} \mathrm{StRing}_{(n\mathrm{Pr},(n+1)\mathrm{Pr},\dots)/}.$$

Proof. For each $n \geq 1$, we have the identification

$$\mathrm{CAlg}((n{+}1)\mathrm{Pr}) = \mathrm{CAlg}(\mathrm{Mod}_{n\mathrm{Pr}}(1\mathrm{Pr})) = \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/} = (\mathrm{CAlg}(1\mathrm{Pr})_{(n-1)\mathrm{Pr}/})_{n\mathrm{Pr}/} = \mathrm{CAlg}(n\mathrm{Pr})_{n\mathrm{Pr}/}.$$

These identifications are compatible with the transition functors. Under them, the forgetful functor

$$\mathrm{CAlg}((n+1)\mathrm{Pr}) \to \mathrm{CAlg}(n\mathrm{Pr})$$

is the projection $\mathrm{CAlg}(n\mathrm{Pr})_{n\mathrm{Pr}/} \to \mathrm{CAlg}(n\mathrm{Pr})$, while the same identification remembers the map from $n\mathrm{Pr}$. Since each omitted lower-degree term in the limit presentation of StRing is forced by the next one, we may start the limit at $\mathrm{CAlg}(2\mathrm{Pr})$ and obtain

$$\begin{aligned} \mathrm{StRing} &\simeq \lim_{\mathrm{CAT}}\big(\cdots \to \mathrm{CAlg}(4\mathrm{Pr}) \to \mathrm{CAlg}(3\mathrm{Pr}) \to \mathrm{CAlg}(2\mathrm{Pr})\big) \\ &\simeq \lim_{\mathrm{CAT}}\big(\cdots \to \mathrm{CAlg}(3\mathrm{Pr})_{3\mathrm{Pr}/} \to \mathrm{CAlg}(2\mathrm{Pr})_{2\mathrm{Pr}/} \to \mathrm{CAlg}(1\mathrm{Pr})_{1\mathrm{Pr}/}\big) \\ &\simeq \big(\lim_{\mathrm{CAT}}\big(\cdots \to \mathrm{CAlg}(3\mathrm{Pr}) \to \mathrm{CAlg}(2\mathrm{Pr}) \to \mathrm{CAlg}(1\mathrm{Pr})\big)\big)_{(1\mathrm{Pr},2\mathrm{Pr},\dots)/} \\ &\simeq \mathrm{StRing}_{(1\mathrm{Pr},2\mathrm{Pr},\dots)/}. \end{aligned}$$

Here we use that slices commute with limits in CAT. The resulting functor is precisely shift by construction. The iterated statement follows by applying the same argument repeatedly. □

5.3 Spectrification and colimits of Stefanich rings

Throughout this section we fix a Stefanich ring A and study colimits in the category $\mathrm{StRing}(A)$ of internal Stefanich rings in A. Limits and $\aleph_1$-filtered colimits in $\mathrm{StRing}(A)$ are computed pointwise, since they are preserved by the transition functors End_n^A in the limit description of Definition 5.7. Arbitrary colimits are harder to describe directly, since they are not (necessarily) preserved by the functors End_n^A.

Definition 5.13. Let
$$\mathcal{P}_A := \prod_{n\geq 1} \mathrm{CAlg}(A_n),$$
and define an endofunctor
$$\mathrm{End}^A_\bullet \colon \mathcal{P}_A \to \mathcal{P}_A, \qquad (D_n)_{n\geq 1} \mapsto (\mathrm{End}^A_n(D_{n+1}))_{n\geq 1}.$$
The category of *internal preStefanich rings in A* is the lax equalizer
$$\mathrm{PreStRing}(A) := \mathcal{P}_A \times_{\mathcal{P}_A\times\mathcal{P}_A} \mathrm{Ar}(\mathcal{P}_A),$$
where $\mathcal{P}_A \to \mathcal{P}_A \times \mathcal{P}_A$ is $(\mathrm{id}, \mathrm{End}^A_\bullet)$ and $\mathrm{Ar}(\mathcal{P}_A) \to \mathcal{P}_A \times \mathcal{P}_A$ is the source-target functor. Thus an internal preStefanich ring is a tuple $D = (D_1, D_2, D_3, \dots)$ with $D_n \in \mathrm{CAlg}(A_n)$, together with structure maps
$$\alpha_n \colon D_n \to \mathrm{End}^A_n(D_{n+1})$$
that are not assumed to be equivalences. By Definition 5.7, the category $\mathrm{StRing}(A)$ identifies with the full subcategory of $\mathrm{PreStRing}(A)$ spanned by those objects for which all α_n are equivalences.

Lemma 5.14. *The category* $\mathrm{PreStRing}(A)$ *admits small colimits, and the projection* $\mathrm{PreStRing}(A) \to \mathcal{P}_A$ *creates them. In particular, colimits in* $\mathrm{PreStRing}(A)$ *are computed pointwise.*

Proof. Let $I \to \mathrm{PreStRing}(A)$ be a diagram, written as (D^i, α^i) for $i \in I$. Let $D := \mathrm{colim}_i D^i$ in $\mathcal{P}_A$, so that $D_n \simeq \mathrm{colim}_i D^i_n$ in $\mathrm{CAlg}(A_n)$. The maps $\alpha^i \colon D^i \to \mathrm{End}^A_\bullet(D^i)$ assemble to a morphism of diagrams $D^\bullet \to \mathrm{End}^A_\bullet(D^\bullet)$. Hence we obtain a structure map
$$D \simeq \mathrm{colim}_i D^i \to \mathrm{colim}_i \mathrm{End}^A_\bullet(D^i) \to \mathrm{End}^A_\bullet(\mathrm{colim}_i D^i) = \mathrm{End}^A_\bullet(D),$$
where the second arrow is the canonical comparison map induced by the cocone $D^i \to D$. This defines an object (D, α) of $\mathrm{PreStRing}(A)$.

For any $(Y, \beta) \in \mathrm{PreStRing}(A)$, the hom anima $\mathrm{Hom}_{\mathrm{PreStRing}(A)}((D, \alpha), (Y, \beta))$ is the homotopy equalizer of the two maps
$$\mathrm{Hom}_{\mathcal{P}_A}(D, Y) \rightrightarrows \mathrm{Hom}_{\mathcal{P}_A}(D, \mathrm{End}^A_\bullet Y)$$
defined by $\beta \circ (-)$ and $\mathrm{End}^A_\bullet(-) \circ \alpha$. Since D is the colimit of the D^i in $\mathcal{P}_A$, this homotopy equalizer identifies with
$$\lim_i \mathrm{Hom}_{\mathrm{PreStRing}(A)}((D^i, \alpha^i), (Y, \beta)).$$
Thus (D, α) has the universal property of the colimit. □

What we will do is to embed $\mathrm{StRing}(A)$ in $\mathrm{PreStRing}(A)$ and then reflect back.

Lemma 5.15. *The inclusion* $\mathrm{StRing}(A) \hookrightarrow \mathrm{PreStRing}(A)$ *admits a left adjoint.*

Proof. Define a functor
$$T \colon \mathrm{PreStRing}(A) \to \mathrm{PreStRing}(A), \qquad (D_n)_{n\geq 1} \mapsto (D'_n)_{n\geq 1},$$

by setting $D'_n = \mathrm{End}_n^A(D_{n+1})$, with structure maps obtained by applying the functors End_n^A to the structure maps of D. This admits a natural transformation $\eta\colon \mathrm{id} \to T$, given by the structure maps of an internal preStefanich ring. For $X \in \mathrm{PreStRing}(A)$, define $X^{(0)} := X$, $X^{(\alpha+1)} := T(X^{(\alpha)})$, and

$$X^{(\lambda)} := \mathrm{colim}_{\alpha<\lambda} X^{(\alpha)}$$

for limit ordinals λ.

We first show that $X^{(\omega_1)}$ lies in $\mathrm{StRing}(A)$. Each End_n^A is the right adjoint of the morphism Mod_n^A of 1Pr from Construction 5.4, hence preserves $\aleph_1$-filtered colimits; therefore the functor T preserves $\aleph_1$-filtered colimits. Hence

$$T(X^{(\omega_1)}) \simeq T(\mathrm{colim}_{\alpha<\omega_1} X^{(\alpha)}) \simeq \mathrm{colim}_{\alpha<\omega_1} T(X^{(\alpha)}) = \mathrm{colim}_{\alpha<\omega_1} X^{(\alpha+1)} \simeq X^{(\omega_1)}.$$

The last equivalence uses that the successor ordinals are cofinal in ω_1. Under these identifications, the structure map $X^{(\omega_1)} \to T(X^{(\omega_1)})$ is the canonical map

$$\mathrm{colim}_{\alpha<\omega_1} X^{(\alpha)} \to \mathrm{colim}_{\alpha<\omega_1} X^{(\alpha+1)},$$

hence is an equivalence. Thus $X^{(\omega_1)}$ is an internal Stefanich ring in A.

It remains to prove the universal property. Let $Y \in \mathrm{StRing}(A)$. We claim by transfinite induction that the natural map

$$\mathrm{Hom}_{\mathrm{PreStRing}(A)}(X^{(\alpha)}, Y) \to \mathrm{Hom}_{\mathrm{PreStRing}(A)}(X, Y)$$

is an equivalence for every ordinal $\alpha \leq \omega_1$. The claim is clear for $\alpha = 0$. For a successor ordinal, it suffices to show that

$$\eta^*_{X^{(\alpha)}}\colon\ \mathrm{Hom}_{\mathrm{PreStRing}(A)}(TX^{(\alpha)}, Y) \to \mathrm{Hom}_{\mathrm{PreStRing}(A)}(X^{(\alpha)}, Y)$$

is an equivalence. Since Y lies in $\mathrm{StRing}(A)$, the map $\eta_Y\colon Y \to TY$ is an equivalence. An inverse to $\eta^*_{X^{(\alpha)}}$ is therefore given by applying T to a map $X^{(\alpha)} \to Y$ and then using η_Y^{-1} to identify TY with Y.

For a limit ordinal λ, the object $X^{(\lambda)}$ is the colimit of the $X^{(\alpha)}$ for $\alpha < \lambda$, so

$$\mathrm{Hom}_{\mathrm{PreStRing}(A)}(X^{(\lambda)}, Y) \simeq \lim_{\alpha<\lambda} \mathrm{Hom}_{\mathrm{PreStRing}(A)}(X^{(\alpha)}, Y) \simeq \mathrm{Hom}_{\mathrm{PreStRing}(A)}(X, Y).$$

Taking $\alpha = \omega_1$ gives

$$\mathrm{Hom}_{\mathrm{StRing}(A)}(X^{(\omega_1)}, Y) \simeq \mathrm{Hom}_{\mathrm{PreStRing}(A)}(X, Y),$$

naturally in X and Y. Thus $X \mapsto X^{(\omega_1)}$ is left adjoint to the inclusion. □

Corollary 5.16. *We may compute colimits in* $\mathrm{StRing}(A)$ *by computing them in* $\mathrm{PreStRing}(A)$ *(where they are computed pointwise) and then reflecting them back into* $\mathrm{StRing}(A)$. □

Remark 5.17. The Stefanich ring $\mathbb{P}$ is initial in StRing, so the slice $\mathrm{StRing}_{\mathbb{P}/}$ is all of StRing. Hence Proposition 5.8 yields an equivalence $\mathrm{StRing} \simeq \mathrm{StRing}(\mathbb{P})$, and we abbreviate $\mathrm{PreStRing} := \mathrm{PreStRing}(\mathbb{P})$. Specializing the above to $A = \mathbb{P}$, colimits in StRing are computed pointwise in PreStRing and then spectrified.

5.4 Tensor products of Stefanich rings

Consider a diagram of the form $B \leftarrow A \to C$. We wish to compute the pushout $B \otimes_A C$ in StRing. By Remark 5.17 (the case $A = \mathbb{P}$ of Corollary 5.16), we can do this by computing the pushout in PreStRing, where colimits are pointwise by Lemma 5.14. This gives the preStefanich ring

$$(B_0 \otimes_{A_0} C_0, B_1 \otimes_{A_1} C_1, \dots),$$

and the actual tensor product is obtained by spectrifying this object.

In certain situations, it is useful to do a relative version of the same construction which uses the slice description from Proposition 5.8.

Construction 5.18. Fix a Stefanich ring $A \in \mathrm{StRing}$. By Proposition 5.8 we have $\mathrm{StRing}_{A/} \simeq \mathrm{StRing}(A)$, and by Corollary 5.16 colimits in $\mathrm{StRing}(A)$ are computed by passing to $\mathrm{PreStRing}(A)$, where they are pointwise, and then spectrifying.

Now let $B, C \in \mathrm{StRing}_{A/}$. Under the equivalence of Proposition 5.8, they correspond to the objects of $\mathrm{PreStRing}(A)$ given by the sequences $((B/A)_m)_{m\geq 0}$ and $((C/A)_m)_{m\geq 0}$, where $(B/A)_m \in \mathrm{CAlg}(A_{m+1})$. Their coproduct in $\mathrm{PreStRing}(A)$ is therefore the pointwise tensor product

$$\big((B/A)_0 \otimes (C/A)_0,\ (B/A)_1 \otimes (C/A)_1,\ \dots\big),$$

where the m-th tensor product is formed in $\mathrm{CAlg}(A_{m+1})$. The coproduct in $\mathrm{StRing}_{A/}$ is obtained by spectrifying this internal preStefanich ring.

Note that this method of computing the tensor product was already used at the end of the previous talk, in the proof of Corollary 4.13.

Construction 5.19. In a yet more refined way, let $g\colon A \to C$ be a morphism of Stefanich rings. The cobase-change functor

$$g^*\colon \mathrm{StRing}_{A/} \to \mathrm{StRing}_{C/}, \qquad B \mapsto B \otimes_A C$$

is computed by passing from $\mathrm{PreStRing}(A)$ to $\mathrm{PreStRing}(C)$. Namely, first apply levelwise extension of scalars

$$g_m^*\colon \mathrm{CAlg}(A_{m+1}) \to \mathrm{CAlg}(C_{m+1})$$

to the sequence attached to B. This produces the internal preStefanich ring in C

$$(g_0^*(B/A)_0,\ g_1^*(B/A)_1,\ g_2^*(B/A)_2,\ \dots).$$

The structure maps are obtained by applying g_m^* to the structure maps of B and then using the natural comparison

$$g_m^*(\mathrm{End}_{m+1}^A(R)) \to \mathrm{End}_{m+1}^C(g_{m+1}^*(R)), \qquad R \in \mathrm{CAlg}(A_{m+2}).$$

In other words, for $R_m = (B/A)_m$, the structure map is the composite

$$g_m^* R_m \to g_m^* \mathrm{End}_{m+1}^A(R_{m+1}) \to \mathrm{End}_{m+1}^C(g_{m+1}^* R_{m+1}).$$

Spectrifying this pre-object in $\mathrm{PreStRing}(C)$ gives $B \otimes_A C$. Thus the displayed tuple should be viewed as the first approximation to the base change; the spectrification step is what forces the transition maps to become equivalences.

During the seminar, this talk also covered the theory of affine maps of Stefanich rings. For the sake of exposition, that material has been collected in Section 6.2 of the following chapter.

6 Properties of maps of Stefanich rings

Talk given by Tashi.

In this talk, we will introduce a variety of properties of maps of Stefanich rings: affine maps, prim maps, proper maps, suave maps, and étale maps.

6.1 Recollections and notations

Before we start, let us recall the notation for the transition maps attached to a morphism of Stefanich rings. First recall the external construction. Let

$$\mathbb{P} = (*, \mathrm{An}, 1\mathrm{Pr}, 2\mathrm{Pr}, \dots)$$

denote the initial Stefanich ring, with $\mathbb{P}_n = (n-1)\mathrm{Pr}$. The absolute transition adjunctions are the adjunctions

$$\mathrm{Mod}^{\mathbb{P}}_n : \mathrm{CAlg}((n-1)\mathrm{Pr}) \rightleftarrows \mathrm{CAlg}(n\mathrm{Pr}) : \mathrm{End}^{\mathbb{P}}_n .$$

Thus a Stefanich ring A consists of objects $A_n \in \mathrm{CAlg}(n\mathrm{Pr})$ together with structure equivalences

$$A_n \xrightarrow{\sim} \mathrm{End}^{\mathbb{P}}_{n+1}(A_{n+1}) = \mathrm{End}_{A_{n+1}}(\mathbb{1}_{A_{n+1}}).$$

By adjunction, these are equivalently encoded by symmetric monoidal left adjoints

$$L^A_n : \mathrm{Mod}_{A_n}(n\mathrm{Pr}) \to A_{n+1}.$$

In the previous talk, we repeated this construction internally to a fixed Stefanich ring A. Namely, for $m \geq 1$ there are internal module-endomorphism adjunctions

$$\mathrm{Mod}^A_m : \mathrm{CAlg}(A_m) \rightleftarrows \mathrm{CAlg}(A_{m+1}) : \mathrm{End}^A_m .$$

These adjunctions give the category $\mathrm{StRing}(A)$ of Stefanich rings internal to A, and Proposition 5.8 identifies it with the slice category $\mathrm{StRing}_{A/}$. Under this equivalence, a map $f : A \to B$ is sent to the internal Stefanich ring B/A whose degree n term is

$$(B/A)_n = f_{n,*}(\mathbb{1}_{B_{n+1}}) \quad \in \quad \mathrm{CAlg}(A_{n+1}),$$

as in Construction 5.3. In particular, B/A comes equipped with internal structure equivalences

$$(B/A)_n \xrightarrow{\sim} \mathrm{End}^A_{n+1}((B/A)_{n+1}).$$

Adjointing these equivalences gives maps in $\mathrm{CAlg}(A_{n+2})$

$$L_n^{B/A}\colon \mathrm{Mod}^A_{n+1}((B/A)_n) = L^A_{n+1}\big(\mathrm{Mod}_{(B/A)_n}(A_{n+1})\big) \to (B/A)_{n+1}.$$

For $B = A$, this gives the corresponding maps $L_n^{A/A}$ for the unit internal Stefanich ring A/A.
Now regard f as a morphism $A/A \to B/A$ in $\mathrm{StRing}(A)$. In degree n, this gives a map

$$(f/A)^*_{n-1}\colon (A/A)_n \to (B/A)_n$$

in $\mathrm{CAlg}(A_{n+1})$; the shifted subscript follows the convention used for pullback functors. The fact that these maps define a morphism of internal Stefanich rings means that they are compatible with the internal structure equivalences. After adjunction, this compatibility says that the square

$$\begin{tikzcd}
\mathrm{Mod}^A_{n+1}((A/A)_n) \arrow[r, "\mathrm{Mod}^A_{n+1}((f/A)^*_{n-1})"] \arrow[d, "L_n^{A/A}"'] & \mathrm{Mod}^A_{n+1}((B/A)_n) \arrow[d, "L_n^{B/A}"] \\
(A/A)_{n+1} \arrow[r, "(f/A)^*_n"'] & (B/A)_{n+1}
\end{tikzcd}$$

commutes. Since Mod^A_{n+1} preserves monoidal units, $L_n^{A/A}$ is an equivalence, and thus we see that the map $(f/A)^*_n$ factors as follows:

$$(A/A)_{n+1} \to \mathrm{Mod}^A_{n+1}((B/A)_n) \xrightarrow{L_n^{B/A}} (B/A)_{n+1}.$$

For later use, we also fix notation for internal Hom objects relative to A. Let $m \geq 1$ and let $D \in \mathrm{CAlg}(A_{m+1})$. For $X, Y \in D$, we write

$$\underline{\mathrm{Hom}}^A_D(X,Y) := \underline{\mathrm{Hom}}_{U^A_m(D)}(X,Y) \quad \in \quad A_m,$$

where $U^A_m\colon A_{m+1} \to \mathrm{Mod}_{A_m}(m\mathrm{Pr})$ is the right adjoint from Construction 5.4. In particular,

$$\mathrm{End}^A_m(D) = \underline{\mathrm{Hom}}^A_D(\mathbb{1}_D, \mathbb{1}_D).$$

6.2 Affine maps

During the seminar, this material was presented as part of the previous talk; it has been collected here for the sake of exposition.

Definition 6.1. Consider $n \geq 0$, and let $A \in \mathrm{StRing}$ be a Stefanich ring. Then $B \in \mathrm{StRing}_{A/}$ is called *n-affine* if it lies in the essential image of the functor $\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{StRing}_{A/}$.

Thus an n-affine object over A is obtained from a single algebra

$$R \in \mathrm{CAlg}(A_{n+1})$$

by inserting it as the relative algebra in degree n and then applying the left adjoint transition functors in all higher degrees. In other words, if B is the n-affine object determined by R, then

$$(B/A)_n \simeq R, \qquad (B/A)_{m+1} \simeq \mathrm{Mod}^A_{m+1}((B/A)_m) \quad (m \geq n),$$

while the lower relative algebras are forced by the right adjoint transitions End^A_m. Equivalently, for an arbitrary $B \in \mathrm{StRing}_{A/}$, the object B is n-affine precisely when the relative structure maps

$$L^{B/A}_m \colon \mathrm{Mod}^A_{m+1}((B/A)_m) \to (B/A)_{m+1}$$

are equivalences for all $m \geq n$.

The fully faithful embedding $\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{StRing}_{A/}$ already says that the recursive construction of an n-affine object does not lose information. We will also need a slightly different recovery statement: in high degrees, the relative algebra $(B/A)_{m+1}$ is obtained by applying L^A_{m+1} to the A_{m+1}-linear category $\mathrm{Mod}_{(B/A)_m}(A_{m+1})$, and we want to recover this linear category from its image in A_{m+2}. This is where Lemma 5.10 enters. If $R \in \mathrm{CAlg}(A_{m+1})$ is countably presented, then $\mathrm{Mod}_R(A_{m+1})$ is dualizable in $\mathrm{Mod}_{A_{m+1}}((m+1)\mathrm{Pr})$. More generally, after writing R as an $\aleph_1$-filtered colimit of countably presented algebras, and using that $R \mapsto \mathrm{Mod}_R(A_{m+1})$ preserves such colimits, $\mathrm{Mod}_R(A_{m+1})$ is an $\aleph_1$-filtered colimit of dualizable objects. On this class, the functor

$$L^A_{m+1} \colon \mathrm{Mod}_{A_{m+1}}((m+1)\mathrm{Pr}) \to A_{m+2}$$

is fully faithful.

Remark 6.2. We have a factorization $\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{CAlg}(A_{n+2}) \hookrightarrow \mathrm{StRing}_{A/}$. In particular, any n-affine Stefanich ring under A is also $(n+1)$-affine.

Here are some basic properties of affineness:

Proposition 6.3. *Let $n \geq 0$ and $A \in \mathrm{StRing}$.*

(1) *The class of n-affines over A is closed under small colimits.*

(2) *If $B \in \mathrm{StRing}_{A/}$ is n-affine, then the comparison map $\mathrm{Mod}_{(B/A)_m}(A_{m+1}) \to B_{m+1}$ is an equivalence for $m \geq n$.*

(3) *Given a map $g \colon A \to C$ in StRing and if $B \in \mathrm{StRing}_{A/}$ is n-affine, then also $g^*B = B \otimes_A C \in \mathrm{StRing}_{C/}$ is n-affine. Moreover, we have*

$$g^*_m(B/A)_m \simeq (B \otimes_A C/C)_m$$

for $m \geq n$.

(4) *Given maps $f \colon A \to B$ and $h \colon B \to C$ in StRing with composite $g \colon A \to C$, if f is n-affine, then g is n-affine if and only if h is.*

Proof. (1) is clear, since $\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{StRing}_{A/}$ is a fully faithful left adjoint.

(2) Let $m \geq n$, and let U^A_{m+1} denote the right adjoint of

$$L^A_{m+1} \colon \mathrm{Mod}_{A_{m+1}}((m+1)\mathrm{Pr}) \to A_{m+2}.$$

The comparison map in question is the unit map

$$\mathrm{Mod}_{(B/A)_m}(A_{m+1}) \to U^A_{m+1} L^A_{m+1}\big(\mathrm{Mod}_{(B/A)_m}(A_{m+1})\big),$$

followed by the identification of the target with B_{m+1}. Indeed, by the recursive description above, since B is n-affine over A, we have

$$\mathrm{Mod}^A_{m+1}((B/A)_m) \simeq (B/A)_{m+1}.$$

Moreover, by the discussion preceding the proposition, after writing $(B/A)_m$ as an $\aleph_1$-filtered colimit of countably presented algebras, the object $\mathrm{Mod}_{(B/A)_m}(A_{m+1})$ is an $\aleph_1$-filtered colimit of dualizable objects. Hence Lemma 5.10 implies that the above unit map is an equivalence. Applying the displayed identification, the target becomes

$$U^A_{m+1}((B/A)_{m+1}) = \mathrm{fgt}((B/A)_{m+1}) \simeq B_{m+1},$$

as desired.

(3) We use the description of cobase change from Construction 5.19. The pre-object whose spectrification computes g^*B is represented in $\mathrm{PreStRing}(C)$ by the sequence

$$(g_0^*(B/A)_0,\ g_1^*(B/A)_1,\ g_2^*(B/A)_2,\ \dots).$$

For $m \geq n$, the recursive description of n-affineness gives

$$(B/A)_{m+1} \simeq \mathrm{Mod}^A_{m+1}((B/A)_m).$$

Extension of scalars is compatible with these transition functors, in the sense that for $R \in \mathrm{CAlg}(A_{m+1})$ we have

$$g^*_{m+1}\mathrm{Mod}^A_{m+1}(R) \simeq \mathrm{Mod}^C_{m+1}(g^*_m R).$$

Thus the transition maps of the above preStefanich ring are already equivalences in all degrees $m \geq n$. Spectrification therefore leaves those degrees unchanged, and the result is the n-affine Stefanich ring over C determined by $g^*_n(B/A)_n$. In particular,

$$g^*_m(B/A)_m \simeq (B \otimes_A C/C)_m$$

for $m \geq n$.

(4) Assume B is n-affine over A, so that it is the image of $(B/A)_n \in \mathrm{CAlg}(A_{n+1})$. Slicing the fully faithful functor

$$\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{StRing}_{A/}$$

at this object gives a fully faithful functor

$$\mathrm{CAlg}(A_{n+1})_{(B/A)_n/} \hookrightarrow \mathrm{StRing}_{B/},$$

whose essential image consists precisely of those objects $C \in \mathrm{StRing}_{B/}$ for which the composite $A \to C$ is n-affine. On the other hand, by part (2),

$$B_{n+1} \simeq \mathrm{Mod}_{(B/A)_n}(A_{n+1}),$$

and therefore

$$\mathrm{CAlg}(B_{n+1}) \simeq \mathrm{CAlg}(A_{n+1})_{(B/A)_n/}.$$

Under this equivalence, the functor $\mathrm{CAlg}(B_{n+1}) \hookrightarrow \mathrm{StRing}_{B/}$ has the same essential image as the sliced functor above. Thus C is n-affine over B if and only if it is n-affine over A. $\square$

Remark 6.4. The converse to part (2) is *not* necessarily true. Let k be a field, let

$$f: * \to B^2\mathbb{G}_m$$

be the base point, and let $A \to B$ be the corresponding map of Stefanich rings. One can show that f is 1-affine, so the comparison in part (2) is an equivalence for $m \geq 1$. It is also an equivalence for $m = 0$: here $A_1 = B_1 = \mathrm{D}(k)$ and $(B/A)_0 = k$, so the comparison is just

$$\mathrm{Mod}_k(\mathrm{D}(k)) \simeq \mathrm{D}(k).$$

Nevertheless f is not 0-affine. Indeed, if it were 0-affine, then B would be determined over A by the algebra $(B/A)_0 \in \mathrm{CAlg}(A_1)$. But $(B/A)_0 = k$ is the unit algebra, and the 0-affine object corresponding to the unit is just the identity object $A \in \mathrm{StRing}_{A/}$. This would force $A \simeq B$, i.e. $* \simeq B^2\mathbb{G}_m$, which is false.

6.3 Prim maps

Definition 6.5. The map $f: A \to B$ is called *n-prim* if for all $i \geq n$ the relative map

$$(f/A)_i^*: (A/A)_{i+1} \to (B/A)_{i+1}$$

admits a right adjoint in A_{i+2}.

Observation 6.6. Assume that $(f/A)_i^*: \mathbb{1} = (A/A)_{i+1} \to (B/A)_{i+1}$ admits a right adjoint

$$(f/A)_{i,*}: (B/A)_{i+1} \to (A/A)_{i+1}$$

in A_{i+2}. For $M \in (B/A)_{i+1}$, this right adjoint is given by the A-internal Hom

$$(f/A)_{i,*}(M) \simeq \underline{\mathrm{Hom}}^A_{(B/A)_{i+1}}(\mathbb{1}_{(B/A)_{i+1}}, M).$$

In particular, we see that

$$(f/A)_{i,*}(f/A)_i^*\,\mathbb{1} \simeq \underline{\mathrm{Hom}}^A_{(B/A)_{i+1}}(\mathbb{1}_{(B/A)_{i+1}}, \mathbb{1}_{(B/A)_{i+1}}) = \mathrm{End}^A_{i+1}((B/A)_{i+1}).$$

By the internal Stefanich-ring structure on B/A, this is just $(B/A)_i$.

Thus, under the equivalence $\mathrm{End}_{A_{i+2}}(\mathbb{1}) \xrightarrow{\sim} A_{i+1}$ given by evaluating at the unit, the algebra $(B/A)_i$ is the value of the monad $(f/A)_{i,*}(f/A)_i^*$ on the unit of $(A/A)_{i+1}$.

Proposition 6.7. *Let $A \in$ StRing.*

(1) *If B/A is n-affine and countably presented, then B/A is n-prim.*

(2) *The n-prim A-Stefanich algebras are closed under $\aleph_0$-filtered colimits.*

(3) *Consider a pushout square of Stefanich rings*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & B \arrow[d] \\ C \arrow[r, "f'"'] \arrow[ur, phantom, "\ulcorner"] & C \otimes_A B. \end{tikzcd}$$

If f is n-prim, then so is f', and we have $g^*(B/A)_m = (B \otimes_A C/C)_m$ for $m \geq n$. (In particular, we do not need to "spectrify".) More precisely, the formation of the right adjoint $(f/A)_{m,*}$ commutes with base change for $m \geq n$.

(4) *The n-prim maps are closed under composition.*

Proof. (4) Let $g\colon B \to C$ be a further n-prim map, with composite $h = g \circ f\colon A \to C$. Fix $i \geq n$. Since g is n-prim, the relative map $(g/B)_i^*\colon \mathbb{1} \to (C/B)_{i+1}$ admits a right adjoint in B_{i+2}. The forgetful functor $B_{i+2} \to A_{i+2}$ is a 2-functor of the underlying 2-categories, and any 2-functor preserves adjunctions; applying it to the previous adjunction exhibits a right adjoint of

$$(B/A)_{i+1} \xrightarrow{(g/A)_i^*} (C/A)_{i+1}$$

in A_{i+2}. Since f is n-prim, the relative map $(f/A)_i^*\colon \mathbb{1} \to (B/A)_{i+1}$ also admits a right adjoint in A_{i+2}. Composing the two right adjoints exhibits a right adjoint of

$$(h/A)_i^*\colon \mathbb{1} \xrightarrow{(f/A)_i^*} (B/A)_{i+1} \xrightarrow{(g/A)_i^*} (C/A)_{i+1}$$

in A_{i+2}. As $i \geq n$ was arbitrary, h is n-prim.

(1) The map $(f/A)_i^*\colon (A/A)_{i+1} \to (B/A)_{i+1}$ factors as

$$(A/A)_{i+1} \to \mathrm{Mod}_{i+1}^A((B/A)_i) \xrightarrow{L_i^{B/A}} (B/A)_{i+1}.$$

If the second map is an equivalence, then this composite has a right adjoint in A_{i+2} if and only if the first map admits a right adjoint. This is something we showed in the previous chapter (cf. Proposition 5.8).

(3) We show that in degrees $i \geq n$ the levels $g^*(B/A)_i$ already assemble into a Stefanich ring, i.e. that the canonical maps $g^*(B/A)_i \to \mathrm{End}_{g^*(B/A)_{i+1}}(\mathbb{1})$ are isomorphisms; this implies that $g^*(B/A)_\bullet$ already agrees with $(C \otimes_A B/C)_\bullet$ in these degrees, so that no spectrification is needed. Fix $i \geq n$. Since f is n-prim, we have the internal adjunction $(f/A)_i^* \dashv (f/A)_{i,*}$ in A_{i+2}, and by the observation preceding the proposition the algebra $(B/A)_i \in \mathrm{CAlg}(A_{i+1})$ is the value of the composite

$$\mathbb{1} \xrightarrow{(f/A)_i^*} (B/A)_{i+1} \xrightarrow{(f/A)_{i,*}} \mathbb{1}$$

under the equivalence $\mathrm{End}_{A_{i+2}}(\mathbb{1}) \xrightarrow{\sim} A_{i+1}$. Applying the cobase-change 2-functor $g_{i+1}^*\colon A_{i+2} \to C_{i+2}$ to this adjunction yields an internal adjunction

$$\mathbb{1} \xrightarrow{g_{i+1}^*((f/A)_i^*)} g_{i+1}^*((B/A)_{i+1}) \xrightarrow{g_{i+1}^*((f/A)_{i,*})} \mathbb{1}$$

in C_{i+2}; in particular $g_{i+1}^*((f/A)_i^*)\colon \mathbb{1} \to g_{i+1}^*((B/A)_{i+1})$ admits the right adjoint $g_{i+1}^*((f/A)_{i,*})$ in C_{i+2}. Computing $\mathrm{End}_{g_{i+1}^*((B/A)_{i+1})}(\mathbb{1}) \in \mathrm{CAlg}(C_{i+1})$ amounts to evaluating the composite of this unit map with its right adjoint, i.e. the object

$$g_{i+1}^*\big((f/A)_{i,*}(f/A)_i^*\big) \quad \in \quad \mathrm{End}_{C_{i+2}}(\mathbb{1}).$$

On endomorphisms of the unit the 2-functor g_{i+1}^* restricts to $g_i^*\colon A_{i+1} \to C_{i+1}$, while $(f/A)_{i,*}(f/A)_i^*$ corresponds to $\mathrm{End}_{(B/A)_{i+1}}(\mathbb{1}) = (B/A)_i \in A_{i+1}$. Hence the above composite evaluates to

$$g_i^*\big((B/A)_i\big) \quad \in \quad C_{i+1}.$$

Thus the canonical maps $g^*(B/A)_i \to \mathrm{End}_{g^*(B/A)_{i+1}}(\mathbb{1})$ are isomorphisms for $i \geq n$, so $g^*(B/A)_m = (C \otimes_A B/C)_m$ for $m \geq n$. Moreover the map $(C/C)_{i+1} \to g^*(B/A)_{i+1} = (C \otimes_A B/C)_{i+1}$ admits the right adjoint $g_{i+1}^*((f/A)_{i,*})$ in C_{i+2}, so f' is n-prim; and this very identification of the right adjoint as $g_{i+1}^*((f/A)_{i,*})$ shows that the formation of the right adjoint $(f/A)_{i,*}$ commutes with base change.

(2) This is a consequence of the following lemma. □

Lemma 6.8. *Consider a functor $X\colon I \to \mathbb{C} \in 2\mathrm{Pr}$ where I is filtered and $0 \in I$ is an initial object. Assume that each $F_i\colon X_0 \to X_i$ has a right adjoint F_i^R. Then the functor $F\colon X_0 \to \operatorname{colim} X$ admits a right adjoint $F^R\colon \operatorname{colim}_i X_i \to X_0$, and we have*

$$F^R F = \operatorname{colim}_{i\in I} F_i^R F_i \quad \in \quad \mathbb{C}(X_0, X_0).$$

Proof. We argue by the (enriched) Yoneda lemma. For each object $Y \in \mathbb{C}$, the represented functor

$$\mathbb{C}(-, Y)\colon \mathbb{C}^{\mathrm{op}} \to 1\mathrm{Pr}$$

sends a 1-morphism that is a left adjoint in $\mathbb{C}$ to a right adjoint functor of categories. Conversely, a 1-morphism G of $\mathbb{C}$ is a left adjoint as soon as $\mathbb{C}(-, Y)(G)$ admits a right adjoint for every Y, and these right adjoints are compatible with the maps $\mathbb{C}(-, Y) \to \mathbb{C}(-, Y')$ induced by morphisms $Y \to Y'$ (the Beck–Chevalley condition); as usual, it suffices to test this on $Y = X_0$ and $Y = \operatorname{colim} X$. We may therefore assume $\mathbb{C} = 1\mathrm{Pr}$.

In $1\mathrm{Pr}$ this is the standard comparison between the colimit presentation $\operatorname{colim}_{i\in I} X_i$ in Pr^{L} and the limit presentation $\lim_{i\in I^{\mathrm{op}}} X_i$ formed along the right adjoints. Writing $\iota_i\colon X_i \to \operatorname{colim} X$ for the structure maps, each ι_i admits a right adjoint ι_i^R; since $0 \in I$ is initial we have $F = \iota_0$, and hence a right adjoint $F^R := \iota_0^R$. Under the limit presentation F^R is computed by the partial right adjoints, and the Beck–Chevalley condition required above holds by construction. Tracing through this description, the composite $F^R F$ is computed as the colimit

$$F^R F = \operatorname{colim}_{i\in I} F_i^R F_i \quad \in \quad \mathbb{C}(X_0, X_0),$$

the colimit being filtered, so that it is again computed in $\mathbb{C}(X_0, X_0)$ compatibly with the structure maps; the unit $\mathrm{id} \to F^R F$ is the inclusion of the initial index $i = 0$, for which $F_0 = \mathrm{id}_{X_0}$. Alternatively, one writes down the unit and counit of the adjunction explicitly and checks the triangle identities directly. $\square$

6.4 Proper maps

Note that one missing feature of the previous proposition is that n-prim maps are not necessarily closed under diagonals.

Example 6.9. Let k be a field of characteristic 0 (for simplicity), and let X be a variety over k. The map of Stefanich rings corresponding to the de Rham stack $X_{\mathrm{dR}} \to \operatorname{Spec}(k)$ is 0-prim over k, but its diagonal is not. Thus n-primness genuinely fails to be stable under passage to diagonals.

This motivates the following definition.

Definition 6.10. A map $f\colon A \to B$ of Stefanich rings is called *n-proper* if f and all its iterated codiagonals

$$\nabla_f\colon B \otimes_A B \to B, \qquad \nabla_{\nabla_f}\colon B \otimes_{B\otimes_A B} B \to B, \qquad \dots$$

are all n-prim.

Proposition 6.11. *Let $A \in \mathrm{StRing}$.*

(1) *If $B \in \mathrm{StRing}_{A/}$ is n-affine and countably presented, then B is n-proper.*

(2) *The full subcategory of $\mathrm{StRing}_{A/}$ spanned by the n-proper A-algebras is closed under countable colimits.*

(3) *The subcategory in (2) is also closed under cobase change.*

(4) *Let $f\colon A \to B$ be n-proper, and let $g\colon B \to C$ be a map. Then g is n-prim if and only if gf is n-prim. Moreover, g is n-proper if and only if gf is n-proper.*

Proof. All four statements follow from Proposition 6.7, together with the fact that n-affine maps are stable under passing to iterated codiagonals. The latter holds because, up to the identifications of Proposition 6.3, the codiagonal of an n-affine map is a cobase change of an n-affine map.

(1) If B is n-affine and countably presented over A, then so is every iterated codiagonal of $f\colon A \to B$, by the stability just mentioned and the fact that countable presentability is preserved under cobase change. Hence each iterated codiagonal is n-prim by Proposition 6.7(1), so f is n-proper.

(2) It suffices to treat sifted colimits and finite coproducts separately. For a sifted colimit, every iterated codiagonal of the colimit map is itself a sifted colimit of cobase changes of the corresponding iterated codiagonals of the diagram, so Proposition 6.7(2) and (3) apply degreewise. Finite coproducts reduce to stability under cobase change and composition, which are parts (3) and (4) below.

(3) The formation of iterated codiagonals commutes with cobase change. Hence, if all codiagonals of f are n-prim, then Proposition 6.7(3) shows that all codiagonals of the cobase change f' are n-prim as well, i.e. f' is n-proper.

(4) An iterated codiagonal of a composite can be written as a composite of cobase changes of the iterated codiagonals of the two individual maps. Together with Proposition 6.7(3) and (4), this shows that n-prim and n-proper maps are closed under composition; in particular, if g is n-prim (resp. n-proper), then so is gf. For the converse, assume f is n-proper and gf is n-prim (resp. n-proper). Factor g as

$$B \to B \otimes_A C \to C,$$

where the first map is the cobase change of $gf\colon A \to C$ along f, hence n-prim (resp. n-proper), and the second map is a cobase change of the codiagonal $\nabla_f\colon B \otimes_A B \to B$, hence n-proper, and in particular n-prim. By the composition statement just proved, g is n-prim (resp. n-proper). □

Being 0-proper is closely related to being *rigid* in the sense of Gaitsgory–Rozenblyum [GR17].

Proposition 6.12. *Let $A \in \mathrm{StRing}$, and let B be a countably presented 1-affine Stefanich A-algebra. Then B is 0-proper over A if and only if $(B/A)_1 \in \mathrm{CAlg}(A_2)$ is* rigid *in A_2, i.e. the following two conditions hold.*

(1) *The unit map $\mathbb{1} \to (B/A)_1$ admits a right adjoint in A_2.*

(2) *The multiplication map $(B/A)_1 \otimes_{A_2} (B/A)_1 \to (B/A)_1$ admits a right adjoint in $\mathrm{Mod}_{(B/A)_1 \otimes_{A_2} (B/A)_1}(A_2)$.*

Proof. Since B is 1-affine and countably presented, it is automatically 1-proper by Proposition 6.11(1). Thus being 0-proper amounts to having, in addition, the relevant right adjoints at the lowest level.

If B is 0-proper, then (1) and (2) hold by applying the 0-prim condition to $f: A \to B$ and to its codiagonal $\nabla_f: B \otimes_A B \to B$, using that by Proposition 6.3(2) we have $(B \otimes_A B)_2 \simeq \mathrm{Mod}_{(B/A)_1 \otimes_{A_2} (B/A)_1}(A_2)$.

Conversely, it suffices to show that (1) and (2) imply that the codiagonal $\nabla_f: B \otimes_A B \to B$ is 0-affine. Indeed, the codiagonal is then certainly 0-proper, and together with $A \to B$ being 0-prim this shows that f is 0-proper. Now the codiagonal is 1-affine and 0-prim, and it admits a section (the diagonal). Hence it is 0-affine by Lemma 6.13. □

The following lemma is often useful to reduce the level of affineness.

Lemma 6.13. *Let $f: A \to B$ be a map of Stefanich rings. Assume that f is $(n+1)$-affine and n-prim, and that there is an $(n+1)$-affine map $g: B \to A$ with $fg = \mathrm{id}_B$. Then f is n-affine.*

Proof. By shifting we may assume $n = 0$. Replacing A by the image of $(B/A)_0 \in \mathrm{CAlg}(A_1)$ in $\mathrm{StRing}_{A/}$, we may assume that $(B/A)_0 = \mathbb{1} \in \mathrm{CAlg}(A_1)$; we must then show that $A \to B$ is an equivalence. By 1-affineness it suffices to show that $\mathbb{1} \to (B/A)_1$ is an equivalence.

Since f is 0-prim, the map $(f/A)_0^*: \mathbb{1} \to (B/A)_1$ admits a right adjoint $(f/A)_{0,*}$ in A_2. Moreover, the unit $\mathrm{id} \to (f/A)_{0,*}(f/A)_0^*$ in $A_1 \simeq \mathrm{End}_{A_2}(\mathbb{1})$ is an equivalence, since it is the map $\mathbb{1} \to (B/A)_0 = \mathbb{1}$. We would like to conclude that $\mathbb{1} \to (B/A)_1$ is an equivalence because g splits it; this is not quite true, but the following weaker statement is. After applying g_1, the map $\mathbb{1} \to (B/A)_1$ becomes the map $(A/B)_1 \to \mathbb{1}$, which is split by $g_0: \mathbb{1} \to (A/B)_1$. Since g_1 preserves the fully faithful adjunction, $\mathbb{1} \to (B/A)_1$ becomes an equivalence after applying g_1. But g is 1-affine, so $g_1: A_2 \simeq \mathrm{Mod}_{(A/B)_1}(B_2) \to B_2$ is conservative, which finishes the proof. □

6.5 Suave maps

Definition 6.14. A map $f: A \to B$ of Stefanich rings is called *n-suave* if the following two conditions are satisfied.

(1) For each $i \geq n$, the map

$$(f/A)_i^*: \mathbb{1} = (A/A)_{i+1} \to (B/A)_{i+1}$$

admits a left adjoint

$$(f/A)_{i,\sharp}: (B/A)_{i+1} \to (A/A)_{i+1}$$

in A_{i+2}. We define $(B/A)_i^! := (f/A)_{i,\sharp}(\mathbb{1})$. The counit of the adjunction defines a map

$$(f/A)_{i-1,!}: (B/A)_i^! \to (A/A)_i$$

in $(A/A)_{i+1}$.

(2) For each i, the map $(f/A)_{i-1,!}: (B/A)_i^! \to (A/A)_i$ admits a right adjoint

$$(f/A)_{i-1}^!: (A/A)_i \to (B/A)_i^!$$

in $(A/A)_{i+1}$.

Note that in this case we have

$$\big((B/A)_i^!\big)^\vee = \mathrm{Hom}_{(A/A)_{i+1}}((f/A)_{i,\sharp}(\mathbb{1}), \mathbb{1}) \simeq \mathrm{End}_{(B/A)_{i+1}}(\mathbb{1}) \simeq (B/A)_i.$$

Note also that the functor $(-)^\vee\colon (A/A)_{i+1} \to (A/A)_{i+1}$ sends the functor $(f/A)_{i-1,!}$ to $(f/A)_{i-1}^*$, so, since it preserves adjunctions, we see that

$$(f/A)_{i-1,\sharp} \simeq ((f/A)_{i-1}^!)^\vee.$$

Proposition 6.15. (1) *The n-suave maps are closed under cobase change. More precisely, if $f\colon A \to B$ is n-suave and $g\colon A \to C$ is any map of Stefanich rings, then the cobase change $f'\colon C \to B \otimes_A C$ is n-suave, and $g_m^*(B/A)_m^! \simeq (B \otimes_A C/C)_m^!$ for $m \geq n$.*

(2) *The n-suave maps are closed under composition.*

Proof. By shifting we may assume $n = 0$. The argument is dual to that of Proposition 6.7; recall from the discussion after Definition 6.14 that under the duality $(-)^\vee$ of $(A/A)_{i+1}$ the suave datum of f corresponds to the prim datum, with $\big((B/A)_i^!\big)^\vee \simeq (B/A)_i$.

(1) For each $i \geq 0$, the first suave condition is an internal adjunction $(f/A)_{i,\sharp} \dashv (f/A)_i^*$ in A_{i+2}. The cobase-change 2-functor $g_{i+1}^*\colon A_{i+2} \to C_{i+2}$ preserves adjunctions, so it carries this to an adjunction in C_{i+2}, exhibiting a left adjoint of $g_{i+1}^*\big((f/A)_i^*\big)$. As in the proof of Proposition 6.7(3), one checks that the base-changed system $g^*(B/A)_\bullet^!$ is already a Stefanich ring, so that no spectrification is needed: applying $(-)^\vee$ identifies the suave adjunction with the prim adjunction, and since $(-)^\vee$ commutes with g_{i+1}^*, the endomorphism computation of Proposition 6.7(3) applies verbatim. Hence $g_{i+1}^*\big((f/A)_i^*\big) = (f'/C)_i^*$, and we obtain

$$g_m^*(B/A)_m^! \simeq (B \otimes_A C/C)_m^! \qquad (m \geq 0).$$

Finally g_{i+1}^* also carries the adjunction $(f/A)_{i-1,!} \dashv (f/A)_{i-1}^!$ of the second suave condition to the corresponding adjunction for f'. Thus f' is 0-suave.

(2) Let $f\colon A \to B$ and $g\colon B \to C$ be 0-suave, with composite $h\colon A \to C$. For $i \geq 0$ the forgetful 2-functor $B_{i+2} \to A_{i+2}$ preserves left adjoints, so the left adjoint of $(g/B)_i^*$ in B_{i+2} yields a left adjoint of the induced map $(B/A)_{i+1} \to (C/A)_{i+1}$ in A_{i+2}. Composing left adjoints along

$$(A/A)_{i+1} \xrightarrow{(f/A)_i^*} (B/A)_{i+1} \to (C/A)_{i+1}$$

produces a left adjoint of $(h/A)_i^*$, which is the first suave condition for h. For the second condition, the resulting maps

$$(C/A)_i^! \to (B/A)_i^! \to (A/A)_i$$

each admit a right adjoint (the first since g is 0-suave, the second since f is), hence so does their composite. Thus h is 0-suave. □

6.6 Étale maps

Definition 6.16. We say that $f\colon A \to B$ is *n-étale* if f and all its codiagonals are n-suave.

Proposition 6.17. *Let $A \in \mathrm{StRing}$ and $n \geq 0$.*

(1) *If $f\colon A \to B$ is n-étale and $g\colon A \to C$ is any map of Stefanich rings, then the cobase change $f'\colon C \to B \otimes_A C$ is n-étale.*

(2) *Let $f\colon A \to B$ be n-étale, and let $g\colon B \to C$ be a map. Then g is n-suave if and only if gf is n-suave, and g is n-étale if and only if gf is n-étale.*

Proof. Since f is n-étale precisely when f and all of its iterated codiagonals are n-suave, both statements follow from Proposition 6.15 by exactly the argument used to deduce Proposition 6.11 from Proposition 6.7. Concretely, the iterated codiagonals of a cobase change (resp. of a composite) are cobase changes (resp. composites of cobase changes) of the iterated codiagonals of the original map(s), and n-suave maps are closed under cobase change and composition by Proposition 6.15. □

6.7 Ambidexterity

In the previous talks we saw that, for $n \geq 1$, all bi-countably presented fpqc stacks give rise to ambidextrous adjunctions, so that the corresponding maps of Stefanich rings are 1-étale and 2-proper. Abstracting that argument, with the general ambidexterity result Proposition 2.17 in place of the geometric input, yields the following.

Proposition 6.18. *Let $f\colon A \to B$ be a map of Stefanich rings.*

(1) *If f and $\nabla_f\colon B \otimes_A B \to B$ are both n-prim, then f is $(n+1)$-suave.*

(2) *If f is n-proper, then f is $(n+1)$-étale.*

(3) *If f and ∇_f are n-suave, then f is $(n+1)$-prim.*

(4) *If f is n-étale, then f is $(n+1)$-proper.*

Moreover, in any of these cases, we have an identification $f_{m,\sharp} \cong f_{m,*}$ for each $m \geq n+1$.

Proof. By shifting we may assume $n = 0$.

(1) Suppose f and ∇_f are 0-prim. Fix $m \geq 1$ and consider the map

$$(f/A)^*_m\colon \mathbb{1} = (A/A)_{m+1} \to (B/A)_{m+1}$$

in the $(\infty,3)$-category A_{m+2} (which lies in $(m+2)\mathrm{Pr}$, hence in $3\mathrm{Pr}$, since $m \geq 1$). As f is 0-prim, $(f/A)^*_m$ admits a right adjoint $(f/A)_{m,*}$ in A_{m+2}; we verify that the unit η and counit ε of this adjunction admit right adjoints, so that Proposition 2.17 applies.

For η, evaluation at the unit gives an equivalence $\mathrm{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$, under which, as in the observation following Definition 6.5, the unit $\eta\colon \mathrm{id} \to (f/A)_{m,*}(f/A)^*_m$ corresponds to the map $(f/A)^*_{m-1}\colon \mathbb{1} \to (B/A)_m$. Since f is 0-prim and $m-1 \geq 0$, this map admits a right adjoint in A_{m+1}.

For ε, the counit lies in $\mathrm{End}_{A_{m+2}}((B/A)_{m+1})$. As in the proof of the ambidexterity property of the functors $f_{n,*}$ in Section 2.4, there is an equivalence

$$\mathrm{End}_{A_{m+2}}\big((B/A)_{m+1}\big) \simeq \mathrm{Mod}_{(B\otimes_A B/A)_m}(A_{m+2}),$$

under which ε corresponds to the relative map of the codiagonal $\nabla_f\colon B \otimes_A B \to B$ at level m. Since ∇_f is 0-prim, this map admits a right adjoint.

Hence Proposition 2.17, applied in A_{m+2}, shows that $(f/A)_{m,*}$ is also a left adjoint $(f/A)_{m,\sharp}$ of $(f/A)^*_m$, and provides a canonical identification $(f/A)_{m,\sharp} \simeq (f/A)_{m,*}$ for $m \geq 1$. By Lemma 6.19 this common functor preserves dualizable objects; in particular $(B/A)^!_m = (f/A)_{m,\sharp}(\mathbb{1})$ is dualizable for $m \geq 1$. As recorded after Definition 6.14, the second suave condition is automatic once $(B/A)^!_m$ is dualizable. Thus f is 1-suave.

(2) If f is 0-proper, then every iterated codiagonal ξ of f has the property that ξ and ∇_ξ are again 0-prim, since the iterated codiagonals of ξ are among those of f. By part (1), each such ξ is 1-suave, so f is 1-étale.

(3) Suppose f and ∇_f are 0-suave. For $m \geq 1$, the map $(f/A)^*_m$ admits a left adjoint $(f/A)_{m,\sharp}$ in A_{m+2}, as f is 0-suave. We apply the dual version of Proposition 2.17 (cf. the remark following it), and so must check that the unit and counit of $(f/A)_{m,\sharp} \dashv (f/A)^*_m$ admit right adjoints. The counit corresponds, via the equivalence $\mathrm{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$, to the map $(f/A)_{m-1,!}$, which admits a right adjoint by the second condition in the definition of 0-suave for f. The unit corresponds, under the relation-object identification used in part (1), to the analogous datum for the codiagonal ∇_f, which admits a right adjoint since ∇_f is 0-suave. Hence the dual ambidexterity result shows that $(f/A)_{m,\sharp}$ is also a right adjoint of $(f/A)^*_m$ for $m \geq 1$, so f is 1-prim.

(4) If f is 0-étale, then every iterated codiagonal ξ of f has ξ and ∇_ξ both 0-suave, so ξ is 1-prim by part (3); hence f is 1-proper.

Finally, the argument above produces the ambidextrous identification for $(f/A)^*_m$ with $m \geq 1$ in the shifted ($n = 0$) picture; unshifting, the canonical identification of Proposition 2.17 reads $(f/A)_{m,\sharp} \simeq (f/A)_{m,*}$ for $m \geq n+1$ in cases (1) and (3), and the same identification holds in cases (2) and (4), applied to f itself. By Lemma 6.19 this common functor preserves dualizable objects. $\square$

The following lemma was used above; in particular, being both n-prim and n-suave forces $f_{m,*}$ and $f_{m,\sharp}$ to preserve dualizable objects.

Lemma 6.19. *Let $\varphi\colon \mathcal{C} \to \mathcal{D}$ be a morphism in $\mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$, and suppose φ admits both a $\mathcal{C}$-linear left adjoint $\varphi_\sharp$ and a $\mathcal{C}$-linear right adjoint φ_*. Then both $\varphi_\sharp$ and φ_* preserve dualizable objects.*

Proof. Let $N \in \mathcal{D}$ be dualizable, with dual $N^\vee$. Since φ is symmetric monoidal, it preserves dualizable objects, and $\mathcal{C}$-linearity of the adjoints yields the projection formulas

$$\varphi_\sharp\big(Y \otimes \varphi(M)\big) \simeq \varphi_\sharp(Y) \otimes M, \qquad \varphi_*\big(Y \otimes \varphi(M)\big) \simeq \varphi_*(Y) \otimes M$$

for $Y \in \mathcal{D}$ and $M \in \mathcal{C}$. Using the adjunction $\varphi_\sharp \dashv \varphi \dashv \varphi_*$ together with these formulas, we compute, naturally in $M \in \mathcal{C}$,

$$\mathrm{Hom}_{\mathcal{C}}(\varphi_\sharp N, M) \simeq \varphi_* \mathrm{Hom}_{\mathcal{D}}(N, \varphi(M)) \simeq \varphi_*\big(N^\vee \otimes \varphi(M)\big) \simeq \varphi_*(N^\vee) \otimes M.$$

Setting $M = \mathbb{1}$ gives $\mathrm{Hom}_{\mathcal{C}}(\varphi_\sharp N, \mathbb{1}) \simeq \varphi_*(N^\vee)$, so the displayed natural equivalence becomes $\mathrm{Hom}_{\mathcal{C}}(\varphi_\sharp N, M) \simeq \mathrm{Hom}_{\mathcal{C}}(\varphi_\sharp N, \mathbb{1}) \otimes M$. Hence $\varphi_\sharp N$ is dualizable, with dual $\varphi_*(N^\vee)$. The symmetric computation, exchanging the roles of the two adjoints, shows that φ_* likewise preserves dualizable objects. $\square$

7 Gestalten

Talk given by Giovanni.

8 Examples

Talk given by Divya.

9 Descendability

Talk given by Shai.

10 Six functors and Gestalten

Talk given by Ou.

11 The Gestalt $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$

Talk given by Marco.

12 Cartier duality

Talk given by Han-Ung.

13 The proof of Cartier duality

Talk given by Sil.

A $\aleph_1$-compact objects

The formalism developed in these notes heavily relies on manipulations with κ-compact objects for uncountable cardinals κ (which we mostly fix to be $\aleph_1$). In this appendix, we record various basic properties of κ-compact objects.

A.1 Compact algebras and modules

We state and prove various basic properties of κ-compact algebras in a κ-presentable category.

Lemma A.1. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$, let $R \in \mathrm{CAlg}(C)$, and let $B \in \mathrm{CAlg}_R(C)$. Then B is κ-compact as an R-algebra if and only if its underlying R-module is κ-compact.*

Proof. The point is that the E_∞-operad is countable. We write

$$\mathrm{Sym}_R \colon \mathrm{Mod}_R(C) \rightleftarrows \mathrm{CAlg}_R(C) \colon U$$

for the free-forgetful adjunction. We will use that κ-small colimits of κ-compact objects are again κ-compact: mapping out of such a colimit gives a κ-small limit of anima, and κ-filtered colimits of animae commute with κ-small limits.

First let $M \in \mathrm{Mod}_R(C)$ be κ-compact. The underlying R-module of the free commutative R-algebra on M is

$$U\,\mathrm{Sym}_R(M) \simeq \coprod_{d\geq 0} (M^{\otimes_R d})_{h\Sigma_d}.$$

Here each tensor power is κ-compact, the homotopy orbits are a countable colimit, and the coproduct is countable. Thus $U\,\mathrm{Sym}_R(M)$ is κ-compact. Also $\mathrm{Sym}_R(M)$ is κ-compact as an algebra, since U preserves κ-filtered colimits.

Since the monad $U\,\mathrm{Sym}_R$ is κ-accessible and preserves κ-compact objects by the formula above, the κ-compact objects in $\mathrm{CAlg}_R(C)$ are generated under κ-small colimits and retracts by free algebras on κ-compact modules. Indeed, for any κ-accessible monad on a κ-presentable category which preserves κ-compact objects, the free algebras on κ-compact objects are κ-compact and generate the algebra category under colimits, so the κ-compact algebras are the retracts of their κ-small colimits.

It remains to check that the class of R-algebras with κ-compact underlying R-module is closed under these operations. Closure under retracts is clear. For κ-small colimits, write such a colimit as the geometric realization of its simplicial replacement; the n-simplices are κ-small coproducts of the algebras in the diagram. A κ-small coproduct of commutative R-algebras is the sifted colimit over its finite sub-coproducts, and the underlying R-module of a finite coproduct is the corresponding finite tensor product in $\mathrm{Mod}_R(C)$. Since U preserves sifted colimits, and since κ-compact R-modules are closed under finite tensor products and κ-small colimits, the underlying module of each simplicial degree, and hence of the realization, is κ-compact. Thus a κ-compact R-algebra has κ-compact underlying R-module.

Conversely, suppose that $U(B)$ is κ-compact. By monadicity, B is the geometric realization of the standard simplicial resolution by free algebras,

$$\cdots \rightrightarrows \mathrm{Sym}_R U \,\mathrm{Sym}_R U(B) \rightrightarrows \mathrm{Sym}_R U(B).$$

By the free-algebra case above, every term in this simplicial object is κ-compact as an algebra. Since the geometric realization is a countable colimit, B is κ-compact in $\mathrm{CAlg}_R(C)$. $\square$

Lemma A.2. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_\kappa)$, and let $A \to B$ be κ-compact in $\mathrm{CAlg}_A(C)$. Then an object $M \in \mathrm{Mod}_B(C)$ is κ-compact as a B-module if and only if its underlying A-module is κ-compact.*

Proof. By Lemma A.1, the underlying A-module of B is κ-compact. We write

$$f^* = B \otimes_A - \colon \mathrm{Mod}_A(C) \rightleftarrows \mathrm{Mod}_B(C) \colon f_*$$

for the extension-restriction adjunction. First suppose that M is κ-compact as a B-module. The κ-compact objects in $\mathrm{Mod}_B(C)$ are generated under κ-small colimits and retracts by the free B-modules $B \otimes X$ with $X \in C^\kappa$. Since f_* preserves colimits and sends such a free module to $B \otimes X$, which is κ-compact as an A-module because the C-action on $\mathrm{Mod}_A(C)$ preserves κ-compact objects, it follows that f_*M is κ-compact.

Conversely, suppose that f_*M is κ-compact as an A-module. The standard monadic resolution

$$M \simeq |\cdots \rightrightarrows f^* f_* f^* f_* M \rightrightarrows f^* f_* M|$$

expresses M as a countable colimit of B-modules of the form f^*N. Since $f_* f^* N \simeq B \otimes_A N$ and the tensor product on $\mathrm{Mod}_A(C)$ preserves κ-compact objects in both variables, the functor $f_* f^*$ preserves κ-compact A-modules. Hence every term in this resolution is of the form f^*N with N κ-compact as an A-module, and is therefore κ-compact as a B-module. Since the realization is a countable, hence κ-small, colimit, M is κ-compact. $\square$

Corollary A.3. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_\kappa)$, and let $g \colon R \to S$ be κ-compact in $\mathrm{CAlg}_R(C)$. Then the restriction functor*

$$g_* \colon \mathrm{Mod}_S(C) \to \mathrm{Mod}_R(C)$$

preserves κ-compact objects.

Proof. This is one direction of Lemma A.2. $\square$

Lemma A.4. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_\kappa)$, and let $A \in \mathrm{CAlg}(C)$.*

(1) The κ-compact objects in $\mathrm{CAlg}_A(C)$ are closed under κ-small colimits and retracts. In particular, they are closed under finite colimits.

(2) *If $B \in \mathrm{CAlg}_A(C)$ is κ-compact and $A \to A'$ is any map in $\mathrm{CAlg}(C)$, then $A' \otimes_A B \in \mathrm{CAlg}_{A'}(C)$ is κ-compact.*

(3) *If $B \in \mathrm{CAlg}_A(C)$ is κ-compact and $D \in \mathrm{CAlg}_B(C)$ is κ-compact, then D is κ-compact as an object of $\mathrm{CAlg}_A(C)$.*

(4) *If $B, D \in \mathrm{CAlg}_A(C)$ are κ-compact and $B \to D$ is a map in $\mathrm{CAlg}_A(C)$, then D is κ-compact as an object of $\mathrm{CAlg}_B(C)$.*

Proof. Part (1) is the general closure property of κ-compact objects in a κ-presentable category. For (2), extension of scalars

$$A' \otimes_A -\colon \mathrm{CAlg}_A(C) \to \mathrm{CAlg}_{A'}(C)$$

is left adjoint to the forgetful functor, and this forgetful functor preserves κ-filtered colimits.

For (3), Lemma A.1 shows that D is κ-compact as a B-module. By Lemma A.2, its underlying A-module is κ-compact, so Lemma A.1 again shows that D is κ-compact as an A-algebra.

For (4), the underlying A-module of D is κ-compact by Lemma A.1. Since B is κ-compact over A, Lemma A.2 implies that D is κ-compact as a B-module. Applying Lemma A.1 once more, D is κ-compact as a B-algebra. □

Lemma A.5. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}}_{\kappa})$, and let $A \to B$ be κ-compact in $\mathrm{CAlg}_A(C)$. If $A \to B$ admits an A-algebra retraction $B \to A$, then A is κ-compact as an object of $\mathrm{CAlg}_B(C)$.*

Proof. Apply Lemma A.4(4) to the map $B \to A$ in $\mathrm{CAlg}_A(C)$: both B and the initial A-algebra A are κ-compact in $\mathrm{CAlg}_A(C)$. □

A.2 Compact objects in sheaf categories

We now collect the compactness facts for presheaves and left exact localizations which are used in the main text.

Lemma A.6. *Let κ be an uncountable regular cardinal, and let C be a small category with finite limits. Then finite limits of κ-small coproducts of representables in $\mathrm{PSh}(C)$ are again κ-small coproducts of representables.*

Proof. The terminal object is representable, so it is enough to prove stability under pullbacks. Let

$$F = \coprod_{a \in A} y(c_a), \qquad G = \coprod_{b \in B} y(d_b), \qquad H = \coprod_{e \in E} y(h_e)$$

with A, B, E κ-small. Consider maps $F \to H$ and $G \to H$. For each $a \in A$, the restriction $y(c_a) \to H$ chooses a summand $e(a) \in E$ and a map $c_a \to h_{e(a)}$ in C. Similarly, for each $b \in B$, the restriction $y(d_b) \to H$ chooses a summand $e'(b) \in E$ and a map $d_b \to h_{e'(b)}$.

Coproducts in a presheaf ∞-topos are disjoint and universal. Therefore the pullback $F \times_H G$ decomposes as

$$F \times_H G \simeq \coprod_{\substack{(a,b) \in A \times B \\ e(a) = e'(b)}} \left(y(c_a) \times_{y(h_{e(a)})} y(d_b) \right).$$

Since the Yoneda embedding preserves finite limits, the summand is represented by $c_a \times_{h_{e(a)}} d_b$. The displayed coproduct is κ-small, because κ is infinite and A and B are κ-small. This proves stability under pullbacks, and hence under all finite limits. $\square$

Lemma A.7. *Let C be a small category, and let $Y_\bullet \in \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{PSh}(C))$ be a simplicial presheaf. For a finite simplicial set K, write*

$$Y(K) := \lim_{\Delta^n \to K} Y_n$$

for the right Kan extension of $Y_\bullet$ from simplices to finite simplicial sets. For $r \geq 0$, set

$$(\mathrm{Ex}^r(Y))_q := Y(\mathrm{sd}^r \Delta^q), \qquad \mathrm{Ex}^\infty(Y) := \mathrm{colim}_{r \in \mathbb{N}} \mathrm{Ex}^r(Y).$$

Then:

(1) Each $(\mathrm{Ex}^r(Y))_q$ is a finite limit of values of $Y_\bullet$.

(2) The natural map $Y_\bullet \to \mathrm{Ex}^\infty(Y)$ induces an equivalence on geometric realizations, and $(\mathrm{Ex}^\infty(Y))_0 \simeq Y_0$.

(3) The simplicial presheaf $\mathrm{Ex}^\infty(Y)$ is a Kan complex in $\mathrm{PSh}(C)$, i.e. the maps

$$\mathrm{Ex}^\infty(Y)(\Delta^n) \to \mathrm{Ex}^\infty(Y)(\partial\Delta^n)$$

are effective epimorphisms, equivalently pointwise surjective on π_0.

(4) If every Y_n is a κ-small coproduct of representables, then each

$$(\mathrm{Ex}^\infty(Y))_{q+1} \times_{(\mathrm{Ex}^\infty(Y))_q} (\mathrm{Ex}^\infty(Y))_{q+1}$$

is the countable colimit, over r, of κ-small coproducts of representables.

Proof. The simplicial set $\mathrm{sd}^r \Delta^q$ is finite. Hence its simplex category is finite, and the formula defining $(\mathrm{Ex}^r(Y))_q$ expresses it as a finite limit of the objects Y_n.

The remaining assertions are the usual internal properties of the Ex^∞ fibrant replacement in an ∞-topos.[1] The degree-zero assertion follows directly from $\mathrm{sd}^r \Delta^0 = \Delta^0$.

For (4), write $Y^{(r)} = \mathrm{Ex}^r(Y)$. By (1) and Lemma A.6, every $Y_q^{(r)}$ is a κ-small coproduct of representables. Since filtered colimits commute with finite limits in presheaves of animae, we have

$$\begin{aligned}(\mathrm{Ex}^\infty(Y))_{q+1} &\times_{(\mathrm{Ex}^\infty(Y))_q} (\mathrm{Ex}^\infty(Y))_{q+1} \\ &\simeq \mathrm{colim}_{r \in \mathbb{N}} \left(Y_{q+1}^{(r)} \times_{Y_q^{(r)}} Y_{q+1}^{(r)} \right).\end{aligned}$$

Each term in the final colimit is again a κ-small coproduct of representables by Lemma A.6. $\square$

Proposition A.8. *Let C be a small category, and let $Y_\bullet \in \mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathrm{PSh}(C))$ be a simplicial presheaf. There is a simplicial object $B_\bullet(Y)$, natural in $Y_\bullet$, with a natural equivalence*

$$|B_\bullet(Y)| \simeq Y_0 \times_{|Y_\bullet|} Y_0.$$

Moreover, if $Y_\bullet^{\mathrm{fib}} := \mathrm{Ex}^\infty(Y)$, then

$$B_q(Y) \simeq Y_{q+1}^{\mathrm{fib}} \times_{Y_q^{\mathrm{fib}}} Y_{q+1}^{\mathrm{fib}},$$

where the two maps $Y_{q+1}^{\mathrm{fib}} \to Y_q^{\mathrm{fib}}$ are the last and first face maps.

[1] See the nLab entry on Kan fibrant replacement, https://ncatlab.org/nlab/show/Kan+fibrant+replacement.

Proof. Set $Y_\bullet^{\mathrm{fib}} := \mathrm{Ex}^\infty(Y)$. By Lemma A.7, the map $Y_\bullet \to Y_\bullet^{\mathrm{fib}}$ induces an equivalence on geometric realizations and identifies degree zero. It is therefore enough to compute

$$Y_0^{\mathrm{fib}} \times_{|Y_\bullet^{\mathrm{fib}}|} Y_0^{\mathrm{fib}}.$$

Let $\mathsf{Dec}_\perp(Y^{\mathrm{fib}})$ and $\mathsf{Dec}_\top(Y^{\mathrm{fib}})$ be the two decalages. Thus both have q-simplices Y_{q+1}^{fib}: the first forgets the last face and degeneracy, and the second forgets the first face and degeneracy. They carry natural maps to $Y_\bullet^{\mathrm{fib}}$, given respectively by the last and first face maps. The augmented simplicial animae

$$\mathsf{Dec}_\perp(Y^{\mathrm{fib}}) \to Y_0^{\mathrm{fib}}, \qquad \mathsf{Dec}_\top(Y^{\mathrm{fib}}) \to Y_0^{\mathrm{fib}}$$

have extra degeneracies, hence their realizations are both equivalent to Y_0^{fib}.

Since $Y_\bullet^{\mathrm{fib}}$ is Kan in $\mathrm{PSh}(C)$, the maps $\mathsf{Dec}_\perp(Y^{\mathrm{fib}}) \to Y_\bullet^{\mathrm{fib}}$ and $\mathsf{Dec}_\top(Y^{\mathrm{fib}}) \to Y_\bullet^{\mathrm{fib}}$ are Kan fibrations in $\mathrm{PSh}(C)$.[2] Thus geometric realization preserves the pullback defining

$$B_\bullet(Y) := \mathsf{Dec}_\perp(Y^{\mathrm{fib}}) \times_{Y_\bullet^{\mathrm{fib}}} \mathsf{Dec}_\top(Y^{\mathrm{fib}}).$$

We obtain

$$|B_\bullet(Y)| \simeq Y_0^{\mathrm{fib}} \times_{|Y_\bullet^{\mathrm{fib}}|} Y_0^{\mathrm{fib}} \simeq Y_0 \times_{|Y_\bullet|} Y_0.$$

The formula for $B_q(Y)$ is immediate from the definition of the two decalages. $\square$

Corollary A.9. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let $Y_\bullet$ be a simplicial object of* $\mathrm{PSh}(C)$ *such that every Y_n is a κ-small coproduct of representables. Then*

$$Y_0 \times_{|Y_\bullet|} Y_0$$

is a κ-small colimit of representables. More precisely, as an object over $Y_0 \times Y_0$, it is the geometric realization of a simplicial object whose q-simplices are countable colimits of κ-small coproducts of representables.

Proof. This follows from Proposition A.8 and part (4) of Lemma A.7. Since κ is uncountable regular, the countable colimits in the simplicial degrees and the countable geometric realization together form a κ-small colimit of representables. $\square$

Proposition A.10. *Let κ be an uncountable regular cardinal, and let C be a small category with finite limits. Then the κ-compact objects of* $\mathrm{PSh}(C)$ *are closed under finite limits.*

Proof. The category $\mathrm{PSh}(C)$ is κ-compactly generated by representables, and its κ-compact objects are κ-small colimits of representables. Indeed, the only point is to absorb retracts: if X is a retract of Y in a category with countable colimits, represented by an idempotent $e\colon Y \to Y$, then

$$X \simeq \operatorname{colim}\left(Y \xrightarrow{e} Y \xrightarrow{e} Y \xrightarrow{e} \cdots\right).$$

The terminal object of $\mathrm{PSh}(C)$ is representable, hence κ-compact. We are thus reduced to pullbacks. Let

$$X \to Y \leftarrow Z$$

[2] For the decalage path fibration, see the nLab entry on decalage, https://ncatlab.org/nlab/show/decalage; for preservation of homotopy pullbacks along such realization fibrations, compare Rezk, *When are homotopy colimits compatible with homotopy base change?*, https://ncatlab.org/nlab/files/RezkHomotopyColimitsBaseChange.pdf.

be a diagram in which all three objects are κ-small colimits of representables. We first reduce to the case where X and Z are representable. Write $X \simeq \operatorname{colim}_i X_i$ and $Z \simeq \operatorname{colim}_j Z_j$ as κ-small colimits of representables, with the induced maps to Y. Since colimits in a presheaf topos are universal, pullback in the slice over Y preserves colimits. Hence

$$X \times_Y Z \simeq \operatorname{colim}_{i,j}(X_i \times_Y Z_j).$$

The indexing category is still κ-small, so it remains to handle the case where X and Z are representable.

Write Y as a κ-small colimit of representables. The simplicial replacement expresses this colimit as the geometric realization of a simplicial object $Y_\bullet$ of $\mathrm{PSh}(C)$ such that each Y_n is a κ-small coproduct of representables. The map $Y_0 \to |Y_\bullet|$ is an effective epimorphism. Since effective epimorphisms in An are the π_0-surjections, maps from representables lift in the slice along effective epimorphisms after evaluation. Thus the two maps $X, Z \to |Y_\bullet|$ admit lifts to Y_0.

Set

$$R := Y_0 \times_{|Y_\bullet|} Y_0.$$

Then

$$X \times_{|Y_\bullet|} Z \simeq (X \times Z) \times_{Y_0 \times Y_0} R,$$

where the map $X \times Z \to Y_0 \times Y_0$ is induced by the chosen lifts. By Corollary A.9, the map $R \to Y_0 \times Y_0$ is the geometric realization, in the slice over $Y_0 \times Y_0$, of a simplicial object $R_\bullet$ such that every R_q is a countable colimit $\operatorname{colim}_s R_{q,s}$ of κ-small coproducts of representables, compatibly over $Y_0 \times Y_0$. Since colimits in a presheaf topos are universal,

$$(X \times Z) \times_{Y_0 \times Y_0} R \simeq |(X \times Z) \times_{Y_0 \times Y_0} R_\bullet|.$$

The q-simplices on the right are

$$\operatorname{colim}_s \left((X \times Z) \times_{Y_0 \times Y_0} R_{q,s}\right).$$

Here $X \times Z$ is representable, and $Y_0 \times Y_0$ is a κ-small coproduct of representables by Lemma A.6. Thus each term $(X \times Z) \times_{Y_0 \times Y_0} R_{q,s}$ is a finite limit of κ-small coproducts of representables, hence is itself a κ-small coproduct of representables. Since $\mathbf{\Delta}^{\mathrm{op}}$ and $\mathbb{N}$ are countable and κ is uncountable regular, the total realization is a κ-small colimit of representables. Therefore $X \times_Y Z$ is a κ-small colimit of representables. ◻

Lemma A.11. *Let κ be an uncountable regular cardinal, and let*

$$L \colon C \rightleftarrows D : \iota$$

be an accessible localization of presentable categories. Assume that C is κ-compactly generated, that C^κ is closed under finite limits, that L preserves finite limits, and that ι preserves κ-filtered colimits. Then D is κ-compactly generated, and D^κ is closed under finite limits.

Proof. Since ι preserves κ-filtered colimits, the left adjoint L preserves κ-compact objects. Thus LK is κ-compact for every $K \in C^\kappa$.

We next show that these objects generate D under κ-filtered colimits. In fact, we will exhibit every object $d \in D$ as an explicit κ-filtered colimit of objects of this form. Observe that the category

$$(C^\kappa)_{/\iota d}$$

is κ-filtered: a κ-small diagram of maps $K_\alpha \to \iota d$ is dominated by the map $\mathrm{colim}_\alpha K_\alpha \to \iota d$, and C^κ is closed under κ-small colimits. Since C is κ-compactly generated, the canonical map

$$\mathrm{colim}_{(K \to \iota d)} K \to \iota d$$

is an equivalence. Applying L gives

$$\mathrm{colim}_{(K \to \iota d)} LK \simeq d.$$

Thus D is κ-compactly generated by the objects LK. Moreover, if d is κ-compact, then the identity of d factors through one of the structure maps $LK \to d$, so d is a retract of some LK.

Now let $F\colon J \to D^\kappa$ be a finite diagram. We apply the previous paragraph not to D itself, but to the induced localization

$$\mathrm{Fun}(J, C) \rightleftarrows \mathrm{Fun}(J, D).$$

Since J is finite, Lemma 1.5 shows that $\mathrm{Fun}(J, C)$ is again κ-compactly generated, with κ-compact generators given by diagrams $J \to C^\kappa$; similarly, F is κ-compact as an object of $\mathrm{Fun}(J, D)$. The right adjoint preserves κ-filtered colimits pointwise. Hence F is a retract, as a diagram, of $LK_\bullet$ for some diagram $K_\bullet\colon J \to C^\kappa$.

Taking limits, $\lim_J F$ is a retract of

$$\lim_J LK_\bullet \simeq L\left(\lim_J K_\bullet\right),$$

where we used that L preserves finite limits. Since C^κ is closed under finite limits, $\lim_J K_\bullet$ lies in C^κ. Therefore $L(\lim_J K_\bullet)$ is κ-compact in D, and so its retract $\lim_J F$ is κ-compact. □

Lemma A.12. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let τ be a Grothendieck topology on C. Suppose that the τ-sheaf condition can be tested on a collection of covering sieves*

$$R \subseteq y(c)$$

which are κ-compact as objects of $\mathrm{PSh}(C)$*. Then τ-sheaves are closed under κ-filtered colimits in* $\mathrm{PSh}(C)$.

In particular, this applies if τ is generated by a pretopology whose covering families have fewer than κ morphisms. More generally, it applies whenever there is such a testing collection in which every sieve is generated by fewer than κ morphisms into the target.

Proof. Let $F \simeq \mathrm{colim}_i F_i$ be a κ-filtered colimit in $\mathrm{PSh}(C)$, where every F_i is a τ-sheaf. Let $R \subseteq y(c)$ be one of the covering sieves in the testing collection. Since $y(c)$ and R are κ-compact in $\mathrm{PSh}(C)$, we have

$$\begin{aligned} F(c) &\simeq \mathrm{Hom}_{\mathrm{PSh}(C)}(y(c), F) \\ &\simeq \mathrm{colim}_i \mathrm{Hom}_{\mathrm{PSh}(C)}(y(c), F_i) \\ &\simeq \mathrm{colim}_i \mathrm{Hom}_{\mathrm{PSh}(C)}(R, F_i) \\ &\simeq \mathrm{Hom}_{\mathrm{PSh}(C)}(R, F). \end{aligned}$$

The middle equivalence uses that each F_i satisfies descent for the covering sieve R. Hence F satisfies descent for every sieve in the testing collection, and is therefore a τ-sheaf.

It remains to justify the final assertion. Let $R \subseteq y(c)$ be the sieve generated by a family

$$\{c_a \to c\}_{a \in A}$$

with A κ-small. Set $U := \coprod_{a \in A} y(c_a)$. The map $U \to y(c)$ factors through an effective epimorphism $U \to R$, and R is the geometric realization of its Čech nerve. Since $R \to y(c)$ is a monomorphism, the terms of this Čech nerve are the corresponding fiber products over $y(c)$:

$$U_n \simeq \coprod_{(a_0, \ldots, a_n) \in A^{n+1}} y(c_{a_0} \times_c \cdots \times_c c_{a_n}).$$

Here the indicated finite limits exist in C. Each U_n is a κ-small coproduct of representables, and the simplicial indexing category is countable. Since κ is uncountable regular, $R \simeq |U_\bullet|$ is a κ-small colimit of representables. Such an object is κ-compact: mapping out of it gives a κ-small limit of evaluations, and κ-filtered colimits of animae commute with κ-small limits. □

Corollary A.13. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let τ be a Grothendieck topology on C. Suppose that τ-sheaves are closed under κ-filtered colimits in* $\mathrm{PSh}(C)$ *(e.g. in the situation of Lemma A.12). Then* $\mathrm{Shv}_\tau(C)$ *is κ-compactly generated, and the κ-compact objects of* $\mathrm{Shv}_\tau(C)$ *are closed under finite limits.*

Moreover, sheafification sends κ-compact presheaves to κ-compact sheaves, and every κ-compact sheaf is a retract of a κ-small colimit of sheafified representables.

Proof. Let

$$L\colon \mathrm{PSh}(C) \rightleftarrows \mathrm{Shv}_\tau(C) : \iota$$

be sheafification and the inclusion. This is an accessible left exact localization, so L preserves finite limits. By assumption, ι preserves κ-filtered colimits. The category $\mathrm{PSh}(C)$ is κ-compactly generated by representables, and Proposition A.10 shows that its κ-compact objects are closed under finite limits. Applying Lemma A.11 gives the first claim.

The same adjunction shows directly that L sends κ-compact presheaves to κ-compact sheaves. The proof of Lemma A.11 also shows that every κ-compact sheaf is a retract of LK for some κ-compact presheaf K. Since κ-compact presheaves are retracts of κ-small colimits of representables, and since such retracts may be absorbed by the usual countable telescope, every κ-compact sheaf is a retract of a κ-small colimit of the objects $Ly(c)$. □

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